On the Partition Function of Yang-Mills Lattice Theories

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Contents

0.1 Acknowledgements . . . 2
0.2 Declaration . . . 2

1 Introduction **3**
1.1 Reading this Thesis . . . 3
1.2 U(N) Lattice Gauge Theory and the Leading Order Formula for the Partition Function . . . 5

2 Summary **15**
2.1 Upper Bound: Localization . . . 16
2.2 Lower Bound . . . 18

3 A-Priori Bounds **20**
3.1 Inputs . . . 20
3.2 A lower bound on the Partition Function . . . 20
3.3 Smallness of the Wilson Action . . . 22

4 Axial Gauge Fixing **25**
4.1 An Upper Bound for the Partition Function Based on Axial Gauge Fixing . . . 30

5 From Lie Group to Lie Algebra **33**

6 Lattice Maxwell Theory **43**
6.1 A Preparation on Gaussian Measures . . . 43
6.2 Lattice Maxwell Theory and Boundary Conditions . . . 45
6.3 Three Key Bounds . . . 50
6.4 Infinite Volume Limit of Lattice Maxwell Theory . . . 63

7 Wilson Action to Maxwell Action **65**
7.1 Linear Algebraic Setting . . . 65
7.2 Quadratic Approximation . . . 65
7.3 Closeness of Wilson action and lattice Hermitian Maxwell action . . . 69

8 Proof of Main Theorem **71**
8.1 Upper Bound . . . 72
8.2 Lower Bound . . . 79

9 Appendix **85**
9.1 U(N) as topological group, and Haar Measure and Integration on U(N) . . . 85
9.2 U(N) as a Smooth Manifold . . . 93
9.3 Limits involving two parameters . . . 96

0.1 Acknowledgements

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Next, I would like to thank Dr. Illia Karabash for being the second examiner and grading some of my exams.

I would also like to thank the mathematics program at the University of Bonn for the opportunity to learn a wide variety of mathematics.

I have many other non-thesis related acknowledgements to give, but I will save these for another time!

0.2 Declaration

In alignment with the rules for Master's theses in Mathematics at Universität Bonn, and out of a personal desire to have the traditional experience of a Master thesis, I generally did not use LLMs for the production of this thesis. I have indicated the one spot where I sought LLM input: see a footnote in Section 6.

1 Introduction

The most compelling introduction to the Yang-Mills Millennium problem can be found on the Clay Institute's website [5]. The goal there is to construct a quantum gauge theory with certain properties. To be physically relevant, a theory must be in the setting of Minkowski spacetime in dimension $d = 4$. On the other hand, due to many theoretical obstructions, it appears to be useful to work on constructing gauge theories in a Euclidean setting. Progress also includes working in lower dimensions, such as $d = 3$. A notable proposed framework for obtaining a Euclidean quantum gauge theory, otherwise known as a Euclidean Yang Mills theory, in dimension $d = 3$ is detailed in [2]. Lattice Yang Mills theories yield a promising approach to obtaining a particular sequence which is required for constructing a Euclidean Yang Mills theory [2, Remark 1.14]. Outside of this particular framework, lattice gauge theories fit into the general idea of solving a continuum problem by discrete approximations.

This thesis is about the partition function of $\mathrm{U}(N)$ lattice gauge theory. Mathematically, the partition function is a normalization constant for a probability measure defined on the lattice. At the time of writing, I am unsure of how to bridge the gap between information on the partition function and information relevant in the direction of [2, Remark 1.14]. Nevertheless, there seems to be agreement that understanding the full asymptotics of the partition function of $\mathrm{U}(N)$ lattice gauge theory would be a major step towards the construction of a Yang Mills theory. Here we study the technique developed by Sourav Chatterjee [4] to compute the leading order term of the free energy, which is the logarithm of the partition function.

1.1 Reading this Thesis

First we note some personal items and apologize for some important shortcomings. This thesis does not include any original results. I rather spent most of my time clarifying various details which did not seem immediate to me in [4]. Also, I did not manage to explain or understand lattice gauge theory in an intuitive or physical sense or derive it from the continuum gauge theory. For example, I could have compared lattice gauge theory to other lattice models, such as the Ising model. In that direction, I note the book [6], which covers classical lattice models (as opposed to quantum ones). Still, I am not sure how to carry the theory in this book over to the quantum setting. There are also the set of notes [7], recommended to me by a physics colleague, which manages to heuristically fit lattice gauge theories into a larger picture consisting of Euclidean field theories, Minkowski field theories, and classical statistical mechanics models. This reference also gives a Wilsonian derivation of $\mathrm{U}(N)$ lattice gauge theories, which one can compare to the derivation presented in [4, Section 3]. There, one learns how the Wilson action arises from the Yang-Mills continuum action on the space of $\mathrm{U}(N)$ connection forms.

Next, some notes on the structure of the thesis and how to navigate it. For the reader who is unaccustomed to $\mathrm{U}(N)$ computations and who would like to review the theory underlying the many quick applications of left or right Haar invariance or the connection to smooth manifold theory, the Appendix entries 9.1 and 9.2 provide my own digestion of such items. There we also present the basic definitions which are assumed throughout the thesis, such as the Hilbert Schmidt norm, and the groups $\mathrm{U}(N)$ and $\mathrm{H}(N)$. Otherwise, we assume these as known.

In subsection 1.2, we introduce the definition of $\mathrm{U}(N)$ lattice gauge theory and the definition of the approximating Gaussian theory: lattice Maxwell theory. The main objective is to state the leading order formula for the partition function, but in a way which makes it clear how a Gaussian theory arises.

In Section 2, we present a compressed and summarized review of Chatterjee's proof of the leading order formula. This section will be helpful for maintaining the thread during later detailed proofs.

The remaining sections prior to the Appendix cover the elements of the proof of the leading order formula. Section 8 fills in the details for tieing all the elements together and dealing rigorously with taking limits in β and n.

Let us record some of the notation that will be used throughout. It is very useful to let $C, C_1, C_2, \cdots$ denote constants depending only on N and d whose values may change from expression to expression. This does not include the constant C_N, which has a definitive value. Sometimes C, C_1, etc. will not actually depend on N or d, but for the purpose of taking limits in the lattice size n and the coupling strength g, it is useful to know that C, C_1, etc. are independent of n and g, and it is easy to indicate this by saying that C, C_1, etc. depend only on N and d.

We have

$$[k] := \{1, \cdots, k\}$$

for a given positive integer k.

I also experimented with the notation for a Borel sigma algebra, for instance, $\mathcal{B}_{\mathrm{U}(N)}$ instead of $\mathcal{B}(\mathrm{U}(N))$. This decision was slightly regretable. The intention was to reduce the number of parentheses.

1.2 $\mathrm{U}(N)$ Lattice Gauge Theory and the Leading Order Formula for the Partition Function

Let d be a positive integer. Let $\mathbb{Z}^d$ be the $d-$dimensional integer lattice. Elements $x \in \mathbb{Z}^d$ are commonly called lattice sites or vertices.

Let $\Lambda \subset \mathbb{Z}^d$ be a subset. Let E_Λ denote the set of positively oriented edges $(x, y) \in \Lambda^2$. Edges are also called links in physics papers. A more explicit formulation of the positively oriented edge set is as follows. For any vector $x = (x_1, \cdots, x_d) \in \mathbb{R}^d$, let $|x|_1 = \sum_{i=1}^d |x_i|$. Then

$$E_\Lambda = \{(x, y) \in \Lambda^2 : |x - y|_1 = 1, x \prec y\}, \tag{1}$$

where $x \prec y$ denotes the lexicographic total ordering on $\mathbb{Z}^d$.

Fix a positive integer $N \geq 1$. The unitary group $\mathrm{U}(N) = \{X \in \mathbb{C}^{N\times N} : UU^* = I = U^*U\}$ is a topological group with a topology induced by the Hilbert Schmidt norm

$$\|X\| = \left(\sum_{1\leq i,j\leq N} |X_{i,j}|^2 \right)^{1/2}.$$

Furthermore, with $\mathcal{B}(\mathrm{U}(N))$, or $\mathcal{B}_{\mathrm{U}(N)}$, the Borel sigma algebra on $\mathrm{U}(N)$, then we have the measure space $(\mathrm{U}(N), \mathcal{B}_{\mathrm{U}(N)}, \sigma)$, where σ denotes Haar probability measure on $\mathrm{U}(N)$. See Appendix 9.1 for a slower introduction to such items.

A unitary configuration on Λ is an assignment of unitary matrices to edges of Λ. The set of all unitary configurations is $\mathrm{U}(N)^{E_\Lambda}$. One has to decide from context whether U indicates a single unitary matrix, that is, $U \in \mathrm{U}(N)$, or whether $U = (U_e)_{e\in E_\Lambda} \in \mathrm{U}(N)^{E_\Lambda}$ is a unitary configuration.

Unitary configurations are random objects. To see this, we introduce some probabilistic notions for product spaces, for which our reference is [9, Chapter 14]. We equip the product set $\mathrm{U}(N)^{E_\Lambda}$ with the product sigma algebra $\bigotimes_{e\in E_\Lambda} \mathcal{B}_{\mathrm{U}(N)}$, which is equal to the Borel sigma algebra on the topological product space $(\mathrm{U}(N)^{E_\Lambda}, \tau)$, where τ denotes the product topology. That is, $\bigotimes_{e\in E_\Lambda} \mathcal{B}_{\mathrm{U}(N)} = \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}}$. We consider product Haar measure $\prod_{e\in E_\Lambda} \sigma$ on $(\mathrm{U}(N)^{E_\Lambda}, \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}})$. Now, let $U = (U_e)_{e\in E_\Lambda} \in \mathrm{U}(N)^{E_\Lambda}$ be a random unitary configuration with distribution $\prod_{e\in E_\Lambda} \sigma$. Here is another way to say the last sentence. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a suitable ambient proability space, and let $U : \Omega \to \mathrm{U}(N)^{E_\Lambda}$ be $\mathcal{F} - \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}}$ measurable with law $U^*\mathbb{P} = \prod_{e\in E_\Lambda} \sigma$. Here, $U^*\mathbb{P}$ denotes the pushforward of $\mathbb{P}$ by the map U. To illustrate this, if we have a Borel subset $B_e \in \mathcal{B}_{\mathrm{U}(N)}$ for each $e \in E_\Lambda$, then the probability of U taking values in (the rectangular cylinder set) $B = \prod_{e\in E_\Lambda} B_e$ is

$$U^*\mathbb{P}(B) = \mathbb{P}(U \in B) = \left(\prod_{e\in E_\Lambda} \sigma \right) \left(\prod_{e\in E_\Lambda} B_e \right) = \prod_{e\in E_\Lambda} \sigma(B_e) \in [0, 1].$$

We are hardly interested, however, in uniformly distributed unitary configurations. To get a more complicated distribution, we introduce the Wilson action.

Let $e_1, \cdots, e_d$ denote the standard basis vectors of $\mathbb{R}^d$. It could happen that e_i denotes an edge rather than a standard basis vector, but it should be decidable based on context. A plaquette p is a closed path in Λ consisting of 4 consecutive positively oriented edges p_1, p_2, p_3, p_4 forming a square in some 2-dimensional subspace of $\mathbb{R}^d$. More explicitly, a plaquette is a triple

$$p = (x, j, k) \in \Lambda \times [d] \times [d] \tag{2}$$

such that $1 \leq j < k \leq d$ and all four of the edges

$$\begin{aligned} p_1 &= (x, x + e_j) \\ p_2 &= (x + e_j, x + e_j + e_k) \\ p_3 &= (x + e_k, x + e_j + e_k) \\ p_4 &= (x, x + e_k) \end{aligned}$$

belong to E_Λ. For each unitary configuration $U \in \mathrm{U}(N)^{E_\Lambda}$ and each plaquette p, we introduce the plaquette variable

$$\begin{aligned} U_p &= U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^* \\ &= U_{p_1} U_{p_2} U_{p_3}^{-1} U_{p_4}^{-1} \in \mathrm{U}(N). \end{aligned}$$

A light analogy is that this is like a total cost for traveling around the loop p, where we incur the inverse cost $U_{p_3}^{-1}$ at p_3 and $U_{p_4}^{-1}$ at p_4, for traveling against the orientation of the edges p_3 and p_4. We denote the set of all plaquettes of the subset Λ by $\mathcal{P}(\Lambda)$. That is,

$$\mathcal{P}(\Lambda) = \{p = (x, j, k) \in \Lambda \times [d] \times [d] : 1 \leq j < k \leq d \text{ and } p_1, p_2, p_3, p_4 \in E_\Lambda\}. \tag{3}$$

The Wilson action for the subset Λ is the function $S_\Lambda : \mathrm{U}(N)^{E_\Lambda} \to [0, \infty)$ given by

$$S_\Lambda(U) = \sum_{p \in \mathcal{P}(\Lambda)} \mathrm{Re}\left(\mathrm{Tr}(I - U_p)\right) \tag{4}$$

for any $U \in \mathrm{U}(N)^{E_\Lambda}$.

The formula for the Wilson action can also be written down as follows. We have

$$\begin{aligned} S_\Lambda(U) &= \sum_{p \in \mathcal{P}(\Lambda)} \frac{1}{2}\|I - U_p\|^2 \\ &= \sum_{p \in \mathcal{P}(\Lambda)} \frac{1}{2}\|I - U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^*\|^2 \end{aligned} \tag{5}$$

The first equality is the content of Lemma 7.2 in [4], which uses nothing other than an observation about the eigenvalues of a unitary matrix. So the Wilson action measures the deviation of the plaquette variables U_p from the identity matrix I.

Lemma 1.1. *Equation 5 holds.*

Proof. Let $U \in \mathrm{U}(N)$. Then U is a normal matrix because $UU^* = I = U^*U$. Therefore U has N eigenvalues $\lambda_1, \cdots, \lambda_N \in \mathbb{C}$. Note that $|\lambda_i| = 1$ for all $i = 1, \cdots, N$. Indeed, fix i and let $\lambda = \lambda_i$. Let $v \in \mathbb{C}^N$ be a non-zero eigenvector associated to λ. Then since unitary matrices are automorphisms of the standard sesequilinear form $\langle \cdot, \cdot \rangle_{\mathbb{C}} : \mathbb{C}^N \times \mathbb{C}^N \to \mathbb{C}$, we have $\langle Uv, Uv \rangle_{\mathbb{C}} = \langle v, v \rangle_{\mathbb{C}}$. But by definition of eigenvector, we have $\langle Uv, Uv \rangle_{\mathbb{C}} = \langle \lambda v, \lambda v \rangle_{\mathbb{C}}$. Then using conjugate linearity in the first slot and linearity in the second slot, we have $\langle \lambda v, \lambda v \rangle_{\mathbb{C}} = |\lambda|^2 \langle v, v \rangle_{\mathbb{C}}$. Putting these observations together, we see that $|\lambda| = 1$, so there exists some $\theta \in [0, 2\pi)$ so that $\lambda = e^{i\theta}$.

Let $\theta_1, \cdots, \theta_N \in [0, 2\pi)$ so that $e^{i\theta_1} = \lambda_1, \cdots, e^{i\theta_N} = \lambda_N$. Since the trace is equal to the sum of the eigenvalues of a matrix, we have $\mathrm{Re}(\mathrm{Tr}(I - U)) = N - \sum_{j=1}^N \cos\theta_j$. On the other hand,

$$\begin{aligned} \frac{1}{2}\|I - U\|^2 &= \frac{1}{2}\,\mathrm{Tr}((I - U)^*(I - U)) \\ &= \frac{1}{2}\,\mathrm{Tr}(I - U^* - U + I) \\ &= \frac{1}{2}(2N - \sum_{j=1}^N (\cos\theta_j + i\sin\theta_j) - \mathrm{Tr}(U^*)). \end{aligned}$$

But the eigenvalues of U^* are $e^{-i\theta_1}, \cdots, e^{-i\theta_N}$, so we have

$$\begin{aligned} \frac{1}{2}\|I - U\|^2 &= \frac{1}{2}(2N - \sum_{j=1}^N (\cos\theta_j + i\sin\theta_j) - \sum_{j=1}^N (\cos(\theta_j) - i\sin(\theta_j)) \\ &= N - \sum_{j=1}^N \cos\theta_j. \end{aligned} \tag{6}$$

Since U_p is unitary for every plaquette p, we see that $\frac{1}{2}\|I - U_p\|^2 = \mathrm{Re}(\mathrm{Tr}(I - U_p))$. □

How does the Wilson action give rise to a more complicated measure on unitary configuration space? It helps to define a Gibbs weight $e^{-\beta S(U)}$ for product Haar measure. Let us proceed rigorously. First we define a measurable function, which will be a density function. Let $g > 0$ be a real number. Let $f : \mathrm{U}(N)^{E_\Lambda} \to [0, \infty)$ be the function defined by $f(U) = \exp\left(-\frac{1}{g^2} S_\Lambda(U)\right)$ for $U \in \mathrm{U}(N)^{E_\Lambda}$. By non-negativity of the Wilson action $S_\Lambda \geq 0$, we verfiy that $f \leq 1$. Furthermore, since $\exp(-x) > 0$, we find that $f > 0$. The Wilson action is continuous with respect to the product topology on configuration space

$\mathrm{U}(N)^{E_\Lambda}$ and the standard topology on $[0,\infty)$. This uses that matrix multiplication is continuous and that the Hilbert Schmidt norm is continuous. Therefore, f is $\mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}} - \mathcal{B}_{[0,\infty)}$ meausurable. Measurability ensures that f can be integrated, which justifies the following. Let $\mu'_{\Lambda,g} : \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}} \to [0,\infty]$ be the measure defined by $\mu'_{\Lambda,g}(B) = \int_B f(U) \prod_{e\in E_\Lambda} d\sigma(U_e)$ for any $B \in \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}}$. Then $\mu'_{\Lambda,g}$ is finite because

$$\begin{aligned}\mu'_{\Lambda,g}(\mathrm{U}(N)^{E_\Lambda}) &= \int_{\mathrm{U}(N)^{E_\Lambda}} f(U) \prod_{e\in E_\Lambda} d\sigma(U_e) \\ &\leq \int_{\mathrm{U}(N)^{E_\Lambda}} 1 \prod_{e\in E_\Lambda} d\sigma(U_e) = 1.\end{aligned}$$

For any $g > 0$, we define $\mu_{\Lambda,g} := \frac{\mu'_{\Lambda,g}}{\mu'_{\Lambda,g}(\mathrm{U}(N)^{E_\Lambda})}$. In other words, $\mu_{\Lambda,g}$ is the probability measure on $(\mathrm{U}(N)^{E_\Lambda}, \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}})$ whose density with respect to product Haar measure $\prod_{e\in E_\Lambda} \sigma$ is

$$\frac{d\mu_{\Lambda,g}}{d\prod_{e\in E_\Lambda}\sigma}(U) = \frac{1}{Z_{\Lambda,g}} \exp\left(-\frac{1}{g^2} S_\Lambda(U)\right) \tag{7}$$

for each $U \in \mathrm{U}(N)^{E_\Lambda}$, where

$$Z_{\Lambda,g} = \int_{\mathrm{U}(N)^{E_\Lambda}} \exp\left(-\frac{1}{g^2} S_\Lambda(U)\right) \prod_{e\in E_\Lambda} d\sigma(U_e) \leq 1 \tag{8}$$

denotes the normalizing constant ensuring that $\mu_{\Lambda,g}$ is a probability measure.

To illustrate this as we did for the uniform probability distribution on configuration space, suppose U is a random unitary configuration with distribution $\mu_{\Lambda,g}$. If $B \in \mathcal{B}_{\mathrm{U}(N)^{E_\Lambda}}$, then the probability of U taking values in B is given by

$$\mathbb{P}(U \in B) = \mu_{\Lambda,g}(B) = \frac{1}{Z_{\Lambda,g}} \int_B \exp\left(-\frac{1}{g^2} S_\Lambda(U)\right) \prod_{e\in E_\Lambda} d\sigma(U_e).$$

The probability measure $\mu_{\Lambda,g}$ is called *lattice gauge theory on* Λ *for the gauge group* $\mathrm{U}(N)$*, with coupling strength* g. For short, it is also called the Yang-Mills measure. The normalization constant $Z_{\Lambda,g}$ is called the *partition function*, and $\log Z_{\Lambda,g}$ is called the *free energy* of the model. Furthermore, we define

$$F_{\Lambda,g} = \frac{\log Z_{\Lambda,g}}{|\Lambda|},$$

which is called the free energy per site, where $|\Lambda|$ denotes the number of vertices in Λ. This thesis is concerned with describing the partition function and free energy when Λ approaches the infinite volume lattice $\mathbb{Z}^d$ and g is close to 0. To make this quantitative, we let

$$\Lambda_n = \{0, 1, \cdots, n-1\}^d \subset \mathbb{Z}^d \tag{9}$$

be the box of width n, for any positive integer n, and we introduce the shorthand notation

$$E_n = E_{\Lambda_n}. \tag{10}$$

We also introduce the parameter $\beta = \frac{1}{g^2}$. This way, as $n, \beta \to \infty$, we have the desired asympotic behavior of $g \to 0$ and $\Lambda_n \to \mathbb{Z}^d$. What can one say about the free energy in this asymptotic regime? The main result will be the leading order formula for the free energy, which is stated ahead in Theorem 1.2. But before getting there, we will present the Gaussian theory which dominates the leading order behavior, and see how it arises from the $\mathrm{U}(N)$ Yang Mills theory. We begin with the following observation and hypothesis.

The density function in (7) bears some resemblance to a Gaussian probability density function because it has an exponential term and a normalization constant. Let us see how far we can take this analogy. Suppose that

$$\frac{1}{Z_{\Lambda_n,g}} \exp\left(-\beta S_{\Lambda_n}(U)\right) \approx \frac{1}{(2\pi)^{k/2}(\det\Sigma)^{1/2}} \exp\left(-\frac{1}{2}\langle x, \Sigma^{-1}x\rangle\right),$$

where the right hand side is a centered Gaussian density function with k degrees of freedom (meaning $x = (x_1, \cdots, x_k)$) and covariance matrix $\Sigma \in \mathbb{R}^{k\times k}$, where $\langle(\cdot), \Sigma^{-1}(\cdot)\rangle : \mathbb{R}^k \times \mathbb{R}^k \to \mathbb{R}$ is some symmetric

bilinear form in the variables $x_1, \cdots, x_k$. If we can figure out what the variables x should be and how the form $\langle (\cdot), \Sigma^{-1}(\cdot)\rangle$ should look, then this is a useful hypothesis for the following reasons. On one hand, if β times the Wilson action is exactly equal to $\frac{1}{2}$ times the form $\langle x, \Sigma^{-1}x\rangle$, then we would have the approximation $Z_{\Lambda_n,g} \approx (2\pi)^{k/2}(\det \Sigma)^{1/2}$ by comparing the two denominators above. Taking this to be valid, then for computing the free energy per site, one would take the logarithm and divide by $|\Lambda_n| = n^d$ and then obtain

$$F_{\Lambda_n,g} \approx \frac{k}{2n^d}\log(2\pi) + \frac{1}{2n^d}\log(\det \Sigma). \tag{11}$$

To make this approximation more accurate however, we would analyze the discrepancy between the Wilson action and the bilinear form, and then take into account the cost in replacing $\exp\left(-\frac{1}{g^2}S_{\Lambda_n}(U)\right)$ by $\exp\left(-\frac{1}{2}\langle x, \Sigma^{-1}x\rangle\right)$.

Unfortunately, it is not clear how to define Gaussian variables on unitary configuration space. For instance, viewing $\mathrm{U}(N)$ as an N^2 dimensional smooth manifold, one could try to define Gaussian variables in coordinate charts, which are subsets of $\mathbb{R}^{N^2}$. But two different configurations could easily have two distinct unitary matrices U_e and V_e on a particular edge, such that U_e and V_e are so far apart that they live inside different coordinate patches.

Fortunately, in the so called axial gauge, a crucial device to be discussed later, we can reduce the configuration space $\mathrm{U}(N)^{E_n}$ to a small neighborhood of the configuration $(I_e)_{e\in E_n}$ with regards to the product topology on $\mathrm{U}(N)^{E_n}$. Another way to say this is that we can argue that $\mathrm{U}(N)$ lattice gauge theory $\mu_{\Lambda_n,g}$ gives high probability to configurations $U \in \mathrm{U}(N)^{E_n}$ satisfying $U_e \approx I$ for all $e \in E_n$. Therefore, on each edge, we can focus on the coordinate patch about the identity matrix. The mathematical device for studying this coordinate patch is the exponential map from the Lie algebra $\mathfrak{u}(N)$, the real vector space of skew-Hermitian matrices, to $\mathrm{U}(N)$, viewed as a Lie group. We note that $\mathrm{u}(N)$ is equal to the imaginary i times the real vector space of Hermitian matrices $\mathrm{H}(N)$.

Motivated by the need for a quadratic form to define a Gaussian measure, we take a quadratic approximation of unitary configurations by replacing U_e on each edge e by the first three terms of its exponential map expansion. That is, we replace U_e by $I + iH_e - H_e^2/2$, where $H_e \in \mathrm{H}(N)$ satisfies $e^{iH_e} = U_e$. Since U_e is in a neighborhood of the identity, this replacement incurs a small error.

Next, we consider what happens when the Wilson action is rewritten using this replacement. A precise statment is as follows. Fix an arbitrary plaquette $p \in \mathcal{P}(B_n)$. Let $H_{p_1}, \cdots, H_{p_4} \in \mathrm{H}(N)$ be four Hermitian matrices satisfying $U_{p_1} = e^{iH_{p_1}}$, $U_{p_2} = e^{iH_{p_2}}$, $U_{p_3} = e^{iH_{p_3}}$, and $U_{p_4} = e^{iH_{p_4}}$. Let $H_p = H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4}$ be another Hermitian matrix. As long as $\|H_{p_j}\| \le \eta$ for some $\eta \le 1$ and each $j = 1, 2, 3, 4$, then

$$\left|\mathrm{Re}\,\mathrm{Tr}(I - U_p) - \frac{1}{2}\|H_p\|^2\right| \le C\eta^3$$

where C is a constant depending only on N. This result is essentially obtained by replacing $U_p = U_{p_1}U_{p_2}U_{p_3}^*U_{p_4}^*$ in the term $\mathrm{Re}\,\mathrm{Tr}(I - U_p)$ by its associated quadratic approximation

$$(I + iH_{p_1} - H_{p_1}^2/2)(I + iH_{p_2} - H_{p_2}^2/2)(I - iH_{p_3} - H_{p_3}^2/2)(I - iH_{p_4} - H_{p_4}^2/2). \tag{12}$$

Then one has

$$I - U_p \approx I - (I + i(H_1 + H_2 - H_3 - H_4) + \text{2nd order terms} + \text{h.o.t}).$$

After taking the real part of the trace, only the second order terms survive (plus the higher order terms which can be estimated away), and we are left with $\frac{1}{2}\|H_p\|^2$.

Performing the same replacement over all plaquettes, we obtain the approximation

$$S(U) \approx \frac{1}{2}\sum_{p\in\mathcal{P}(B_n)}\|H_p\|^2.$$

for configurations $(U_e)_{e\in E_n}$ in a neighborhood of the identity.

Observe that $\sum_{p\in\mathcal{P}(B_n)}\|H_p\|^2$ is already a quadratic form on the real vector space of Hermitian configurations $\mathrm{H(N)}^{E_n}$. To see this, observe that

$$\mathrm{H}(N) \times \mathrm{H}(N) \ni (H, R) \mapsto \mathrm{Re}\,\mathrm{Tr}(H^*R) \in \mathbb{R}$$

defines a symmetric bilinear form on $\mathrm{H}(N)$, and then summing over plaquettes, we have that

$$\mathrm{H(N)}^{E_n} \times \mathrm{H(N)}^{E_n} \ni (H, R) \mapsto \sum_{p \in \mathcal{P}(B_n)} \mathrm{Re}\,\mathrm{Tr}\big(H_p^* R_p\big) \in \mathbb{R}$$

defines a symmetric bilinear form on $\mathrm{H(N)}^{E_n}$ whose diagonal is precisely the map M_n^{H} given by

$$\mathrm{H(N)}^{E_n} \ni H \mapsto \sum_{p \in \mathcal{P}(B_n)} \|H_p\|^2 \in \mathbb{R}.$$

The map M_n^{H} is called the lattice Maxwell action. It acts on Hermitian configurations H with regards to the lattic Λ_n. For a well-defined Gaussian measure, however, we need a positive definite quadratic form. We note that M_n^{H} is not positive definite on $\mathrm{H}(N)^{E_n}$. A simple counterexample is setting $N = 1, d = 2, n = 2$ and all Hermitian matrix edge variables to $1 \in \mathrm{H}(1)$.

Once again, the axial gauge provides the remedy. Let T_n be the particular spanning tree of Λ_n with root $(0, \cdots, 0) \in \Lambda_n$ and containing an edge $e = (x, y) \in E_n$ if and only if $x = (x_1, \cdots, x_j, 0, \cdots, 0)$ and $y = (x_1, \cdots, x_j + 1, 0, \cdots, 0)$ for some $1 \leq j \leq d$. We let E_n^0 denote the edges of T_n. Furthermore, we define $E_n^1 = E_n \setminus E_n^0$. Let $\mathrm{H}(N)^{E_n^1}$ denote the subset $\{H \in \mathrm{H}(N)^{E_n} : H_e = 0 \text{ for all } e \in E_n^0\}$. By a variant of Lemma 6.2, M_n^{H} defines a positive definite quadratic form on $\mathrm{H}(N)^{E_n^1}$.

Let λ denote Lebesgue measure on $\mathrm{H}(N)$. In more detail, let

$$\Psi_H : \mathbb{R}^{N^2} \to \mathrm{H}(N)$$

be the isometry

$$\begin{aligned}(y, z, w) &= \left((y_{jj})_{1 \leq j \leq N}, (z_{jk})_{1 \leq j < k \leq N}, (w_{jk})_{1 \leq j < k \leq N}\right) \\ &\mapsto \begin{bmatrix} y_{11} & & \frac{z_{jk} + i w_{jk}}{\sqrt{2}} \\ & \ddots & \\ \frac{z_{jk} - i w_{jk}}{\sqrt{2}} & & y_{NN} \end{bmatrix}.\end{aligned} \tag{13}$$

Then $d\lambda(H)$ is equal to the pushforward of l_{N^2}, N^2-dimensional Lebesgue measure, by the map Ψ_H:

$$\lambda = \Psi_H^* l_{N^2} = l_{N^2} \circ \Psi_H^{-1}.$$

Furthermore, let δ_0 denote the Dirac measure centered at the matrix $0 \in \mathrm{H}(N)$. Then by the measure $\lambda_{E_n^1}$ on $\mathrm{H}(N)^{E_n^1}$ we mean the product measure

$$d\lambda_{E_n^1}(H) = \left(\prod_{e \in E_n^0} d\delta_0(H_e)\right) \otimes \left(\prod_{e \in E_n^1} d\lambda(H_e)\right).$$

A random Hermitian Maxwell configuration is a configuration $H \in \mathrm{H}(N)^{E_n}$ with distribution

$$d\mu_{\mathrm{Herm},n}(H) = \frac{1}{Z_H(B_n)} \exp\left(-\frac{1}{2} M_n^{\mathrm{H}}(H)\right) d\lambda_{E_n^1}(H)$$

where

$$Z_H(B_n) = \int_{\mathrm{H}(N)^{E_n}} \exp\left(-\frac{1}{2} M_n^{\mathrm{H}}(H)\right) d\lambda_{E_n^1}(H).$$

By definition of the measure $\lambda_{E_n^1}$, integrating over $\mathrm{H}(N)^{E_n}$ gives the same result as integrating over $\mathrm{H}(N)^{E_n^1}$.

Therefore we have a probability space $(\mathrm{H}(N)^{E_n}, \mathcal{B}_{\mathrm{H}(N)^{E_n}}, \mu_{\mathrm{Herm},n})$. Let $\Psi_H^{E_n}$ denote the product map

$$\Psi_H^{E_n}((x_e)_{e \in E_n}) = (\Psi_H(x_e))_{e \in E_n} \in \mathrm{H}(N)^{E_n}$$

for any $(x_e)_{e \in E_n} \in \mathbb{R}^{N^2 |E_n|}$. Let us identify $\mathbb{R}^{N^2 |E_n^1|}$ with the subset

$$\{(x_e)_{e \in E_n} : x_e = 0 \in \mathbb{R}^{N^2} \text{ for all } e \in E_n^0\} \subset \mathbb{R}^{N^2 |E_n|}.$$

Let $l_{N^2|E_n^1|}$ denote the pullback of the measure $\lambda_{E_n^1}$ via the map $\Psi_H^{E_n}$. Then we have

$$dl_{N^2|E_n^1|}(x) = \prod_{e\in E_n^0} d\delta_0(x_e) \prod_{e\in E_n^1} dl_{N^2}(x_e)$$

where in this case, δ_0 denotes the Dirac measure on $\mathbb{R}^{N^2}$ centered at $0 \in \mathbb{R}^{N^2}$.

Let $\mu_{\mathbb{R}^{N^2|E_n^1|}}$ denote the pullback of the measure $\mu_{\mathrm{Herm},n}$ by the map $\Psi_H^{E_n}$. This means that for any Borel subset $B \in \mathcal{B}_{\mathbb{R}^{N^2|E_n|}}$, we have

$$\mu_{\mathbb{R}^{N^2|E_n^1|}}(B) = \mu_{\mathrm{Herm},n}(\Psi_H^{E_n}(B)).$$

Therefore

$$\begin{aligned} d\mu_{\mathbb{R}^{N^2|E_n^1|}}(x) &= \frac{1}{Z_H(B_n)} \exp\left(-\frac{1}{2} M_n^{\mathrm{H}}(\Psi_H^{E_n}(x))\right) dl_{N^2|E_n^1|}(x) \\ &= \frac{1}{Z_H(B_n)} \exp\left(-\frac{1}{2} M_n^{\mathrm{H}}(\Psi_H^{E_n}(x))\right) \prod_{e\in E_n^0} d\delta_0(x_e) \prod_{e\in E_n^1} dl_{N^2}(x_e). \end{aligned}$$

where $x \in \mathbb{R}^{N^2|E_n|}$.

Note that

$$M_n^{(\mathbb{R}^{N^2})} = M_n^{\mathrm{H}} \circ \Psi_H^{E_n}$$

defines a positive definite quadratic form on $(\mathbb{R}^{N^2})^{E_n^1} = \mathbb{R}^{N^2|E_n^1|}$.

We find that

$$\exp\left(-\frac{1}{2} M_n^{(\mathbb{R}^{N^2})}(x)\right) = \prod_{j=1}^{N^2} \exp\left(-\frac{1}{2} \sum_{p\in\mathcal{P}(\Lambda_n)} (x_p^{(j)})^2\right),$$

for all $x \in \mathbb{R}^{N^2|E_n|}$ and where $x_p^{(j)} = x_{p_1}^{(j)} + x_{p_2}^{(j)} - x_{p_3}^{(j)} - x_{p_4}^{(j)}$ for any plaquette $p \in \mathcal{P}(\Lambda_n)$ and any index $j \in [N^2]$.

The quadratic form

$$M_n^{(\mathbb{R})} : \mathbb{R}^{E_n} \to \mathbb{R}$$

given by

$$M_n^{(\mathbb{R})}(t) = \sum_{p\in\mathcal{P}(\Lambda_n)} (t_p)^2$$

is the so called scalar lattice Maxwell action. When restricted to $\mathbb{R}^{E_n^1}$, which we identify with $\{t \in \mathbb{R}^{E_n} : t_e = 0 \text{ for all } e \in E_n^0\}$, then $M_n^{(\mathbb{R})}$ becomes a positive definite quadratic form.

Let τ_n denote the probability measure on $(\mathbb{R}^{E_n}, \mathcal{B}_{\mathbb{R}^{E_n}})$ given by

$$d\tau_n(t) = \frac{1}{Z_M(\Lambda_n)} \exp\left(-\frac{1}{2} M_n^{(\mathbb{R})}(t)\right) \prod_{e\in E_n^0} d\delta_0(t_e) \prod_{e\in E_n^1} dl_1(t_e),$$

where

$$Z_M(\Lambda_n) = \int_{\mathbb{R}^{E_n}} \exp\left(-\frac{1}{2} M_n^{(\mathbb{R})}(t)\right) \prod_{e\in E_n^0} d\delta_0(t_e) \prod_{e\in E_n^1} dl_1(t_e).$$

A random scalar Maxwell configuration is a scalar configuration $t \in \mathbb{R}^{E_n}$ with law τ_n.

We compute that $Z_M(\Lambda_n)^{N^2} = Z_H(\Lambda_n)$ and as a result,

$$\mu_{\mathbb{R}^{N^2|E_n^1|}} = \bigotimes_{j=1}^{N^2} \tau_n.$$

This implies that the law of Hermitian Maxwell configurations $\mu_{\mathrm{Herm},n}$ is equal to the pushforward of the joint law of N^2 independent scalar Maxwell configurations through the product map $\Psi_H^{E_n}$.

Since τ_n is a Gaussian measure on $\mathbb{R}^{E_n^1}$, the normalization constant $Z_M(\Lambda_n)$ can be written in the canonical form

$$(2\pi)^{|E_n^1|/2} (\det \Sigma_n^0)^{-1/2}$$

where $\Sigma_n^0 \in \mathbb{R}^{E_n^1 \times E_n^1}$ denotes the matrix representation of the form $M_n^{(\mathbb{R})}$ resticted to $\mathbb{R}^{E_n^1}$.

Additionally, since $\mu_{\mathbb{R}^{N^2|E_n^1|}}$ is a Gaussian measure on $\mathbb{R}^{N^2|E_n^1|}$, we can write $Z_H(\Lambda_n)$ in the format

$$(2\pi)^{N^2|E_n^1|/2}(\det \Xi_n^0)^{-1/2},$$

where Ξ_n^0 denotes the matrix representation of the quadratic form $M_n^{(\mathbb{R}^{N^2})}$ on $\mathbb{R}^{N^2|E_n^1|}$. This matrix is block diagonal with N^2 blocks Σ_n^0. Therefore,

$$\det \Xi_n^0 = (\det \Sigma_n^0)^{N^2}.$$

Let

$$K_{n,d} = -\frac{\log \det \Sigma_n^0}{2n^d}.$$

In other words,

$$\begin{aligned}\frac{\log Z_H(\Lambda_n)}{n^d} &= \frac{N^2|E_n^1|}{2n^d}\log(2\pi) - N^2\frac{\log\det\Sigma_n^0}{2n^d} \\ &= \frac{N^2|E_n^1|}{2n^d}\log(2\pi) + N^2 K_{n,d}.\end{aligned}$$

Lemma 8.2 states that that $|E_n^1| = (d-1)n^d - dn^{d-1} + 1$. Therefore,

$$\frac{N^2|E_n^1|}{2n^d}\log(2\pi) \to \frac{N^2(d-1)}{2}\log(2\pi)$$

as $n \to \infty$. Based on the simple-minded guess in display (11) with $k = N^2|E_n^1|$ degrees of freedom, we have

$$F_{\Lambda_n,g} \to \frac{N^2(d-1)}{2}\log(2\pi) + N^2 \lim_{n\to\infty} K_{n,d}.$$

The following result shows that this guess holds some truth.

Theorem 1.2 ([4], Theorem 2.1)**.** *Take any $d \geq 2$ and $N \geq 1$. Let $K_{n,d}$ and $F_{\Lambda_n,g}$ be defined as above. Then the limit*

$$K_d := \lim_{n\to\infty} K_{n,d}$$

exists, is finite, and

$$\lim_{\substack{n\to\infty \\ g\to 0}} \left(F_{\Lambda_n,g} + \frac{|E_n^1|}{2n^d} N^2 \log\left(\frac{1}{g^2}\right)\right) = (d-1)\log C_N + \frac{N^2(d-1)}{2}\log(2\pi) + N^2 K_d$$

where $|E_n^1| = (d-1)n^d - dn^{d-1} + 1$ and $C_N = \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N(N+1)/2}}$.

The terms in the leading order formula shown in the above theorem can be interpreted as follows. The term $\frac{|E_n^1|}{2n^d}N^2\log\left(\frac{1}{g^2}\right)$ is a renormalization term which arises when making a change of variable in $\mathbb{R}^{|E_n^1|N^2}$ in order to get β out of the exponent of the Gibbs weight. Meanwhile, $(d-1)\log C_N$ arises when exchanging integration over Lie Group $\mathrm{U}(N)$ for integration over Lie algebra $\mathrm{i}\cdot\mathrm{H}(N)$. In particular, the pushforward of Haar measure by coordinates for $\mathrm{U}(N)$ near the identity matrix has a Radon Nikodym derivative which is approximately equal to C_N. Lastly, as already alluded to, the Gaussian terms $\frac{N^2(d-1)}{2}\log(2\pi) + N^2 K_d$ arise from N^2 independent scalar lattice Maxwell partition functions, one for each dimension of the Lie algebra $\mathrm{i}\cdot\mathrm{H}(N)$.

Remark 1.3. I wonder how the above formula would look for other gauge groups G besides $\mathrm{U}(N)$. For example, an $\mathrm{SU}(2)$ model is discussed in [3]. In contrast to $\mathrm{U}(2)$, the group $\mathrm{SU}(2)$ has 3 parameters instead of $N^2 = 2^2 = 4$ parameters. The Gaussian terms might therefore take the form $\frac{3(d-1)}{2}\log(2\pi) + 3K_d$ and the renormalization term might take the form $\frac{|E_n^1|}{2n^d}3\log\left(\frac{1}{g^2}\right)$. Also, since C_N in Theorem 1.2 is the reciprocal of the volume of $\mathrm{U}(N)$ (cf. [4, Section 11]), it might be the case that C_N would be replaced by the reciprocal of the volume of $\mathrm{SU}(2)$.

Next we discuss the continuum limit on the scaled lattice. It is helpful in this direction to note that (by simple arithmetic) Theorem 1.2 can also be written

$$\lim_{\substack{n\to\infty\\ g\to 0}} \left(F_{\Lambda_n,g} - \frac{1}{2}\left(d-1-\frac{d}{n}+\frac{1}{n^d}\right) N^2 \log(g^2)\right) = (d-1)\log\left(\frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2}}\right) + N^2 K_d. \tag{14}$$

Let us denote the lattice spacing by ϵ. The scaled box is given by

$$\epsilon\Lambda_n = \{0, \epsilon, \epsilon 2, \cdots, \epsilon(n-1)\}^d.$$

The positively oriented edge set is

$$E_{\epsilon\Lambda_n} = \{(x,y) \in (\epsilon\Lambda_n)^2 : \frac{1}{\epsilon}x \prec \frac{1}{\epsilon}y\}.$$

The space of configurations is

$$\mathrm{U}(N)^{E_{\epsilon\Lambda_n}}.$$

The set of plaquettes is

$$\begin{aligned}\mathcal{P}(\epsilon\Lambda_n) = \{p = (p_1,p_2,p_3,p_4) \in (E_{\epsilon\Lambda_n})^4 : \exists j,k \in [d] \text{ and } x \in \epsilon\Lambda_n \text{ such that } j<k \text{ and}\\ p_1 = (x, x+\epsilon e_j)\\ p_2 = (x+\epsilon e_j, x+\epsilon(e_j+e_k))\\ p_3 = (x+\epsilon e_k, x+\epsilon(e_j+e_k))\\ p_4 = (x, x+\epsilon e_k)\}.\end{aligned}$$

We have the plaquette variable

$$\forall p \in \mathcal{P}(\epsilon\Lambda_n) : U_p = U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^*.$$

And the Wilson action is given by

$$S_{\epsilon\Lambda_n}(U) = \sum_{p\in\mathcal{P}(\epsilon\Lambda_n)} \mathrm{Re}\,\mathrm{Tr}(I - U_p).$$

Define the parameter g_ϵ via

$$g_\epsilon = g_0^2 \epsilon^{4-d}$$

where $g_0 > 0$ denotes the coupling constant of the theory. The $\mathrm{U}(N)$ lattice gauge theory measure takes the form

$$d\mu_{\epsilon\Lambda_n,g_0}(U) = \frac{1}{Z_{\epsilon\Lambda_n,g_0}} \exp\left(-\frac{1}{g_\epsilon^2} S_{\epsilon\Lambda_n}(U)\right) \prod_{e\in E_{\epsilon\Lambda_n}} d\sigma(U_e)$$

where

$$Z_{\epsilon\Lambda_n,g_0} = \int_{\mathrm{U}(N)^{E_{\epsilon\Lambda_n}}} \exp\left(-\frac{1}{g_\epsilon^2} S_{\epsilon\Lambda_n}(U)\right) \prod_{e\in E_{\epsilon\Lambda_n}} d\sigma(U_e)$$

is the partition function, and $\log Z_{\epsilon\Lambda_n,g_0}$ is the free energy. It is apparent that

$$Z_{\Lambda_n,g_\epsilon} = Z_{\epsilon\Lambda_n,g_0}$$

where the left hand side is the partition function on the unscaled box (8) with coupling strength g_ϵ, and the right hand side is the partition function on the scaled box with coupling strength g_0. Therefore, Theorem 1.2 applies to

$$F_{\epsilon\Lambda_n,g_0} = \frac{\log Z_{\epsilon\Lambda_n,g_0}}{n^d} = \frac{\log Z_{\Lambda_n,g_\epsilon}}{n^d} = F_{\Lambda_n,g_\epsilon}$$

with g replaced by g_ϵ. That is,

$$\lim_{\substack{n\to\infty\\ g_\epsilon\to 0}} \left(F_{\epsilon\Lambda_n,g_0} - \frac{1}{2}\left(d-1-\frac{d}{n}+\frac{1}{n^d}\right) N^2 \log(g_\epsilon^2)\right) = (d-1)\log\left(\frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2}}\right) + N^2 K_d. \tag{15}$$

All the previous objects are said to be in the continuum limit in $\mathbb{R}^d$ when ϵ goes to 0 and n goes to infinity faster than $\frac{1}{\epsilon}$. In other words, $\lim_{j\to\infty} \frac{1}{n_j\epsilon_j} = 0$ where $n : \mathbb{N} \to \mathbb{N}$ and $\epsilon : \mathbb{N} \to (0,\infty)$ are sequences going to ∞ and 0, respectively.

On the other hand, when n grows like $\frac{R}{\epsilon}$ for some fixed $R > 0$, then this corresponds to the continuum limit in the box of side length R.

Corollary 1.4 (Theorem 2.2 [4]). *Take any $N \geq 1$ and take $d = 2$ or 3. Suppose that $n \to \infty$ and $\epsilon \to 0$ simultaneously such that $n\epsilon$ stays bounded away from zero. Then for any $g_0 > 0$,*

$$\log Z_{\epsilon\Lambda_n, g_0} = n^d \left(\frac{1}{2}(d-1)N^2 \log\left(g_0^2 \epsilon^{4-d}\right) + (d-1)\log\left(\frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2}}\right) + N^2 K_d \right) + o(n^d),$$

where K_d is as in Theorem 1.2, and $o(n^d)$ denotes a term that tends to zero when divided by n^d.

Note that Corollary 1.4 is silent on which continuum limit it refers to. If $\lim_{n\to\infty} \frac{1}{n\epsilon} = 0$, then certainly $n\epsilon$ stays bounded away from 0 and the continuum limit in $\mathbb{R}^d$ is achieved. But also, if $n = \frac{R}{\epsilon}$, then $n\epsilon$ also stays bounded away from zero, and then the continuum limit in a box of side-length R is achieved.

Proof. Fix $g_0 > 0$. Consider two sequences $n : \mathbb{N} \to \mathbb{N}$, $j \mapsto n_j$ and $\epsilon : \mathbb{N} \to (0, \infty)$, $j \mapsto \epsilon_j$, such that $n_j \epsilon_j \geq M$ for some positive constant M and $n_j \to \infty$, $\epsilon_j \to 0$ as $j \to \infty$. Since $d < 4$, the sequence $g_{\epsilon_j} = g_0^2 \epsilon_j^{4-d} \to 0$ as $j \to \infty$. Therefore

$$\begin{aligned}
&\lim_{j\to\infty} \left(F_{\epsilon_j \Lambda_{n_j}, g_0} - \frac{1}{2}\left(d - 1 - \frac{d}{n_j} + \frac{1}{n_j^d}\right) N^2 \log\left(g_{\epsilon_j}^2\right) \right) \\
&= \lim_{\substack{n_j \to \infty \\ g_{\epsilon_j} \to 0}} \left(F_{\epsilon_j \Lambda_{n_j}, g_0} - \frac{1}{2}\left(d - 1 - \frac{d}{n_j} + \frac{1}{n_j^d}\right) N^2 \log\left(g_{\epsilon_j}^2\right) \right) \\
&= (d-1)\log\left(\frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2}}\right) + N^2 K_d,
\end{aligned} \tag{16}$$

where the second equality is by 15.

Since $n\epsilon$ stays bounded away from zero, it follows that

$$\frac{1}{n_j} \log\left(g_{\epsilon_j}^2\right) \to 0, \qquad \frac{1}{n_j^d} \log\left(g_{\epsilon_j}^2\right) \to 0$$

as $j \to \infty$. Therefore,

$$\begin{aligned}
&\lim_{j\to\infty} \left(F_{\epsilon_j \Lambda_{n_j}, g_0} - \frac{1}{2}\left(d - 1 - \frac{d}{n_j} + \frac{1}{n_j^d}\right) N^2 \log\left(g_{\epsilon_j}^2\right) \right) \\
&= \lim_{j\to\infty} \left(F_{\epsilon_j \Lambda_{n_j}, g_0} - \frac{1}{2}(d-1)N^2 \log\left(g_{\epsilon_j}^2\right) \right)
\end{aligned} \tag{17}$$

Displays 17 and 16 show that

$$\lim_{j\to\infty} \frac{\log Z_{\epsilon_j \Lambda_{n_j}, g_0} - {n_j}^d \left(\frac{1}{2}(d-1)N^2 \log\left(g_0^2 {\epsilon_j}^{4-d}\right) + (d-1)\log\left(\frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2}}\right) + N^2 K_d \right)}{{n_j}^d} = 0,$$

as desired. □

2 Summary

In this summary, we aim to present the main ideas that are present in this thesis. This way, the rest of the thesis can be devoted to filling in the details and showing how the mathematical manipulations really work. This summary draws upon Section 4 in [4].

The main goal is to compute the limit of the free energy per site as $n \to \infty$ and $g \to 0$. This can be done by computing an upper bound and a lower bound, and then showing that they coincide in the limit.

The proof for the upper bound consists of two main procedures. The first is a localization procedure. This means that the partition function integral, which is an integral of the Gibbs weight $e^{-\beta S(U)}$ over configuration space $\mathrm{U}(N)^{E_n}$, is upper bounded by a localized integral in the axial gauge (see Theorem 4.11):

$$\int_{U_\epsilon} e^{-\beta S(U)} \prod_{e \in E_n^1} d\sigma(U_e) \prod_{e \in E_n^0} d\delta_I(U_e), \tag{18}$$

where $U_\epsilon := \{U \in \mathrm{U}(N)^{E_n} : \|I - U_e\| \leq \epsilon \, \forall e \in E_n\}$, and ϵ scales with n and/or g. Since the localization procedure for the upper bound is particularly involved, we summarize it in the next subsection, and skip now to the second procedure, which is also used for the lower bound. This second step is a Gaussian approximation procedure, which has been previously alluded to. Here, we upper bound the local integral 18 by the Gaussian integral

$$\int_{\mathrm{H}(N)} e^{-\frac{1}{2} M_n^{\mathrm{H}}(H)} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e),$$

namely the partition function of Hermitian Maxwell theory in the axial gauge.

Let us now emphasize that this Gaussian approximation procedure accounts for all the terms in the leading order formula, Theorem 1.2. Starting with the local integral 18 we switch from Lie group $\mathrm{U}(N)$ to Lie algebra $\mathrm{i} \cdot \mathrm{H}(N)$ using Theorem 5.2. The way this works it to pushforward Haar measure via the local inverse of i times the exponential map, and to compute the corresponding Radon Nikodym derivative with resepect to Lebesgue measure on $\mathrm{H}(N)$. This derivative is computed by considering how small balls in $\mathrm{U}(N)$ scale under the exponential map. The result is that the derivative, in the $\epsilon \to 0$ limit, is C_N. The pushforward change of variables formula, applied to each edge $e \in E_n^1$, then yields

$$\begin{aligned} &\int_{U_\epsilon} e^{-\beta S(U)} \prod_{e \in E_n^1} d\sigma(U_e) \prod_{e \in E_n^0} d\delta_I(U_e) \\ &\approx C_N^{|E_n^1|} \int_{H_\epsilon} e^{-\beta S(e^{\mathrm{i}H})} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e), \end{aligned}$$

where H_ϵ is some small neighborhood of the Hermitian configuration equal to $0 \in \mathrm{H}(N)$ on each edge. For the upper bound, we can replace H_ϵ by $\mathrm{H}(N)^{E_n}$ because the integrand is non-negative.

Next, we replace the Wilson action by the Maxwell action using Theorem 7.4, which is described by the logic we alluded to in equation 12. The error vanishes in the limit, so we write

$$\begin{aligned} &C_N^{|E_n^1|} \int_{\mathrm{H}(N)^{E_n}} e^{-\beta S(e^{\mathrm{i}H})} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e) \\ &\approx C_N^{|E_n^1|} \int_{\mathrm{H}(N)^{E_n}} e^{-\beta \frac{1}{2} M_n^{\mathrm{H}}(H)} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e). \end{aligned}$$

The next step is to take β out of the exponential weight in the integral. The inverse temperature β pulls into the Maxwell action by $\beta M_n^{\mathrm{H}}(H) = M_n^{\mathrm{H}}(\sqrt{\beta})$. Then a change of variable $G_e = \sqrt{\beta} H_e$ on each edge

$e \in E_n^1$ has the Jacobian $\left(\frac{1}{\sqrt{\beta}}\right)^{N^2}$ on each edge. This implies that

$$\begin{aligned}
&C_N^{|E_n^1|} \int_{\mathrm{H}(N)^{E_n}} e^{-\beta \frac{1}{2} M_n^{\mathrm{H}}(H)} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e) \\
&= C_N^{|E_n^1|} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2 |E_n^1|} \int_{\mathrm{H}(N)^{E_n}} e^{-\frac{1}{2} M_n^{\mathrm{H}}(H)} \prod_{e \in E_n^1} d\lambda(G_e) \prod_{e \in E_n^0} d\delta_0(G_e) \\
&= C_N^{|E_n^1|} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2 |E_n^1|} Z_{H,\Lambda_n}.
\end{aligned}$$

The Hermitian Maxwell theory partition function breaks up into a product of N^2 scalar Maxwell theory partition functions, one for each parameter of the N^2 dimensional Lie algebra $\mathrm{i} \cdot \mathrm{H}(N)$. See in particular Lemma 8.1. Collecting what we have:

$$Z_{\Lambda_n, g} \leq \text{Error} \times C_N^{|E_n^1|} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2 |E_n^1|} Z_{M,\Lambda_n}^{N^2},$$

and then taking the logarithm, dividing by n^d, and taking the limit yields the leading order formula. Note that some care has to be taken with the error term, but that is discussed in Section 8. Next we discuss the localization argument for the upper bound, then we describe the particularities of the lower bound.

2.1 Upper Bound: Localization

The localization procedure for the upper bound is covered in Sections 3 and 4. Here we summarize those results. The goal is to obtain an integral of the form 18. We begin with the observation that the Wilson action S is small with high probability under the Yang Mills measure $\mu_{\Lambda_n, g}$. Let's say that S is ϵ-small, for some small parameter ϵ scaling with n and β. The precise result for this observation is Theorem 3.4. This result is obtained using the a-priori bound on the partition function, Theorem 3.3, and Markov's inequality.

Since the Wilson action is ϵ-small on average, we can deduce that

$$Z_{\Lambda_n, g} \approx \int_{S \leq \epsilon} e^{-\beta S(U)} \prod_{e \in E_n} d\sigma(U_e),$$

where $\{S \leq \epsilon\}$ denotes configurations for which the Wilson action is small. This is because the measure

$$\mathcal{B}(\mathrm{U}(N)^{E_n}) \ni A \mapsto \int_A e^{-\beta S(U)} \prod_{e \in E_n} d\sigma(U_e) \in [0, \infty)$$

scales with (i.e. is proportional to) the Yang Mills measure $\mu_{\Lambda_n, g}$. The precise result is Corollary 3.5.

We already have that $S \leq \epsilon$ is quite near to the condition that on each edge $e \in E_n$, $U_e \approx I$. Indeed, we have

$$S(U) = \sum_{p \in \mathcal{P}(\Lambda_n)} \frac{1}{2} \| I - U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^* \|^2 \leq \epsilon$$

implies that $\| I - U_p \| \leq \epsilon$ for each plaquette $p \in \mathcal{P}(\Lambda_n)$. However, it is of course possible that two edges, p_1, p_3, say, of the plaquette p are not both near to the identity, but still $U_p \approx I$. For example, in $\mathrm{U}(1)$, we could have $U_{p_1} = -1$, $U_{p_2} = 1$, $U_{p_3} = -1$, and $U_{p_4} = 1$, and nevertheless, $U_p = 1$, in this example. On the other hand, "the closeness of $[U_p]$ to I implies that any three of the four unitary matrices attached to the edges of the plaquette approximately determine the fourth" [4, Section 4]. To illustrate this, suppose that U_{p_1} is unknown and $\| U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^* - I \| < \epsilon$. Then you can essentially solve for U_{p_1}. Since $U_{p_2} U_{p_3}^* U_{p_4}^*$ is a unitary matrix, we factor it out from $\| U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^* - I \|$ and apply invariance of the Hilbert Schmidt norm under unitary multiplication to conclude that

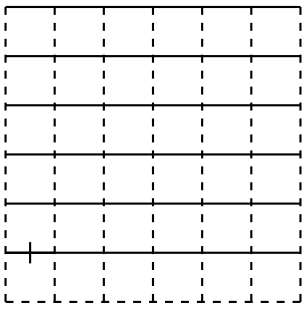
$$\| U_{p_1} - (U_{p_2} U_{p_3}^* U_{p_4}^*)^* \| < \epsilon. \tag{19}$$

Therefore $U_{p_1} \approx (U_{p_2} U_{p_3}^* U_{p_4}^*)^*$. This is only a useful fact, however, if each plaquette has three edge variables which are already approximately (or exactly) equal to the identity matrix. Fortunately, the

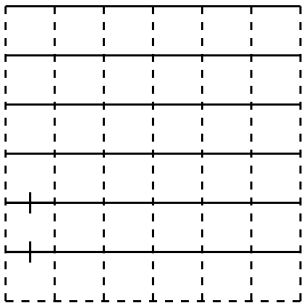
axial gauge gives us enough "boundary" such that we can inductively argue that in each plaquette, there are three edges which are close to the identity matrix, as long as n is not too large. This argument can be found in the proof of Lemma 4.10.

Here we wish to provide a sort of pictorial argument that the E_n^0 edge variables, in the axial gauge, determine that the E_n^1 edges variables all satisfy $U_e \approx I$. We illustrate this in the case $d = 2$. Let $p_1 = ((0,0),(1,0))$, $p_2 = ((1,0),(1,1))$, $p_3 = ((0,1),(1,1))$, and $p_4 = ((0,0),(0,1))$, and let p be the plaquette bounded by these four edges. This is the plaquette in the bottom left corner in Figure 1a or 1b. Notice that p_3 is the only edge in E_n^1. We approximately solve for U_{p_3} by using the assumption that U_p is close to I. We obtain $\|U_{p_1}^* U_{p_2}^* U_{p_4} - U_{p_3}^*\| \leq \epsilon$, which is similar in spirit to display 19. We have now determined U_{p_3} up to some small error, so we add a tick mark in Figure 1a as a sign that this edge variable is settled.

Next, we see that the plaquette directly above p contains one matrix that is unknown on the edge $e = ((0,2),(1,2))$. Therefore, we use the three matrices $U_{(0,1),(0,2)}^*, U_{(1,1),(1,2)}$, and U_{p_3} to approximate U_e. We add a tick mark in Figure 1b to indicate that we have an approximation for U_e. Continuing in this manner, we can determine all the matrices on the thickly drawn edges (E_n^1 edges), assuming that the matrices on the dashed edges (E_n^0 edges) are already known. One can imagine that the approximation gets increasingly imprecise as the induction goes on because the error builds up from the first approximation onwards. Therefore, n must be relatively small for this to work.



(a) The tick mark represents determining $U_{(0,1),(1,1)}$ based on the three other edge variables in the bottom left plaquette.



(b) Where $e = ((0,2),(1,2))$, the upper tick represents determining U_e.

Figure 1: The inductive procedure of determining all E_n matrices from matrices on E_n^0 edges. The dashed lines are for E_n^0 edges. The thick lines are for E_n^1 edges. Here we have $n = 7$ and $d = 2$.

The main upshot of the above discussion is that if n is small, then any configuration in the axial gauge $U \in \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0}$ such that $S(U) \leq \epsilon$, must also be in a neighborhood of the identity configuration, that is, $I \approx U_e$ for all $e \in E_n$. In other words,

$$\{S \leq \epsilon\} \subset U_\epsilon$$

in the axial gauge. This is slightly inaccurate because the scaling of ϵ in U_ϵ must also take into account $\max_{x \in \Lambda_n} |x|_1 = n$ (see Lemma 4.10), but it anyways expresses the idea of how the combination of small n, the axial gauge, and smallness of the Wilson action combine to give us a local set of configurations.

Before discussing why we can assume n is small, we note that we have obtained localization because

$$\begin{aligned} Z_{\Lambda_n, g} &\leq \text{Const} \times \int_{S \leq \epsilon} e^{-\beta S(U)} \prod_{e \in E_n} d\sigma(U_e) \\ &\leq \text{Const} \times \int_{U_\epsilon} e^{-\beta S(U)} \prod_{e \in E_n^1} d\sigma(U_e) \prod_{e \in E_n^0} d\delta_I(U_e), \end{aligned}$$

where the second step uses axial gauge fixing and gauge invariance of the Wilson action, which is explained in Section 4.

Now we comment on the argument that n can be taken to be small, relative to β. The result that states this is Lemma 8.4, which says

$$F_{\Lambda_n, g} \leq \left(1 - \frac{Cm}{n}\right) F_{\Lambda_m, g}$$

for $m \leq n$, that is, the free energy per site of $\mathrm{U}(N)$ lattice gauge theory in the box Λ_n is upper bounded by the free energy per site in a smaller box Λ_m, up to some factor. The argument for this is to break up Λ_n into disjoint translates of the smaller box Λ_m and then "remove the plaquettes that touch the boundaries of multiple boxes" [4, Section 4] from the Wilson action:

$$S(U) \geq \sum_{\text{translates}} \sum_{\text{plaquettes}} \frac{1}{2}\|I - U_p\|^2.$$

Then since there are no longer plaquettes which are common to multiple translates, then the partition function integral breaks up into a product integral. See the proof of Lemma 8.4 for the details. This technique does not apply to the lower bound, because dropping plaquettes decreases the Wilson action, which increases the Gibbs weight $e^{-\beta S(U)}$.

2.2 Lower Bound

For the upper bound, we had to work to be able to localize the paritition function integral. But for the lower bound, since the Gibbs weight $e^{-\beta S(U)}$ is non-negative, we can localize immediately:

$$Z_{\Lambda_n, g} \geq \int_{U_\delta^0} e^{-\beta S(U)} \prod_{e \in E_n^1} d\sigma(U_e), \tag{20}$$

where U_δ^0 denotes configurations U that are in a δ neighborhood of the identity configuration, and with the axial boundary condition. The next step is to apply the Gaussian approximation procedure to the integral on the right hand side of 20: exchange Lie Group for Lie Algebra using Theorem 5.2, exchange Wilson action for Maxwell action using Theorem 7.4, and make a change of variable in $\mathrm{H}(N)^{E_n^1}$ or equivalently in $\mathbb{R}^{|E_n^1|N^2}$ to get β out of the exponent.

After these steps, we are left with an N^2-fold product of the following localized Gaussian integral:

$$\int_{|t_e| \leq \epsilon \, \forall e} e^{-\frac{1}{2} M_n^{(\mathbb{R})}(s)} \prod_{e \in E_n^1} ds_e.$$

We want to extract the partition function of Maxwell theory (Z_M) from this integral, so we multiply by 1:

$$\begin{aligned}
&\int_{|t_e| \leq \epsilon \, \forall e} e^{-\frac{1}{2} M_n^{(\mathbb{R})}(s)} \prod_{e \in E_n^1} ds_e \\
&= Z_M \int_{|t_e| \leq \epsilon \, \forall e} \frac{1}{Z_M} e^{-\frac{1}{2} M_n^{(\mathbb{R})}(s)} \prod_{e \in E_n^1} ds_e \\
&= Z_M \tau_n(\{|t_e| \leq \epsilon \, \forall e\}).
\end{aligned}$$

Let us call $p = \tau_n(\{|t_e| \leq \epsilon \, \forall e\})$. We want for

$$p \geq \exp\bigl(-o(n^d)\bigr), \tag{21}$$

because then after taking the logarithm and multipling by $\frac{1}{n^d}$, we are left with

$$\lim_{n \to \infty} \frac{\log Z_M}{n^d} - \frac{o(n^d)}{n^d} = \lim_{n \to \infty} \frac{\log Z_M}{n^d},$$

which is what we need to match the upper bound.

Note that $p \geq \exp\bigl(-o(n^d)\bigr)$ is stronger than $p \geq \exp\bigl(-Cn^d\bigr)$, and that $p \geq \exp\bigl(-Cn^d\bigr)$ is not sufficient for our goal of matching the upper bound. On the other hand, since p is the probability that $|E_n^1| = O(n^d)$ edges have prescribed scalars attached to them, we would expect that $p \approx \exp\bigl(-Cn^d\bigr)$. Indeed, τ_n is a Gaussian measure on $\mathbb{R}^{|E_n^1|}$, and therefore each component of a τ_n-distributed vector $(t_e)_{e \in E_n^1}$ is one-dimensionally Gaussian distributed, with prescribed values in $[-\epsilon, \epsilon]$. It turns out that ϵ does not depend on n, so this probability is roughly e^{-C}. Taking into account the $O(n^d)$ factors, we expect that $p \approx \exp\bigl(-O(n^d)\bigr)$.

It turns out, however, that we can achieve the stronger condition $p \geq \exp\big(-o(n^d)\big)$ by using conditioning to view p as the probability of a certain set of $o(n^d)$ edges taking prescribed values. This is Theorem 6.7 and its proof.

The way this is done is by splitting up the box Λ_n into subboxes, which are translates of Λ_m, and which intersect at common faces. Then the faces consist of $O(n^d m^{-1})$ edges, which is $o(n^d)$. Here, m satisfies $2 \leq m \leq \sqrt{n}$. In each subbox we define a conditional lattice Maxwell theory Gaussian measure which comes with a mean μ and covariance Σ. In each subbox, the associated Gaussian measure is conditioned on the face edges of that particular subbox. By a combination of Lemma 6.1 and Lemma 6.2 we have the covariance bound

$$\lambda_{\max}(\Sigma) \leq C m^{d+2},$$

where $\lambda_{\max}(\Sigma)$ is the top eigenvalue of the covariance matrix. Meanwhile, the maximum entry of the mean vector μ is determined by the covariance bound and by the scalars on the boundary/face edges, which we denote here by θ. That is:

$$\mu_e \leq C m^{d+2} \|\theta\|.$$

See equation 43 in the proof of Theorem 6.6.

Next, introducing another smallness parameter $\epsilon' < \epsilon$ and by conditional probability, we have

$$\begin{aligned} p &= \tau_n(|t_e| \leq \epsilon \, \forall e) \\ &\geq \tau_n(|t_e| \leq \epsilon \text{ for all interior edges and } |\theta_e| = |t_e| \leq \epsilon' < \epsilon \text{ for all face edges}) \\ &= \tau_n(|t_e| \leq \epsilon \text{ for all interior edges} \mid |\theta_e| = |t_e| \leq \epsilon' \text{ for all face edges}) \\ &\times \tau_n(|\theta_e| = |t_e| \leq \epsilon' \, \forall e \text{ a face edge}). \end{aligned} \tag{22}$$

By an interior edge, we mean an edge which is not on the face of one of the subboxes, but is rather contained inside of one of the subboxes.

The Gaussian measure for each subbox dictates that each interior edge is unlikely to deviate far from the mean of the subbox Gaussian measure. This is a property of Gaussian measures, stated and proved in Lemma 6.4. In other words, on average, $|t_e| \leq \max_e\{\mu_e\} + \delta$, for some small $\delta > 0$ and for the interior edges e.

Now assume that $\epsilon \approx m^{d+2}$. It turns out that ϵ' can be chosen so small that $\|\theta\| = O(1)$, and so, on average,

$$|t_e| \leq \max_e\{\mu_e\} + \delta \leq C m^{d+2} \|\theta\| + \delta \approx O(1) m^{d+2} \approx \epsilon,$$

where we are being very loose. What we are trying to indicate is that, conditioned on $|\theta_e| \leq \epsilon'$ for all face edges, we have with high probability that $|t_e| \leq \epsilon$ for all interior edges. With $\epsilon' = n^{-d/2}$, as is taken in the proof of Theorem 6.7, this implies that

$$\begin{aligned} &\tau_n(|t_e| \leq \epsilon \text{ for all interior edges} \mid |\theta_e| = |t_e| \leq \epsilon' \text{ for all face edges}) \\ &\geq e^{-O(n^d/m^d)} \end{aligned}$$

because the probability for each subbox is $\geq \frac{1}{2} \geq e^{-1}$, and there are $O(n^d/m^d)$ subboxes.

On the other hand, $\tau_n(|\theta_e| = |t_e| \leq \epsilon' \, \forall e \text{ a face edge})$ is the probability of $o(n^d)$ edges taking prescribed values, and this probability is lower bounded by $\exp\big(-o(n^d)\big)$, see Theorem 6.3. Since $e^{-O(n^d/m^d)} \times \exp\big(-o(n^d)\big) = \exp\big(-o(n^d)\big)$, by 22 we can conclude that $p \geq \exp\big(-o(n^d)\big)$, as desired.

3 A-Priori Bounds

3.1 Inputs

We begin by quoting two results on small ball probabilities for Haar measure. These results will first be used to derive some a-priori bounds for $\mathrm{U}(N)$ lattice gauge theory. This refers to bounds obtained before gauge fixing. Later on, these small ball probabilities will show up when computing the Radon Nikodym derivative when we pass from the partition function as an integral over a local neighborhood of the identity in the Lie group $\mathrm{U}(N)$, to an integral over the Lie Algebra $\mathrm{i} \cdot \mathrm{H}(N)$.

Theorem 3.1 (Theorem 6.1 [4]). *Let σ denote the Haar measure on $\mathrm{U}(N)$. Then*

$$\lim_{\delta \to 0} \frac{\sigma(\{U : \|I - U\| \leq \delta\})}{\delta^{N^2}} = \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2} 2^{N^2/2} \Gamma\left(N^2/2 + 1\right)}.$$

Theorem 3.1 is proved using random matrix theory. First one uses a formula due to Weyl to switch from the study of the Haar measure of small balls in $\mathrm{U}(N)$ to the study of eigenvalues. Then one uses the joint density of eigenvalues in the GUE ensemble to produce the above formula.

Corollary 3.2 (Corollary 6.3 [4]). *Let σ denote the Haar measure on $\mathrm{U}(N)$. There are positive constants C_1 and C_2 depending only on N, such that for any $\delta \in (0, \sqrt{N})$,*

$$C_1 \delta^{N^2} \leq \sigma(\{U : \|I - U\| \leq \delta\}) \leq C_2 \delta^{N^2}.$$

A note on the omitted proof of the corollary. Suppose momentarily that $C = \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2} 2^{N^2/2} \Gamma(N^2/2+1)}$, which clearly depends only on N. Then by definition of limit and Theorem 3.1, we have for $\epsilon = C/2$ that there is some δ_* such that for all $0 < \delta < \delta_*$,

$$\delta^2 (C/2) \leq \sigma(\{U : \|I - U\| \leq \delta\}).$$

The one difficulty in obtaining the corollary is convincing oneself that the threshold $\delta_* > 0$ truly depends only on N.

3.2 A lower bound on the Partition Function

For an unitary configuration $U \in \mathrm{U}(N)^{E_n}$ and any $\delta > 0$, let

$$\begin{aligned} b_{\mathrm{U}(N)^{E_n}}(U, \delta) &= \{V \in \mathrm{U}(N)^{E_n} : V_e \in b_{\mathrm{U}(N)}(U_e, \delta) \, \forall e \in E_n\} \\ &= \{V \in \mathrm{U}(N)^{E_n} : \|V_e - U_e\|_{\mathrm{HS}} < \delta \, \forall e \in E_n\} \\ &= \prod_{e \in E_n} b_{\mathrm{U}(N)}(U_e, \delta). \end{aligned}$$

Theorem 3.3 (Theorem 7.1 [4]). *For β large, in particular for $\beta \geq 2$, there is a positive constant C depending only on N and d, such that for any n,*

$$Z_{\Lambda_n, g} \geq e^{-C n^d \log \beta}.$$

The idea of the proof below is to obtain a lower bound by localizing the partition function integral to configurations which are δ small on each edge, and then pull out the Gibbs weight, which has a uniform lower bound in this local setting, based on an upper bound on the Wilson action. Pulling out the Gibbs weight allows to use the small ball probability lower bound of the previous section. Setting δ in terms of β is the final ingredient.

Proof. By Corollary 3.2, we have that the product Haar measure of a δ neighborhood of the identity configuration I is bounded below:

$$\begin{aligned} \sigma_{\Lambda_n}\left(\prod_{e \in E_n} b_{\mathrm{U}(N)}(I, \delta)\right) &= \prod_{e \in E_n} \sigma(b_{\mathrm{U}(N)}(I, \delta)) \\ &\geq (C_1 \delta^{N^2})^{|E_n^1|} \\ &\geq (C_1 \delta)^{C_3 n^d} \end{aligned}$$

where the last line uses that $N^2|E_n| \leq C_3 n^d$ and $C_1 \delta^{N^2} < 1$, and the constant C_1 changes from the second to the third line to account for the power of N^2.

Let $U \in b_{\mathrm{U}(N)^{E_n}}(I, \delta)$. Let $p \in \mathcal{P}(\Lambda_n)$ be arbitary. Then $\|I - U_{p_j}\| \leq \delta$ for each $j \in [4]$ by definition of the set $b_{\mathrm{U}(N)^{E_n}}(I, \delta)$ and by definition of the plaquette variables U_{p_j}, where the $p_j \in E_n$ are positively oriented. In addition,

$$\|I - U_p\| \leq 4\delta.$$

Indeed, by telescoping and using that unitary matrices are isometries, we have

$$\begin{aligned}
\|I - U_{p_1}U_{p_2}U_{p_3}^*U_{p_4}^*\| =& \|I - U_{p_1} + U_{p_1} - U_{p_1}U_{p_2} + \cdots + U_{p_1}U_{p_2}U_{p_3}^* - U_{p_1}U_{p_2}U_{p_3}^*U_{p_4}^*\| \\
\geq& \|I - U_{p_1}\| + \|U_{p_1} - U_{p_1}U_{p_2}\| + \|U_{p_1}U_{p_2} - U_{p_1}U_{p_2}U_{p_3}^*\| \\
&+ \|U_{p_1}U_{p_2}U_{p_3}^* - U_{p_1}U_{p_2}U_{p_3}^*U_{p_4}^*\| \\
=& \|I - U_{p_1}\| + \|U_{p_1}(I - U_{p_2})\| + \|U_{p_1}U_{p_2}(I - U_{p_3}^*)\| \\
&+ \|U_{p_1}U_{p_2}U_{p_3}^*(I - U_{p_4}^*)\| \\
=& \|I - U_{p_1}\| + \|I - U_{p_2}\| + \|I - U_{p_3}^*\| \\
&+ \|I - U_{p_4}^*\|.
\end{aligned}$$

We use the isometry property to smuggle U_{p_3} into $\|I - U_{p_3}^*\|$ and see that $\|I - U_{p_3}^*\| = \|U_{p_3}(I - U_{p_3}^*)\| = \|U_{p_3} - I\|$, and similarly $\|I - U_{p_4}^*\| = \|U_{p_4} - I\|$. Hence,

$$\|I - U_{p_1}U_{p_2}U_{p_3}^*U_{p_4}^*\| \leq 4\delta,$$

as desired.

Next, recalling the defininition of the Wilson action, we have

$$\begin{aligned}
S_{\Lambda_n}(U) &= \sum_{p \in \mathcal{P}(\Lambda_n)} \frac{1}{2}\|I - U_p\|^2 \\
&\leq \sum_{p \in \mathcal{P}(\Lambda_n)} \frac{1}{2}(4\delta)^2 \\
&= |\mathcal{P}(\Lambda_n)| 8\delta^2
\end{aligned}$$

whenever $U \in b_{\mathrm{U}(N)^{E_n}}(I, \delta)$ is a unitary configuration in a δ neighborhood of the identity configuration.

Therefore we have the set inclusion

$$\begin{aligned}
b_{\mathrm{U}(N)^{E_n}}(I, \delta) &\subset \{U \in \mathrm{U}(N)^{E_n} : S_{\Lambda_n}(U) \leq 8|\mathcal{P}(\Lambda_n)|\delta^2\} \\
&= \{U \in \mathrm{U}(N)^{E_n} : e^{-\beta S_{\Lambda_n}(U)} \geq e^{-\beta \cdot 8|\mathcal{P}(\Lambda_n)|\delta^2}\} \\
&=: D_{\delta,\beta},
\end{aligned}$$

where we have introduced the notation $D_{\delta,\beta}$ in the last line. This says that the Gibbs weight $e^{-\beta S_{\Lambda_n}(U)}$ is uniformly bounded below throughout a small neighborhood of the identity configuration Indeed, the lower bound $e^{-\beta \cdot 8|\mathcal{P}(\Lambda_n)|\delta^2}$ does not depend on U.

Recall that $e^{-(\,\cdot\,)}$ is non-negative, which implies that integrating over a subset decreases the integral. Therefore we have

$$\begin{aligned}
Z_{\Lambda_n, g} &= \int_{\mathrm{U}(N)^{E_n}} e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} \sigma(dU_e) \\
&\geq \int_{\mathrm{U}(N)^{E_n}} 1_{D_{\delta,\beta}}(U)\, e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} \sigma(dU_e) \\
&\geq \int_{\mathrm{U}(N)^{E_n}} 1_{D_{\delta,\beta}}(U)\, e^{-\beta \cdot 8|\mathcal{P}(\Lambda_n)|\delta^2} \prod_{e \in E_n} \sigma(dU_e) \\
&\geq \int_{\mathrm{U}(N)^{E_n}} 1_{b_{\mathrm{U}(N)^{E_n}}(I,\delta)}(U)\, e^{-\beta \cdot 8|\mathcal{P}(\Lambda_n)|\delta^2} \prod_{e \in E_n} \sigma(dU_e) \\
&= e^{-\beta \cdot 8|\mathcal{P}(\Lambda_n)|\delta^2} \sigma_{\Lambda_n}(b_{\mathrm{U}(N)^{E_n}}(I, \delta)) \\
&\geq e^{-\beta \cdot 8|\mathcal{P}(\Lambda_n)|\delta^2} (C_1\delta)^{C_3 n^d}.
\end{aligned}$$

Fix $\beta \geq 2$ and let $\delta = \frac{1}{\sqrt{8\beta}}$. Then we have

$$Z_{\Lambda_n,g} \geq e^{-|\mathcal{P}(\Lambda_n)|}(C_1\beta^{-1/2})^{C_3 n^d},$$

where one instance of a $\sqrt{8}$ was absorbed into C_1. We have $|\mathcal{P}(\Lambda_n)| \leq C_2 n^d$, so $e^{-|\mathcal{P}(\Lambda_n)|} \geq e^{-C_2 n^d} = (\frac{1}{e^{C_2}})^{n^d}$. Even though C_1 is possibly extremely small, there there is still a positive constant C depending only on N and d such that

$$\min\{\frac{1}{e^{C_2}}, C_1^{C_3}\} \geq \left(\frac{1}{\beta}\right)^C,$$

which would not be the case if $\beta \leq 1$. Therefore,

$$\begin{aligned} Z_{\Lambda_n,g} &\geq (\frac{1}{e^{C_2}})^{n^d} \cdot (C_1)^{C_3 n^d} \cdot (\beta^{-1})^{C_4 n^d} \\ &\geq \left(\frac{1}{\beta}\right)^{Cn^d} \cdot \left(\frac{1}{\beta}\right)^{Cn^d} \cdot (\beta^{-1})^{C_4 n^d} \\ &\geq (\beta^{-1})^{C_5 n^d} = e^{-Cn^d \log \beta}, \end{aligned}$$

where the meaning of C in the last line has again changed. □

3.3 Smallness of the Wilson Action

The following theorem shows that with high probability under the Yang Mills measure $\mu_{\Lambda_n,g}$, the Wilson action is of order $n^d \frac{\log \beta}{\beta}$.

Theorem 3.4 (Theorem 8.1 [4])**.** *Suppose that $\beta \geq 2$. Then there is a positive constant C depending only on N and d such that for any $t > 0$ and $n \geq 2$,*

$$\mu_{\Lambda_n,g}(\{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) \geq Cn^d \log \beta + t\}) \leq e^{-t}.$$

Proof. Note that

$$\begin{aligned} &\{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) \geq Cn^d \log \beta + t\} = \\ &\{U \in \mathrm{U}(N)^{E_n} : e^{-\beta S_{\Lambda_n}(U)} \leq e^{-Cn^d \log \beta} e^{-t}\}. \end{aligned}$$

Denote this set of configurations by D. We have

$$\begin{aligned} \mu_{\Lambda_n,g}(D) &= \int_D \mu_{\Lambda_n,g}(dU) \\ &= \int_D \frac{e^{-\beta S_{\Lambda_n}(U)}}{Z_{\Lambda_n,g}} \prod_{e \in E_n} \sigma(dU_e) \\ &\leq \int_D \frac{e^{-\beta S_{\Lambda_n}(U)}}{e^{-Cn^d \log \beta}} \prod_{e \in E_n} \sigma(dU_e) \qquad \text{(Theorem 3.3)} \\ &\leq \int_D \frac{e^{-Cn^d \log \beta} e^{-t}}{e^{-Cn^d \log \beta}} \prod_{e \in E_n} \sigma(dU_e) \quad \text{(Definition of } D\text{)} \\ &= e^{-t} \int_D \prod_{e \in E_n} \sigma(dU_e) \\ &\leq e^{-t}, \end{aligned}$$

where the last line uses that the product Haar measure of D is less than or equal to 1. This completes the proof. □

We note that a slight variant of the proof above can be viewed as an application of Markov's inequality, namely $\mathbb{P}[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$ for the non-negative random variable $X = e^{\beta S(\,\cdot\,)}$, and with $a = e^{Cn^d \log \beta} e^t$, and working in the probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathrm{U}(N)^{E_n}, \mathcal{B}(\mathrm{U}(N)^{E_n}), \mu_{\Lambda_n,g})$. The difference to the proof above is that D above involves a negative sign in front of β, but the proof using Markov's inequality does not have a negative sign in front of β in the definition of X. We leave out the details.

The following corollary shows that we can upper bound the partition function integral by an integral over configurations for which the Wilson action is small. This is not good enough for the localization step of the upper bound as discussed in Section 2, because certain edge variables U_e could still be far from the identity, despite the Wilson action being small.

Corollary 3.5 (Corollary 8.2 [4]). *Suppose that $\beta \geq 2$. Then there is a constant C_0 depending only on N and d such that for any $n \geq 2$,*

$$Z_{\Lambda_n,g} \leq 2 \int_A e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} \sigma(dU_e) \leq 2 Z_{\Lambda_n,g}$$

where

$$A = \{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) \leq C_0 n^d \log \beta\}.$$

Proof. To obtain the upper bound, we note that

$$\begin{aligned} Z_{\Lambda_n,g} &= \int_{\mathrm{U}(N)^{E_n}} e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} \sigma(dU_e) \\ &\geq \int_A e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} \sigma(dU_e), \end{aligned}$$

because we integrate a nonnegative function over a smaller set. The lower bound in the corollary holds if and only if the same relation divided by $Z_{\Lambda_n,g}$ holds:

$$1 \leq 2 \int_A \frac{e^{-\beta S_{\Lambda_n}(U)}}{Z_{\Lambda_n,g}} \prod_{e \in E_n} \sigma(dU_e) = 2 \cdot \mu_{\Lambda_n,g}(A).$$

where we have noted that the right hand side of the inequality is equal to twice the probability of A under $\mathrm{U}(N)$ lattice gauge theory $\mu_{\Lambda_n,g}$. Therefore, the lower bound of the corollary holds if and only if $\mu_{\Lambda_n,g}(A) \geq \frac{1}{2}$.

Fix $\beta \geq 2$, and take $t = \log(2)$ in Theorem 3.4 to obtain

$$\mu_{\Lambda_n,g}(\{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) \geq C n^d \log \beta + \log(2)\}) \leq \frac{1}{2},$$

where C is a positive constant depending only on N and d. Therefore the complement event has probability greater than one half:

$$\mu_{\Lambda_n,g}(\{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) < C n^d \log \beta + \log(2)\}) \geq \frac{1}{2}. \tag{23}$$

Since $n \geq 2$ and $\beta \geq 2$, we have

$$\begin{aligned} C n^d \log \beta + \log(2) &\leq C n^d \log \beta + C n^d \log \beta \\ &\leq C_0 n^d \log \beta, \end{aligned}$$

where we take, e.g. $C_0 \geq 2C$. We have

$$\{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) < C n^d \log \beta + \log(2)\} \subset A$$

because the defining property of the left set is tighter than the defining property of the right set. Combining this with 23, we have $\mu_{\Lambda_n,g}(A) \geq \frac{1}{2}$, which is enough to conclude the lower bound of the corollary by previous remarks. □

4 Axial Gauge Fixing

In view of the fact that the unitary matrices on the E_m^0 edges approximately determine the matrices on the other edges, and that we need at most $|E_m^1|$ non-trivial Hermitians for a well-defined Gaussian measure, it is therefore desirable that any unitary configuration is equivalent to a configuration with $U_e = I$ for all $e \in E_m^0$. In this section, we discuss how this equivalence works through the axial gauge.

Let $G = \mathrm{U}(N)$. Fix $n \geq 1$ and $d \geq 1$. Recall that E_n is the positively oriented edge set of the lattice $\Lambda_n = \{0, 1, \cdots, n-1\}^d$, and that E_n^0 is the edge set of the spanning tree T_n from the introduction.

A gauge transform is an element $\theta \in G^{\Lambda_n}$. That is, it is an assignment of unitary matrices to vertices of the box Λ_n.

Lemma 4.1. *G^{Λ_n} is a group under componentwise multiplication.*

Proof. Fix two gauge transforms θ^1 and θ^2. Then $\theta^1\theta^2$ is given by $(\theta^1\theta^2)_x = \theta^1_x\theta^2_x$ for all $x \in \Lambda_n$. That G^{Λ_n} is a group then follows from the fact that G is a group. □

Let $\theta \in G^{\Lambda_n}$ be an arbitrary gauge transform, and let $U \in G^{E_n}$ be an arbitary unitary configuration. Then for each $(x, y) \in E_n$, we define

$$(\theta U)_{(x,y)} = \theta(x)U(x, y)\theta(y)^*.$$

In particular, $\theta U \in G^{E_n}$.

Lemma 4.2. *This defines a group action of the group of gauge transformations G^{Λ_n} on the set of unitary configurations $\mathrm{U}(N)^{E_n}$.*

Proof. Clearly

$$G^{\Lambda_n} \times \mathrm{U}(N)^{E_n} \ni (\theta, U) \mapsto \theta U \in \mathrm{U}(N)^{E_n}$$

is a well-defined function. Let us verify that $(\theta^1\theta^2)U = \theta^1(\theta^2 U)$. Fix $(x, y) \in E_n$. We have

$$\begin{aligned}
[\theta^1(\theta^2 U)]_{(x,y)} &= \theta^1(x)[\theta^2 U]_{(x,y)}\theta^1(y)^* \\
&= \theta^1(x)\theta^2(x)U_{(x,y)}\theta^2(y)^*\theta_1(y)^* \\
&= (\theta^1\theta^2)(x)U_{(x,y)}(\theta^1\theta^2)(y)^* \\
&= [(\theta^1\theta^2)U]_{(x,y)},
\end{aligned}$$

as desired. Last, let $\theta_x = I$ for all $x \in \Lambda_n$. Then we have $\theta U = U$. This finishes the proof. □

Before discussing axial gauge fixing, we note the gauge invariance of the Wilson action.

Lemma 4.3. *Let $\theta \in \mathrm{U}(N)^{\Lambda_n}$ and $U \in \mathrm{U}(N)^{E_n}$. Then $S_{\Lambda_n}(U) = S_{\Lambda_n}(\theta U)$.*

Proof. Fix a plaquette $p \in \mathcal{P}(\Lambda_n)$. Let $x \in \Lambda_n$ be the first vertex of both edges p_1 and p_4. It is easy to see that

$$\begin{aligned}
(\theta U)_p &= (\theta U)_{p_1}(\theta U)_{p_2}(\theta U)^*_{p_3}(\theta U)^*_{p_4} \\
&= \theta_x U_{p_1} U_{p_2} U^*_{p_3} U^*_{p_4} \theta^*_x.
\end{aligned}$$

Since the matrix trace is invariant under conjugation by unitary matrices, we have

$$\begin{aligned}
\operatorname{Re}\operatorname{Tr}(I - U_p) &= \operatorname{Re}\operatorname{Tr} I - \operatorname{Re}\operatorname{Tr} U_p \\
&= \operatorname{Re}\operatorname{Tr} I - \operatorname{Re}\operatorname{Tr} \theta_x U_p \theta^*_x \\
&= \operatorname{Re}\operatorname{Tr} I - \operatorname{Re}\operatorname{Tr}(\theta U)_p \\
&= \operatorname{Re}\operatorname{Tr}(I - (\theta U)_p).
\end{aligned}$$

Summing over all plaquettes yields the equivalence. □

Next, we define gauge fixing, which is a way to set the gauge field on the vertices in a convenient way. The particular method of gauge fixing that is used in $\mathrm{U}(N)$ lattice gauge theory is so-called axial gauge fixing. But there are other gauge fixing procedures, such as unitary gauge fixing, which is used for an SU(2) model considered in [3]. We do not define unitary gauge fixing.

Recall that, when given $U \in G^{E_n}$, the orbit of U is the set

$$\{\theta U : \theta \in G^{\Lambda_n}\}.$$

Gauge fixing means selecting one element from each orbit. That is, gauge fixing is a function

$$\phi : G^{E_n} \to G^{E_n}$$

such that for each $U \in G^{E_n}$, we have that $\phi(U) \in \{\theta U : \theta \in G^{\Lambda_n}\}$.

Next, we specify a map ϕ_{path} which we will use to specify the axial gauge ϕ_{ax}.

For any vertex $x \in \Lambda_n$ such that $x \neq 0$, let $q(x) = (0, q_1), (q_1, q_2), \cdots, (q_k, x)$ be the path of edges in the spanning tree T_n between the root $0 \in T_n$ and the vertex x. Furthermore, let

$$U_{q(x)} = U_{(0,q_1)} U_{(q_1,q_2)} \cdots U_{(q_k,x)}.$$

In particular $U_{q(x)}$ is a product of unitary matrices attached to E_n^0 edges.

Now let $\phi_{path} : G^{E_n} \to G^{\Lambda_n}$ be the function defined by

$$\phi_{path}(U)(x) = \begin{cases} U_{q(x)} & \text{if } x \neq 0 \\ I & \text{if } x = 0 \end{cases}$$

for any $x \in \Lambda_n$ and any unitary configuration $U \in G^{E_n}$.

Let us alternatively define ϕ_{path} as in [4, Section 9] and verify that the definitions are equivalent. Fix $U \in G^{E_n}$. We define $\phi_{path}(U) \in G^{\Lambda_n}$ inductively. First let $\phi_{path}(U)(0) = I$. Now assume inductively that $\phi_{path}(U)(y)$ has been defined for all $y \prec x$ for some $x = (x_1, \cdots, x_d) \in \Lambda_n$, where $y \prec x$ means that y comes before x in the lexicographic ordering. Let j be the largest index such that $x_j \neq 0$. Let $y := x - e_j$. Then $y \in \Lambda_n$ and $y \prec x$. Thus, $\phi_{path}(U)(y)$ is already defined. Let

$$\phi_{path}(U)(x) = \phi_{path}(U)(y) U(y, x).$$

By induction, $\phi_{path}(U)(x)$ is defined for all $x \in \Lambda_n$.

Lemma 4.4. *These definitions coincide.*

Proof. Let ϕ'_{path} denote the definition in [4, Section 9]. Fix $U \in G^{E_n}$. We proceed by induction. The base case is that $\phi_{path}(U)(0) = I = \phi'_{path}(U)(0)$.

Now assume inductively that $\phi_{path}(U)(y) = \phi'_{path}(U)(y)$ for all $y \prec x$ for some $x \in \Lambda_n$, where $x \neq 0$. Let j be the largest index such that $x_j \neq 0$. Let $y := x - e_j$. Then

$$\phi'_{path}(U)(x) = \phi'_{path}(U)(y) U(y, x).$$

Observe that $(y, x) \in E_n^0$. Therefore the path from 0 to x in T_n is given by $q(y), (y, x)$, where $q(y)$ is the path from 0 to y in the spanning tree T_n. We have

$$\phi_{path}(U)(x) = U_{q(x)} = U_{q(y)} U(y, x) = \phi_{path}(U)(y) U(y, x),$$

unless $y = 0$, in which case $\phi_{path}(U)(x) = U(y, x)$. By the induction hypothesis, we have that $\phi'_{path}(U)(y) = \phi_{path}(U)(y)$, which implies that $\phi_{path}(U)(x) = \phi'_{path}(U)(x)$. This closes the induction. □

Let

$$\phi_{\mathrm{ax}}(U) = \phi_{path}(U) U$$

for any $U \in G^{E_n}$. We see that $\phi_{\mathrm{ax}}(U) \in \{\theta U : \theta \in G^{\Lambda_n}\}$ for all $U \in G^{E_n}$, so ϕ_{ax} is indeed an instance of gauge fixing.

The key property is as follows.

Lemma 4.5. *We have*

$$\mathrm{Im}(\phi_{ax}) = \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0},$$

where

$$\mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0} = \{U \in \mathrm{U}(N)^{E_n} : U_e = I \text{ for all } e \in E_n^0\}.$$

Proof. Fix $U \in \mathrm{U}(N)^{E_n}$. We want to show that $\phi_{\mathrm{ax}}(U) = \phi_{path}(U)U \in \mathrm{U}(N)^{E_n^1}$. Fix $e = (x, y) \in E_n^0$. Then we have

$$\begin{aligned}[\phi_{path}(U)U]_{(x,y)} &= \phi_{path}(U)(x)U(x,y)\phi_{path}(U)(y)^* \\ &= U_{q(x)}U(x,y)U_{q(y)}^*.\end{aligned}$$

Observe that the path from 0 to y, namely $q(y)$, is equal to the path $q(x), (x, y)$, that is, the path from 0 to x with the extra edge (x, y). Therefore,

$$U_{q(y)}^* = U(x,y)^* U_{q(x)}^*.$$

Then

$$U_{q(x)}U(x,y)U_{q(y)}^* = U_{q(x)}U(x,y)U(x,y)^*U_{q(x)}^* = I.$$

Now assume that $U \in \mathrm{U}(N)^{E_n^1}$. One can check that $\phi_{\mathrm{ax}}(U) = U$ in this case. This concludes the proof. □

Suppose $U = (U_e)_{e\in E_n}$ is distributed as product Haar measure $\prod_{e\in E_n} \sigma$. The next developments are about answering the following question: how is $\phi_{\mathrm{ax}}(U)$ distributed on the space $\mathrm{U}(N)^{E_n}$? In other words, which measure is $(\phi_{\mathrm{ax}})_* \prod_{e\in E_n} \sigma$ equal to, where $(\phi_{\mathrm{ax}})_* \prod_{e\in E_n} \sigma$ denotes the pushforward of the product Haar measure by the axial gauge map ϕ_{ax}?

Since $\mathrm{Im}(\phi_{\mathrm{ax}}) = \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0}$, it follows that $\phi_{\mathrm{ax}}(U)_e = I$ for all $e \in E_n^0$. Thus, $\phi_{\mathrm{ax}}(U)_e$ is distributed as δ_I for all $e \in E_n^0$, where δ_I denotes the Dirac measure on $\mathrm{U}(N)$ centered at the identity matrix I. It also follows that the joint law of the family of random unitary matrices $(\phi_{\mathrm{ax}}(U)_e)_{e\in E_n^0}$ has the product distribution

$$\prod_{e\in E_n^0} \delta_I,$$

which can be verified by proving that $(\phi_{\mathrm{ax}}(U)_e)_{e\in E_n^0}$ is an independent family. It takes more work to show that the family of random unitary matrices $(\phi_{\mathrm{ax}}(U)_e)_{e\in E_n^1}$ is jointly distributed as product Haar measure on $\mathrm{U}(N)^{E_n^1}$:

$$(\phi_{\mathrm{ax}}(U)_e)_{e\in E_n^1} \sim \prod_{e\in E_n^1} \sigma.$$

But once we have this result, then it follows that the law of $\phi_{\mathrm{ax}}(U)$ on $\mathrm{U}(N)^{E_n}$ is

$$\prod_{e\in E_n^0} \delta_I \otimes \prod_{e\in E_n^1} \sigma$$

because the product of the Dirac measures is independent of the family of measures on the E_n^1 edges. In other words,

$$(\phi_{\mathrm{ax}})_* \left(\prod_{e\in E_n} \sigma \right) = \prod_{e\in E_n^0} \delta_I \otimes \prod_{e\in E_n^1} \sigma. \tag{24}$$

Lemma 4.6. *Let $e = (x, y) \in E_n^1$ be arbitrary and let $V_e = [\phi_{ax}(U)]_e$, where U is distributed as $\prod_{e\in E_n} \sigma$. Then V_e is Haar distributed.*

Proof. We will show that

$$\mathbb{E}[1_B(V_e)] = \sigma(B)$$

for an arbitrary Borel set $B \in \mathcal{B}_{\mathrm{U}(N)}$. This suffices because $\mathbb{E}[1_B(V_e)] = \mathbb{P}[V_e \in B]$. We have

$$\mathbb{E}[1_B(V_e)] = \int_{\mathrm{U}(N)^{E_n}} 1_B([\phi_{\mathrm{ax}}(U)]_e) \prod_{e\in E_n} d\sigma(U_e).$$

Note that $[\phi_{\mathrm{ax}}(U)]_e = U_{q(x)}U_eU_{q(y)}^*$, and that $U_{q(x)}$ and $U_{q(y)}^*$ are products of matrices that do not include U_e. Therefore we have

$$\begin{aligned}\mathbb{E}[1_B(V_e)] &= \int_{\mathrm{U}(N)^{E_n}} 1_B(U_{q(x)}U_eU_{q(y)}^*) \prod_{e\in E_n} d\sigma(U_e) \\ &= \int_{\mathrm{U}(N)^{E_n\setminus\{e\}}} \left(\int_{\mathrm{U}(N)^{\{e\}}} 1_B(U_{q(x)}U_eU_{q(y)}^*) d\sigma(U_e) \right) \prod_{e'\in E_n\setminus\{e\}} d\sigma(U_{e'}).\end{aligned}$$

Due to left and right Haar invariance and that $U_{q(x)}$ and $U^*_{q(y)}$ can be viewed as fixed with regards to the innermost integral, we see that the inner integral above is

$$\int_{\mathrm{U}(N)^{\{e\}}} 1_B(U_{q(x)}U_eU^*_{q(y)})d\sigma(U_e) = \int_{\mathrm{U}(N)^{\{e\}}} 1_B(U_e)d\sigma(U_e) = \sigma(B).$$

Therefore,

$$\int_{\mathrm{U}(N)^{E_n\setminus\{e\}}} \left(\int_{\mathrm{U}(N)^{\{e\}}} 1_B(U_{q(x)}U_eU^*_{q(y)})d\sigma(U_e)\right) \prod_{e'\in E_n\setminus\{e\}} d\sigma(U_{e'}) = \sigma(B),$$

which uses that Haar measure is a probability measure. This concludes the proof. □

Lemma 4.7. *Suppose $U = (U_e)_{e\in E_n}$ is product Haar distributed. Let $V_e = [\phi_{ax}(U)]_e$ for all $e \in E^1_n$. For each $e \in E^1_n$, let $f_e : \mathrm{U}(N) \to \mathbb{R}$ be an arbitrary measurable function. Then*

$$\mathbb{E}[\prod_{e\in E^1_n} f_e(V_e)] = \prod_{e\in E^1_n} \mathbb{E}[f_e(V_e)].$$

Proof. We have

$$\mathbb{E}[\prod_{e\in E^1_n} f_e(V_e)] = \int_{\mathrm{U}(N)^{E_n}} \prod_{(x,y)=e\in E^1_n} f_e(U_{q(x)}U(x,y)U^*_{q(y)}) \prod_{e\in E_n} d\sigma(U_e).$$

Note that V_e can be written as $U_{q(x)}U(x,y)U^*_{q(y)}$ because if $e \in E^1_n$, then e is not incident to the origin. Then by Fubini's theorem, we have that the right hand side in the previous expression is equal to

$$\int_{\mathrm{U}(N)^{E^0_n}} \left(\int_{\mathrm{U}(N)^{E^1_n}} \prod_{(x,y)=e\in E^1_n} f_e(U_{q(x)}U(x,y)U^*_{q(y)}) \prod_{e\in E^1_n} d\sigma(U_e)\right) \prod_{e\in E^0_n} d\sigma(U_e)$$

Since $U_{q(x)}$ and $U^*_{q(y)}$ are products of E^0_n-attached matrices, they can be viewed as fixed with regards to the inner integral over $\mathrm{U}(N)^{E^1_n}$. Then by left and right Haar invariance, we have that the previous expression is equal to

$$\int_{\mathrm{U}(N)^{E^0_n}} \left(\int_{\mathrm{U}(N)^{E^1_n}} \prod_{(x,y)=e\in E^1_n} f_e(U(x,y)) \prod_{e\in E^1_n} d\sigma(U_e)\right) \prod_{e\in E^0_n} d\sigma(U_e).$$

Then pulling the inner integral outside as a constant and using that Haar measure is a probability measure, we arrive at the equivalent expression

$$\int_{\mathrm{U}(N)^{E^1_n}} \prod_{(x,y)=e\in E^1_n} f_e(U(x,y)) \prod_{e\in E^1_n} d\sigma(U_e).$$

Then by Fubini's theorem and tracing back to the beginning, we have

$$\mathbb{E}[\prod_{e\in E^1_n} f_e(V_e)] = \prod_{(x,y)=e\in E^1_n} \int_{\mathrm{U}(N)^e} f_e(U(x,y))d\sigma(U_e).$$

The right hand side is equal to $\prod_{e\in E^1_n} \mathbb{E}[f_e(U_e)]$. But by Lemma 4.6, V_e is Haar-distributed when $e \in E^1_n$, so we have $\mathbb{E}[f_e(U_e)] = \mathbb{E}[f_e(V_e)]$ for each $e \in E^1_n$. This concludes the proof. □

Now that we know how Haar distributed unitary configurations are distributed under the axial gauge, we can use this to simplify the integral which defines the partition function.

Lemma 4.8 (Corollary 9.4 [4])**.** *We have*

$$\int_{\mathrm{U}(N)^{E_n}} e^{-\beta S(U)} \prod_{e\in E_n} d\sigma(U_e) = \int_{\mathrm{U}(N)^{E^1_n}\times\{I\}^{E^0_n}} e^{-\beta S(U)} \prod_{e\in E^0_n} d\delta_I(U_e) \prod_{e\in E^1_n} d\sigma(U_e).$$

Proof. By gauge invariance of the Wilson action, we obtain

$$\int_{\mathrm{U}(N)^{E_n}} e^{-\beta S(U)} \prod_{e\in E_n} d\sigma(U_e) = \int_{\mathrm{U}(N)^{E_n}} e^{-\beta S(\phi_{\mathrm{ax}}U)} \prod_{e\in E_n} d\sigma(U_e).$$

Then using a change of variables formula for pushforward measure, we obtain

$$\begin{aligned}\int_{\mathrm{U}(N)^{E_n}} e^{-\beta S(\phi_{\mathrm{ax}}U)} \prod_{e\in E_n} d\sigma(U_e) &= \int_{\phi_{\mathrm{ax}}(\mathrm{U}(N)^{E_n})} e^{-\beta S(U)} (\phi_{\mathrm{ax}})_* \left(\prod_{e\in E_n} \sigma\right)(dU) \\ &= \int_{\mathrm{U}(N)^{E_n^1}\times\{I\}^{E_n^0}} e^{-\beta S(U)} \prod_{e\in E_n^0} d\delta_I(U_e) \prod_{e\in E_n^1} d\sigma(U_e),\end{aligned}$$

as desired. □

The following proposition will not be used for the main theorem, Theorem 1.2, but I wanted to give a statement for the setting of $\mathrm{U}(N)$ lattice gauge theory that is analagous to [3, Lemma 5.3], which gives the law of configurations after so-called unitary gauge fixing.

Proposition 4.9. *Suppose the law of $U = (U_e)_{e\in E_n}$ is lattice gauge theory $\mu_{\Lambda_n,g}$. Then the law of $\phi_{ax}(U)$ is*

$$\frac{e^{-\beta S_{\Lambda_n}(U)}}{Z_{\Lambda_n,g}} \prod_{e\in E_n^0} d\delta_I(U_e) \prod_{e\in E_n^1} d\sigma(U_e).$$

In the above proposition we interpret

$$\phi_{\mathrm{ax}}(\,\cdot\,) : (\mathrm{U}(N)^{E_n}, \mathcal{B}(\mathrm{U}(N)^{E_n}), \mu_{\Lambda_n,g}) \to (\mathrm{U}(N)^{E_n}, \mathcal{B}(\mathrm{U}(N)^{E_n}))$$

as a random configuration from a probability space other than the usual probability space $(\mathrm{U}(N)^{E_n}, \mathcal{B}(\mathrm{U}(N)^{E_n}), \prod_{e\in E_n}\sigma)$.

Proof. We will show that for an arbitrary bounded measurable function $f : \mathrm{U}(N)^{E_n} \to \mathbb{R}$, that

$$\mathbb{E}_{\mu_{\Lambda_n,g}}[f(\phi_{\mathrm{ax}}(U))] = \int_{\mathrm{U}(N)^{E_n}} f(U) \frac{e^{-\beta S_{\Lambda_n}(U)}}{Z_{\Lambda_n,g}} \prod_{e\in E_n^0} d\delta_I(U_e) \prod_{e\in E_n^1} d\sigma(U_e), \tag{25}$$

because then for $f = 1_B$ (for $B \in \mathcal{B}(\mathrm{U}(N)^{E_n})$) we have

$$\mu_{\Lambda_n,g}(\phi_{\mathrm{ax}}^{-1}(B)) = \mathbb{E}_{\mu_{\Lambda_n,g}}[1_B(\phi_{\mathrm{ax}}(U))] = \int_B \frac{e^{-\beta S_{\Lambda_n}(U)}}{Z_{\Lambda_n,g}} \prod_{e\in E_n^0} d\delta_I(U_e) \prod_{e\in E_n^1} d\sigma(U_e)$$

where the leftmost expression is by definition the law of $\phi_{\mathrm{ax}}(U)$ under $\mu_{\Lambda_n,g}$, and the right most expression means that this law is described as in the proposition.

Now we proceed with 25. We have

$$\begin{aligned}\mathbb{E}_{\mu_{\Lambda_n,g}}[f(\phi_{\mathrm{ax}}(U))] &= \int_{\mathrm{U}(N)^{E_n}} f(\phi_{\mathrm{ax}}(U)) d\mu_{\Lambda_n,g}(U) \\ &= \int_{\mathrm{U}(N)^{E_n}} f(\phi_{\mathrm{ax}}(U)) \frac{e^{-\beta S_{\Lambda_n}(U)}}{Z_{\Lambda_n,g}} \prod_{e\in E_n} d\sigma(U_e) \\ &= \int_{\mathrm{U}(N)^{E_n}} f(\phi_{\mathrm{ax}}(U)) \frac{e^{-\beta S_{\Lambda_n}(\phi_{\mathrm{ax}}(U))}}{Z_{\Lambda_n,g}} \prod_{e\in E_n} d\sigma(U_e) \\ &= \int_{\phi_{\mathrm{ax}}(\mathrm{U}(N)^{E_n})} f(V) \frac{e^{-\beta S_{\Lambda_n}(V)}}{Z_{\Lambda_n,g}} (\phi_{\mathrm{ax}})_* \left(\prod_{e\in E_n}\right)(dV) \\ &= \int_{\mathrm{U}(N)^{E_n^1}\times\{I\}^{E_n^0}} f(V) \frac{e^{-\beta S_{\Lambda_n}(V)}}{Z_{\Lambda_n,g}} \prod_{e\in E_n^0} d\delta_I(V_e) \prod_{e\in E_n^1} d\sigma(V_e) \\ &= \int_{\mathrm{U}(N)^{E_n}} f(V) \frac{e^{-\beta S_{\Lambda_n}(V)}}{Z_{\Lambda_n,g}} \prod_{e\in E_n^0} d\delta_I(V_e) \prod_{e\in E_n^1} d\sigma(V_e).\end{aligned}$$

The first equality is the definition of $\mathbb{E}_{\mu_{\Lambda_n,g}}$. The second equality is the definition of the density of $\mu_{\Lambda_n,g}$ with respect to product Haar measure. The third equality is the invariance of the Wilson action under the axial gauge. The fourth equality is the change of variables formula for pushforward measure. The fifth equality uses Lemma 4.5 to see that $\mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0} = \phi_{\mathrm{ax}}(\mathrm{U}(N)^{E_n})$ and equation 24 for the pushforward of product Haar by the axial gauge. The sixth equality uses a property of the Dirac measure centered at $I \in \mathrm{U}(N)$. □

4.1 An Upper Bound for the Partition Function Based on Axial Gauge Fixing

A configuration $U \in \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0}$ if and only if

$$U_e = \begin{cases} U_e & \text{if } e \in E_n^1 \\ I & \text{if } e \in E_n^0, \end{cases}$$

for all $e \in E_n$. Furthermore, given $x \in \mathbb{R}^d$, we have $|x|_1 = \sum_{j=1}^d |x_j|$. Observe that for a given $n \geq 1$ and $x \in \Lambda_n$, that $|x|_1$ coincides with the graph distance between 0 and x, or the number of edges in a shortest path between 0 and x.

Lemma 4.10 (Lemma 10.2 [4]). *Take any $n \geq 2$. For any $U \in \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0}$ and any $(x,y) \in E_n$,*

$$\|I - U_{(x,y)}\|_{\mathrm{HS}} \leq (2|x|_1 S_{\Lambda_n}(U))^{1/2}.$$

Proof. Fix $n \geq 2$ and $U \in \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0}$.

Claim: Let $k \geq 0$ be a given integer. Suppose $x \in \Lambda_n$ and $|x|_1 = k$. Then for any $y \in \Lambda_n$ such that $(x,y) \in E_n$, there exists a subset of plaquettes $P(x,y) \subset \mathcal{P}(\Lambda_n)$ such that $|P(x,y)| \leq k$ and

$$\|I - U_{(x,y)}\| \leq \sum_{p \in P(x,y)} \|I - U_p\|. \tag{26}$$

Proof of Claim. We proceed by induction on k. For the base case, suppose $k = 0$. Then $x = 0 \in \Lambda_n$. Let $y \in \Lambda_n$ such that $(x,y) \in E_n$. Then $(x,y) \in E_n^0$ by definition of E_n^0. By definition of the configuration U, we have $U_{(x,y)} = I$, so the empty set of plaquettes suffices, because $\|I - U_{(x,y)}\| = 0$.

Now assume the induction hypothesis for some $k \geq 0$. We will prove the statement for $k+1$. Suppose $x = (x_1, \cdots, x_d) \in \Lambda_n$ with $|x|_1 = k+1$. Let $y \in \Lambda_n$ such that $(x,y) \in E_n$. By definition of E_n, there exists some direction $j \in [d]$ such that $y = x + e_j$. Let

$$l = \max\{h \in [d] : x_h > 0\}$$

be the last coordinate direction at which x is non-zero. By definition of E_n^0, we have that $(x,y) \in E_n^0$ whenever $j \geq l$. In this case, $\|I - U_{(x,y)}\| = 0$, which means even the empty set of plaquettes $|P(x,y)| = |\emptyset| \leq k+1$ does the job. On the other hand, suppose $j < l$. Note that, on the path in the axial tree T_n from the root 0 to the vertex x, the last edge before obtaining x is the edge $e_* = (x - e_l, x)$, just by definition of E_n^0. The next step is to identify a plaqeutte which contains both the edge $(x,y) = (x, x+e_j)$ and the edge e_*. The plaquette $p = (x - e_l, j, l)$ is the plaquette containing such edges. It satisfies $p_1 = (x - e_l, x - e_l + e_j)$, $p_2 = (x - e_l + e_j, x + e_j)$, $p_3 = (x, x + e_j)$, and $p_4 = (x - e_l, x)$. The next observation is that $p_2 \in E_n^0$ and $p_4 \in E_n^0$. Hence,

$$\begin{aligned} U_p &= U_{p_1} U_{p_2} U_{p_3}^* U_{p_4}^* \\ &= U_{p_1} U_{p_3}^*, \end{aligned}$$

by the assumption on the configuration U in the Lemma. This was a crucial step which could not be done without the axial gauge. The variable we want to estimate is $U_{p_3} = U_{(x,y)}$. What we have at our disposal is that the edge p_1 satisfies the induction hypthesis, because $|x - e_l| = k$. We have

$$\begin{aligned} \|I - U_{p_3}\| &= \|I - U_p^* U_{p_1}\| \\ &= \|U_p^*(U_p - U_{p_1})\| \\ &= \|U_p - U_{p_1}\| && \text{(Isometry prop.)} \\ &= \|U_p - I + I - U_{p_1}\| \\ &\leq \|U_p - I\| + \|I - U_{p_1}\| \\ &\leq \|U_p - I\| + \sum_{p' \in P(x - e_l, x - e_l + e_j)} \|I - U_{p'}\|, && \text{(Ind. Hyp.)} \end{aligned}$$

and we thus see that $\|I - U_{(x,y)}\|$ is bounded by a sum over $\leq k+1$ plaquettes. This closes the induction. □

We have shown that for each $x \in \Lambda_n$, and $(x,y) \in E_n$, there exists a set of plaquettes $P(x,y) \subset \mathcal{P}(\Lambda_n)$ such that $|P(x,y)| \leq |x|_1$ and

$$\|I - U_{(x,y)}\| \leq \sum_{p \in P(x,y)} 1 \cdot \|I - U_p\|.$$

By the Cauchy Schwarz inequality on $\mathbb{R}^{|P(x,y)|}$, we obtain

$$\|I - U_{(x,y)}\| \leq \sqrt{(\sum_{p \in P(x,y)} 1^2)} \sqrt{\sum_{p \in P(x,y)} \|I - U_p\|} \leq \sqrt{(2|x|_1 S_{\Lambda_n}(U))},$$

where the last step uses that $|P(x,y)| \leq |x|_1$ and the alternative characterization of the Wilson action. □

Theorem 4.11 (Theorem 10.1 [4]). *There is a constant C_1 depending only on N and d such that the following is true. For any $n, \beta \geq 2$,*

$$F_{\Lambda_n, g} \leq \frac{\log 2}{n^d} + \frac{1}{n^d} \log \int_{U_0^\beta(\Lambda_n)} e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n^0} d\delta_I(U_e) \prod_{e \in E_n^1} d\sigma(U_e)$$

where

$$U_0^\beta(\Lambda_n) := \left\{ U \in \mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0} : \|I - U_e\| \leq C_1 n^{(d+1)/2} \left(\frac{\log \beta}{\beta}\right)^{1/2} \text{ for all } e \in E_n \right\}.$$

Proof. Fix $n, \beta \geq 2$. We begin where we left off with Corollary 3.5. Let C_0 depend only on N and d such that

$$Z_{\Lambda_n, g} \leq 2 \int_A e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} d\sigma(U_e),$$

where $A = \{U \in \mathrm{U}(N)^{E_n} : \beta S_{\Lambda_n}(U) \leq C_0 n^d \log \beta\}$. By the invariance of the Wilson action under the axial gauge, we see that $1_A(\,\cdot\,) = 1_A(\phi_{\mathrm{ax}}(\,\cdot\,))$. Then switching to the axial gauge, we have

$$\begin{aligned}\int_{\mathrm{U}(N)^{E_n}} 1_A(U) e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n} d\sigma(U_e) &= \int_{\mathrm{U}(N)^{E_n}} 1_A(\phi_{\mathrm{ax}}(U)) e^{-\beta S_{\Lambda_n}(\phi_{\mathrm{ax}}(U))} \prod_{e \in E_n} d\sigma(U_e) \\ &= \int_{\mathrm{U}(N)^{E_n^1} \times I^{E_n^0}} 1_A(V) e^{-\beta S_{\Lambda_n}(V)} \prod_{e \in E_n^0} d\delta_I(V_e) \prod_{e \in E_n^1} d\sigma(V_e),\end{aligned}$$

where the second step repeats the ideas presented in the previous section. Let

$$A_0 = A \cap \left(\mathrm{U}(N)^{E_n^1} \times I^{E_n^0}\right),$$

such that the integral on the bottom right in the previous display is taken over A_0. Let $U \in A_0$. Then for any $(x,y) \in E_n$, we have by Lemma 4.10 that

$$\begin{aligned}\|I - U(x,y)\| &\leq (2|x|_1 S_{\Lambda_n}(U))^{1/2} \\ &\leq \left(2|x|_1 C_0 n^d \frac{\log \beta}{\beta}\right)^{1/2} \qquad \text{(def of } A\text{)} \\ &\leq \left(2nC_2 n^d \frac{\log \beta}{\beta}\right)^{1/2} \qquad (|x|_1 \leq dn) \\ &= C_1 n^{(d+1)/2} \left(\frac{\log \beta}{\beta}\right)^{1/2}.\end{aligned}$$

Therefore, $A_0 \subset U_0^\beta(\Lambda_n)$, so by collecting what we obtained earlier and then integrating over a larger set, we have

$$\begin{aligned}Z_{\Lambda_n, g} &\leq 2 \int_{A_0} e^{-\beta S_{\Lambda_n}(V)} \prod_{e \in E_n^0} d\delta_I(V_e) \prod_{e \in E_n^1} d\sigma(V_e) \\ &\leq 2 \int_{U_0^\beta(\Lambda_n)} e^{-\beta S_{\Lambda_n}(V)} \prod_{e \in E_n^0} d\delta_I(V_e) \prod_{e \in E_n^1} d\sigma(V_e).\end{aligned}$$

Last, we take the logarithm of both sides and we divide by n^d to obtain the result of the theorem. □

5 From Lie Group to Lie Algebra

The setting for this section is the probability space $(\mathrm{U}(N), \mathcal{B}_{\mathrm{U}(N)}, \sigma)$, where σ denotes Haar measure, and the measure space $(\mathrm{H}(N), \mathcal{B}_{\mathrm{H(N)}}, \lambda)$, where λ denotes Lebesgue measure on $\mathrm{H}(N)$. We will define a measure ν on the measurable space $(\mathrm{H}(N), \mathcal{B}_{\mathrm{H}(N)})$ which we can think of as Haar measure σ in coordinates.

We denote a ball in $\mathrm{H}(N)$ centered at H with radius r by

$$b_{\mathrm{H}(N)}(H, r).$$

Meanwhile, we denote a ball in $\mathrm{U}(N)$ centered at $V \in \mathrm{U}(N)$ with radius r by

$$b_{\mathrm{U}(N)}(V, r).$$

Recall the map

$$\psi(\cdot) := e^{i(\cdot)} : \mathrm{H}(N) \to \mathrm{U}(N)$$

defined by

$$H \mapsto I + \mathrm{i}H - \frac{H^2}{2} + \cdots = \sum_{j=0}^{\infty} \frac{(\mathrm{i}H)^j}{j!}.$$

In order to obtain an approximation theorem which allows for exchanging integration with respect to unitary matrices (or configurations) with integration using Hermitian matrices (or configurations), we begin with the following assumption. We postulate the existence of some threshold $r_0 > 0$, depending only on N, describing the radius of some sufficiently small ball about $0 \in \mathrm{H}(N)$, and with the following five properties.

First, ψ restricted to $b_{\mathrm{H}(N)}(0, 3r_0)$ is a homeomorphism onto its image. Using this first property, we define the measure ν in the rest of this paragraph. Since this restriction of ψ is bi-continuous, the map

$$\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,3r_0))} : \psi(b_{\mathrm{H}(N)}(0, 3r_0)) \to b_{\mathrm{H}(N)}(0, 3r_0)$$

is measurable (because it is continuous). Therefore,

$$\left(\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,3r_0))}\right)_* \sigma : \mathcal{B}(b_{\mathrm{H}(N)}(0, 3r_0)) \to [0, 1],$$

the pushforward of Haar measure σ by the map ψ^{-1} with domain $\psi(b_{\mathrm{H}(N)}(0, 3r)) \subset \mathrm{U}(N)$, is a well-defined measure on the measurable space $(b_{\mathrm{H}(N)}(0, 3r_0), \mathcal{B}_{b_{\mathrm{H}(N)}(0,3r_0)})$. Note that

$$\left(\left(\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,3r_0))}\right)_* \sigma\right)(A) = \sigma\left(\left(\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,3r_0))}\right)^{-1}(A)\right) = \sigma(\psi(A)),$$

for any $A \in \mathcal{B}_{b_{\mathrm{H}(N)}(0,3r_0)}$, where the first equality is the definition of pushforward measure, and the second equality is due to the fact that $\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,3r_0))}$ is a bijection. Let us denote this measure by $\psi_*^{-1}\sigma$, with the understanding that ψ^{-1} may only be well-defined and measurable on the restricted domain $b_{\mathrm{H}(N)}(0, 3r_0)$. Therefore,

$$\nu : \mathcal{B}_{\mathrm{H}(N)} \to [0, 1]$$

defined by

$$\nu(A) = \psi_*^{-1}\sigma(A \cap b_{\mathrm{H}(N)}(0, 3r_0))$$

for any $A \in \mathcal{B}_{\mathrm{H}(N)}$ is a well-defined and finite measure, which is zero outside of $b_{\mathrm{H}(N)}(0, 3r_0)$.

Second, we require that r_0 is sufficiently small so that ν is absolutely continuous with respect to Lebesgue measure λ on $\mathrm{H}(N)$. We denote the Radon-Nikodym derivative of ν with respect to λ by

$$\rho(H) = \frac{d\nu}{d\lambda}(H)$$

for $H \in \mathrm{H}(N)$.

Third, we require that r_0 is sufficiently small such that when $r \in (0, r_0]$, we have that

$$(2 - e^{6r})^{N^2} C_N \leq \rho(H) \tag{27}$$

for almost all $H \in b_{\mathrm{H}(N)}(0,r)$, where

$$C_N := \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N(N+1)/2}}.$$

According to [4, Section 11], the reciprocal of C_N equals the volume of $\mathrm{U}(N)$. Fourth, we require that

$$\rho(H) \leq e^{6rN^2} C_N \tag{28}$$

for almost all $H \in b_{\mathrm{H}(N)}(0,3r)$, whenever $r \in (0, r_0]$. Requirements 27 and 28 say that the Radon-Nikodym derivative approaches C_N with a logarithmically negligible error as $r \to 0$. Indeed, we have

$$e^{-Cr} C_N \leq \rho(H) \leq e^{Cr} C_N$$

and hence

$$-Cr + \log(C_N) \leq \log \rho(H) \leq Cr + \log(C_N)$$

for some constant C depending only on N, whenever H is close enough to the 0 matrix. To see the former relation, note the following.

Remark 5.1. There exists a constant $C > 0$ and a threshold $\theta > 0$ such that $e^{-Cx} < 2 - e^{6x}$ for all $x \in (0, \theta]$. The threshold θ must be strictly less than $\frac{\log(2)}{6}$, where $2 - e^{6x}$ intercepts the x-axis. Furthermore, $C > 6$.

Fifth, we assume that

$$e^{\mathrm{i} b_{\mathrm{H}(N)}(0,r)} \subset b_{\mathrm{U}(N)}(I, 2r) \subset e^{\mathrm{i} b_{\mathrm{H}(N)}(0,3r)} \tag{29}$$

for any $r \in (0, r_0]$. This says that ψ does not take a small ball to a large ball, as long as ψ is sufficiently restricted.

Assuming that $r_0 > 0$ (depending only on N) exists such that ν as defined above is well-defined and absolutely continuous with Radon Nikodym derivative ρ satisfying the bounds as in 27 and 28, and that

$$\psi\left(b_{\mathrm{H}(N)}(0,r)\right) \subset b_{\mathrm{U}(N)}(I, 2r) \subset \psi\left(b_{\mathrm{H}(N)}(0,3r)\right)$$

for any $r \in (0, r_0]$, then we obtain the following theorem. Proving the existence of r_0 will be done at the end of this section. We will see that taking $r_0 = \frac{\log(4/3)}{18}$ works.

Theorem 5.2 (Theorem 11.1 [4])**.** *Let $f : \mathrm{U}(N) \to [0, \infty)$ be a Borel measurable function. Then for any $r \leq \frac{\log(4/3)}{18} = r_0$,*

$$(2 - e^{6r})^{N^2} C_N \int_{b_{\mathrm{H}(N)}(0,r)} f(e^{\mathrm{i}H}) d\lambda(H) \leq \int_{b_{\mathrm{U}(N)}(I,2r)} f(U) d\sigma(U) \leq e^{6rN^2} C_N \int_{b_{\mathrm{H}(N)}(0,3r)} f(e^{\mathrm{i}H}) d\lambda(H)$$

where

$$C_N := \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N(N+1)/2}}$$

More generally, if for some $n \geq 1$, $f : U(N)^n \to [0, \infty)$ is a Borel measurable function, then for any $r \leq r_0$,

$$\begin{aligned}
&(2 - e^{6r})^{N^2 n} C_N^n \int_{b_{\mathrm{H}(N)}(0,r)^n} f(e^{\mathrm{i}H_1}, \ldots, e^{\mathrm{i}H_n}) \prod_{j \in [n]} d\lambda(H_j) \\
&\leq \int_{b_{\mathrm{U}(N)}(I,2r)^n} f(U_1, \ldots, U_n) \prod_{j \in [n]} d\sigma(U_j) \\
&\leq e^{6rN^2 n} C_N^n \int_{b_{\mathrm{H}(N)}(0,3r)^n} f(e^{\mathrm{i}H_1}, \ldots, e^{\mathrm{i}H_n}) \prod_{j \in [n]} d\lambda(H_j).
\end{aligned}$$

Proof assuming r_0 exists with the properties outlined above. Take $r \leq r_0$. By the third property, inequality (27), we have

$$(2 - e^{6r})^{N^2} C_N \int_{b_{\mathrm{H}(N)}(0,r)} f(e^{\mathrm{i}H}) d\lambda(H) \leq \int_{b_{\mathrm{H}(N)}(0,r)} f(e^{\mathrm{i}H}) \rho(H) d\lambda(H).$$

Then since $\rho(H) = \frac{d\nu}{d\lambda}(H)$,

$$\int_{b_{\mathrm{H}(N)}(0,r)} f(e^{\mathrm{i}H})\rho(H)d\lambda(H) = \int_{b_{\mathrm{H}(N)}(0,r)} f(e^{\mathrm{i}H})d\nu(H).$$

As $b_{\mathrm{H}(N)}(0,r) \subset b_{\mathrm{H}(N)}(0,3r_0)$, ν is given by a pushforward of Haar measure, so we have by the pushforward change of variables formula that

$$\int_{b_{\mathrm{H}(N)}(0,r)} f(e^{\mathrm{i}H})d\nu(H) = \int_{\psi(b_{\mathrm{H}(N)}(0,r))} f(U)d\sigma(U).$$

By the fifth property, 29, and non-negativity of f, we have

$$\int_{\psi(b_{\mathrm{H}(N)}(0,r))} f(U)d\sigma(U) \leq \int_{b_{\mathrm{U}(N)}(0,2r)} f(U)d\sigma(U).$$

This achieves the lower bound of the theorem.

Continuing, again by the fifth property, 29, and non-negativity of f, we see that

$$\int_{b_{\mathrm{U}(N)}(0,2r)} f(U)d\sigma(U) \leq \int_{\psi(b_{\mathrm{H}(N)}(0,3r))} f(U)d\sigma(U).$$

Since $b_{\mathrm{H}(N)}(0,3r) \subset b_{\mathrm{H}(N)}(0,3r_0)$, then ν is given by a pushforward of Haar measure, and we have

$$\int_{\psi(b_{\mathrm{H}(N)}(0,3r))} f(U)d\sigma(U) = \int_{b_{\mathrm{H}(N)}(0,3r)} f(e^{\mathrm{i}H})d\nu(H).$$

Rewriting the right hand side in the previous display using the Radon Nikodym derivative ρ and applying the bound 28, we see that

$$\int_{b_{\mathrm{H}(N)}(0,3r)} f(e^{\mathrm{i}H})d\nu(H) \leq e^{6rN^2} C_N \int_{b_{\mathrm{H}(N)}(0,3r)} f(e^{\mathrm{i}H})d\lambda(H),$$

which proves the upper bound of the present theorem.

The case of general n is achieved using an iterative argument and is omitted. □

Now we begin the search for r_0.

Lemma 5.3 (Lemma 11.2 [4]). *Take $r \in [0,\infty)$ and suppose that $H_1, H_2 \in \bar{b}_{\mathrm{H}(N)}(0,r)$. Then*

$$(2-e^r)\left\|H_1 - H_2\right\| \leq \left\|\psi\left(H_1\right) - \psi\left(H_2\right)\right\| \leq e^r \left\|H_1 - H_2\right\|.$$

In particular, ψ is continuous.

Remark 5.4. Since the proof of 5.3 in [4] is completely transparent, we do not include it here. We note that it does not use properties of the Hermitian matrices H, rather properties of $e^{\mathrm{i}(\,\cdot\,)}$, namely its power series expansion. This suggests a possibility of extending the results of this section to matrix subgroups of $\mathrm{U}(N)$ because the same lemma would hold with H replaced by another Lie algebra element. The other input needed for this section is Theorem 3.1 and its Corollary 3.2, for the small ball Haar measure estimates. If one obtained similar estimates for matrix subgroups of $\mathrm{U}(N)$, then one would likely obtain the main result of this section, Theorem 5.2. This remark mirrors the remark in [4].

Observe that injectivity of ψ follows from the quantity $2-e^r$ being strictly greater than 0. Therefore, taking the following restriction suffices, despite not being optimal.

Lemma 5.5. *The function ψ is injective when restricted to $b_{\mathrm{H}(N)}(0,\log(2)/2)$. As a consequence,*

$$\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,\log(2)/2))} : \psi(b_{\mathrm{H}(N)}(0,\log(2)/2)) \to b_{\mathrm{H}(N)}(0,\log(2)/2)$$

is well-defined.

Proof. If $H_1 \neq H_2$ are elements of $b_{\mathrm{H}(N)}(0,\log(2)/2)$, then by Lemma 5.3, we see that

$$\|\psi(H_1) - \psi(H_2)\| \geq (2-\sqrt{2})\|H_1 - H_2\| > 0.$$

Therefore, $\psi|_{b_{\mathrm{H}(N)}(0,\log(2)/2)}$ is injective. □

Lemma 5.6. *The function $\psi^{-1}|_{\psi(b_{\mathrm{H}(N)}(0,\log(2)/2))}$ is continuous as a map from a topological subspace of $\mathrm{U}(N)$ to a topological subspace of $\mathrm{H}(N)$.*

Proof. Fix $\psi(H_1) \in \psi(b_{\mathrm{H}(N)}(0, \log(2)/2))$. We prove that (the restriction of) ψ^{-1} is continuous at $\psi(H_1)$. Fix $\epsilon > 0$ and choose $\delta = \epsilon/2$. Let $\psi(H_2) \in \psi(b_{\mathrm{H}(N)}(0, \log(2)/2))$ such that $\|\psi(H_2) - \psi(H_1)\| \leq \delta$. We have

$$\|\psi^{-1}(\psi(H_2)) - \psi^{-1}(\psi(H_1))\| = \|H_1 - H_2\|.$$

For the next step, we observe that $H_1, H_2 \in b_{\mathrm{H}(N)}(0, \log(2)/2)$. Then by Lemma 5.3, we have

$$\begin{aligned}\|H_1 - H_2\| &\leq \frac{\|\psi(H_1) - \psi(H_2)\|}{2 - \sqrt{2}} \\ &\leq 2\delta \leq \epsilon.\end{aligned}$$

This concludes the proof. □

Thus, as long as $3r_0 \leq \frac{\log(2)}{2}$, then ψ restricted to $b_{\mathrm{H}(N)}(0, 3r_0)$ is a homeomorphism onto its image, which settles the question of the first property. By the discussion following from the first property at the beginning of this section, we obtain a measure ν, but let us write ν_{r_0} instead, to emphasize that it depends on r_0. We recall

$$\nu_{r_0} : \mathcal{B}_{\mathrm{H}(N)} \to [0, 1]$$

is given by

$$\mathcal{B}_{\mathrm{H}(N)} \ni A \mapsto \sigma(\psi(A \cap b_{\mathrm{H}(N)}(0, 3r_0))) \in [0, 1].$$

Lemma 5.7. *Suppose that $r_0 \in [0, \frac{\log(2)}{6}]$ is fixed. Then ν_{r_0} is absolutely continuous with respect to Lebesgue measure λ on $\mathrm{H}(N)$.*

Proof. If $r_0 = 0$, then ν_{r_0} is always 0, in which case it is absolutely continuous with respect to Lebesgue measure.

Thus, it remains to consider the case $r_0 > 0$. By definition of absolute continuity, it suffices to prove that, if $A \in \mathcal{B}_{\mathrm{H}(N)}$ and $\lambda(A) = 0$, then $\nu_{r_0}(A) = 0$. Thus, we start by taking an arbitrary $A \in \mathcal{B}_{\mathrm{H}(N)}$ with $\lambda(A) = 0$. We will prove that $\nu_{r_0}(A) < \epsilon$ for any $\epsilon > 0$, which suffices for $\nu_{r_0}(A) = 0$.

Take $\epsilon > 0$. By the standard construction of Lebesgue (outer) measure (which uses an infimum over countable coverings by balls), there exists a countable sequence of Euclidean balls

$$b_{\mathrm{H}(N)}(H_1, \delta_1), \cdots, b_{\mathrm{H}(N)}(H_k, \delta_k), \cdots$$

where $H_k \in \mathrm{H}(N)$ and $\delta_j > 0$ for all $j \in \mathbb{N}$, such that

$$A \subset \bigcup_{j=1}^{\infty} b_{\mathrm{H}(N)}(H_j, \delta_j)$$

and

$$\sum_{j=1}^{\infty} \lambda(b_{\mathrm{H}(N)}(H_j, \delta_j)) < \epsilon.$$

It suffices to prove that $\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \leq C\lambda(b_{\mathrm{H}(N)}(H_j, \delta_j))$ for any j, where C depends only on N, and not on ϵ nor j nor the covering taken. This suffices because then we have

$$\begin{aligned}\nu_{r_0}(A) &\leq \nu_{r_0}\left(\bigcup_{j=1}^{\infty} b_{\mathrm{H}(N)}(H_j, \delta_j)\right) \\ &\leq \sum_{j=1}^{\infty} \nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \\ &\leq \sum_{j=1}^{\infty} C\lambda(b_{\mathrm{H}(N)}(H_j, \delta_j)) < C\epsilon,\end{aligned}$$

which shows that $\nu_{r_0}(A) = 0$, since ϵ is arbitrary, and C depended only on N.[1]

[1] If C depended on the covering, say, then it could be that for each different choice of descreasing ϵ, a larger and larger C was needed, which would diasallow the conclusion.

Fix $j \in \mathbb{N}$. We want to show that $\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \leq C\lambda(b_{\mathrm{H}(N)}(H_j, \delta_j))$ for some constant C depending only on N. As a first observation, we note that the Lebesgue measure of a ball of radius δ_j in $\mathbb{R}^{N^2}$ is given by a constant C depending only on N, multiplied by $\delta_j^{N^2}$. Thus, $\lambda(b_{\mathrm{H}(N)}(H_j, \delta_j)) = C\delta_j^{N^2}$. Therefore, it suffices to prove that $\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \leq C\delta_j^{N^2}$ for another constant C depending only on N.

If δ_j is large, say $\delta_j > r_0$, then we proceed as follows. We have

$$\begin{aligned}\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \leq \sigma(\mathrm{U}(N)) = 1 &\leq \frac{r_0^{N^2}}{r_0^{N^2}} \\ &\leq \left(\frac{1}{r_0^{N^2}}\right)\delta_j^{N^2},\end{aligned}$$

where $C = \frac{1}{r_0^{N^2}}$ depends only on N. This is where we use that r_0 is already larger than 0, otherwise this step of the proof would fail.

Now we assume that $\delta_j \leq r_0$, and we consider various cases for the size of H_j.

Case 1. If $\|H_j\| \geq 4r_0$, then $b_{\mathrm{H}(N)}(H_j, \delta_j)$ does not intersect $b_{\mathrm{H}(N)}(0, 3r_0)$, which implies that $\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) = 0$.

Case 2. Suppose $\|H_j\| < 4r_0$. The idea is to use Lemma 5.3 to find some small ball in $\mathrm{U}(N)$ about the matrix $e^{\mathrm{i}H_j}$ which contains $\psi(b_{\mathrm{H}(N)}(H_j, \delta_j))$, and then to apply Corollary 3.2. To this end, we let $\psi(G) \in \psi(b_{\mathrm{H}(N)}(H_j, \delta_j))$. Then $G \in b_{\mathrm{H}(N)}(H_j, \delta_j)$, so both G and H are in the ball $b_{\mathrm{H}(N)}(0, 4r_0 + \delta_j) \subset b_{\mathrm{H}(N)}(0, 5r_0)$. Lemma 5.3 implies that

$$\begin{aligned}\|\psi(G) - \psi(H)\| &\leq e^{5r_0}\|G - H\| \\ &\leq e^{5r_0}\delta_j.\end{aligned}$$

Hence,

$$\psi(b_{\mathrm{H}(N)}(H_j, \delta_j)) \subset \bar{b}_{\mathrm{U}(N)}(e^{iH_j}, e^{5r_0}\delta_j).$$

Considering ν_{r_0} again, we note that

$$\psi(b_{\mathrm{H}(N)}(H_j, \delta_j) \cap b_{\mathrm{H}(N)}(0, 3r_0)) \subset \psi(b_{\mathrm{H}(N)}(H_j, \delta_j)) \subset \bar{b}_{\mathrm{U}(N)}(e^{iH_j}, e^{5r_0}\delta_j).$$

By monotonicity of Haar measure σ, we have

$$\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) = \sigma(\psi(b_{\mathrm{H}(N)}(H_j, \delta_j) \cap b_{\mathrm{H}(N)}(0, 3r_0))) \leq \sigma(\bar{b}_{\mathrm{U}(N)}(e^{iH_j}, e^{5r_0}\delta_j))$$

Since $e^{-iH_j} \cdot \bar{b}_{\mathrm{U}(N)}(e^{iH_j}, e^{5r_0}\delta_j) = \bar{b}_{\mathrm{U}(N)}(I, e^{5r_0}\delta_j)$ and $e^{-iH_j} \in \mathrm{U}(N)$, then by left invariance of Haar measure, we have

$$\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \leq \sigma(\bar{b}_{\mathrm{U}(N)}(e^{iH_j}, e^{5r_0}\delta_j)) = \sigma(\bar{b}_{\mathrm{U}(N)}(I, e^{5r_0}\delta_j)).$$

Since $e^{5r_0} \leq e^{6r_0} \leq 2$, we have $\sigma(\bar{b}_{\mathrm{U}(N)}(I, e^{5r_0}\delta_j)) \leq \sigma(\bar{b}_{\mathrm{U}(N)}(I, 2\delta_j))$. Since $2\delta_j \leq 2r_0 \leq \log(2)/3 \leq 1 \leq \sqrt{N}$ for any $N \geq 1$, then we apply Corollary 3.2 with $\delta = 2\delta_j$ to obtain

$$\nu_{r_0}(b_{\mathrm{H}(N)}(H_j, \delta_j)) \leq \sigma(\bar{b}_{\mathrm{U}(N)}(I, 2\delta_j)) \leq C_2(2\delta_j)^{N^2},$$

where $C = C_2 2^{N^2}$ is a constant depending only on N. This concludes case 2 and also concludes the proof. □

Lemma 5.8. *Suppose $0 < r < \log(2)/6$. Suppose $H \in b_{\mathrm{H}(N)}(0, r)$, and let $\delta \leq r$. Then*

$$b_{\mathrm{U}(N)}(e^{iH}, (2 - e^{6r})\delta) \subset \psi(b_{\mathrm{H}(N)}(H, \delta)).$$

Remark 5.9. This proof follows the lines of Corollary 11.3 [4]. We quote: "Therefore, if r is sufficiently small (depending only on N), the above inequality implies that $1 - \cos(\theta_j) \geq \theta_j^2/8$ for each j" [4, P. 25 Arxiv Version]. To the best of our knowledge, the inequality holds independently of r, because $\theta_j \in [-\pi, \pi]$ by the beginning of the proof, and therefore, r_0 is independent of N here.

Proof. Let

$$U \in b_{\mathrm{U}(N)}(e^{iH}, (2 - e^{6r})\delta). \tag{30}$$

We want to find a matrix $G \in \mathrm{H}(N)$ such that G is δ-close to H and $U = \psi(G)$.

Here is how G is found. "Let $e^{\mathrm{i}\theta_1}, \cdots, e^{\mathrm{i}\theta_N}$ be the eigenvalues of U, with θ_j's chose such that $-\pi \leq \theta_j < \pi$ for each j. Then U has a spectral decomposition VDV^*, where V is a unitary matrix and D is a diagonal matrix with diagonal entries $e^{\mathrm{i}\theta_1}, \cdots, e^{\mathrm{i}\theta_N}$. Let Λ be the diagonal matrix with diagonal entries $\theta_1, \cdots, \theta_N$. Then $G := V\Lambda V^*$ is a Hermitian matrix, and $U = e^{\mathrm{i}G}$" [4]. Therefore, it suffices to prove that

$$G \in b_{\mathrm{H}(N)}(H, \delta) \tag{31}$$

because then $U = \psi(G) \in \psi(b_{\mathrm{H}(N)}(H, \delta))$, which implies the desired set inclusion, based on 30.

Here is another observation that is helpful. If

$$G, H \in b_{\mathrm{H}(N)}(0, 6r), \tag{32}$$

then we have obtained the lemma. Indeed, we have by Lemma 5.3 and 30

$$(2 - e^{6r})\|G - H\| \leq \|e^{\mathrm{i}G} - e^{\mathrm{i}H}\| \leq (2 - e^{6r}) \cdot \delta.$$

Since $r < \log(2)/6$, we divide by $2 - e^{6r} > 0$ to obtain

$$\|G - H\| \leq \delta,$$

which is 31.

It remains to show that $G, H \in b_{\mathrm{H}(N)}(0, 6r)$. By the hypothesis of the lemma, we have $H \in b_{\mathrm{H}(N)}(0, r) \subset b_{\mathrm{H}(N)}(0, 6r)$. On the other hand, showing that G is $6r$-close to the zero matrix involves bounding its eigenvalues, which are related to the eigenvalues of U through the exponential map.

In particular, we have

$$\begin{aligned}
\|G\|^2 &= \mathrm{Tr}(G^*G) \\
&= \sum_{j=1}^{N} \theta_j^2 \\
&\leq \sum_{j=1}^{N} 8(1 - \cos\theta_j) \\
&\leq 4\|I - U\|^2,
\end{aligned}$$

where the third inequality follows from the fact that $8(1 - \cos\theta_j) \geq \theta_j^2$ for all $\theta_j \in [-\pi, \pi)$, and the fourth inequality follows by 6 from the introduction. Therefore it remains to show that $\|I - U\|^2 \leq 9r^2$, because then we have $\sqrt{\|G\|^2} \leq \sqrt{4 \cdot 9r^2} = 6r$.

We have

$$\begin{aligned}
\|I - U\| &\leq \|I - e^{iH}\| + \|e^{iH} - U\| & \\
&= \|e^{\mathrm{i}0} - e^{iH}\| + \|e^{iH} - U\| & \\
&\leq e^r\|H\| + \|e^{iH} - U\| & H \in b_{\mathrm{H}(N)}(0, r) \text{ and Lemma 5.3} \\
&\leq e^r\|H\| + (2 - e^{6r})\delta & U \in b_{\mathrm{U}(N)}(e^{iH}, (2 - e^{6r})\delta) \\
&\leq e^r r + (2 - e^{6r})r & \delta \leq r \\
&\leq e^r r + r & r \geq 0 \Rightarrow 2 - e^{6r} \leq 1. \\
&\leq 3r & r \in [0, \log(2)] \Rightarrow e^r \cdot r \leq 2r.
\end{aligned}$$

By taking squares on both sides of the above display, we attain $\|I - U\|^2 \leq 9r^2$, which, by previous remarks, concludes the proof. □

Lemma 5.10. *Let $r_0 \in [0, \frac{\log 2}{6}]$. Then for λ-almost all $H \in \mathrm{H}(N)$, we have that the Radon Nikodym derivative of ν_{r_0} with respect to λ at H satisfies*

$$\rho(H) = \lim_{a \to 0} \frac{\Gamma(N^2/2 + 1)\nu_{r_0}(b_{\mathrm{H}(N)}(H, a))}{\pi^{N^2/2} a^{N^2}}. \tag{33}$$

Proof. Recall the isometry $\Psi_H : \mathbb{R}^{N^2} \mapsto \mathrm{H}(N)$ defined in equation 13. Note that $(\Psi_H^{-1})_*\nu_{r_0}$ (the pushforward of ν_{r_0}) is absolutely continuous with respect to Lebesgue measure l_{N^2} on $\mathbb{R}^{N^2}$ by the proof of Lemma 5.7. Recall that for l_{N^2}-almost all $x \in \mathbb{R}^{N^2}$,

$$\begin{aligned}\frac{d(\Psi_H^{-1})_*\nu_{r_0}}{dl_{N^2}}(x) &= \lim_{r\to 0} \frac{(\Psi_H^{-1})_*\nu_{r_0}(b_{\mathbb{R}^{N^2}}(x,r))}{l_{N^2}(b_{\mathbb{R}^{N^2}}(x,r))} \\ &= \lim_{r\to 0} \frac{\Gamma(N^2/2+1)(\Psi_H^{-1})_*\nu_{r_0}(b_{\mathbb{R}^{N^2}}(x,r))}{\pi^{N^2/2}r^{N^2}}\end{aligned}$$

where the first equality is the small ball formula for the Radon Nikodym derivative, see [4, Section 11] and the reference therein, and the second equality is obtained by substitution of the formula for the Lebesgue measure of the ball of radius r in $\mathbb{R}^{N^2}$.

Then for any bounded measurable function $f : \mathrm{H}(N) \to \mathbb{R}$, we have

$$\begin{aligned}\int_{\mathrm{H}(N)} f(H)\rho(H)d\lambda(H) &= \int_{\mathrm{H}(N)} f(H)d\nu_{r_0}(H) \\ &= \int_{\mathbb{R}^{N^2}} f(\Psi_H(x))\, d(\Psi_H^{-1})_*\nu_{r_0}(x) \\ &= \int_{\mathbb{R}^{N^2}} f(\Psi_H(x))\, \frac{d(\Psi_H^{-1})_*\nu_{r_0}}{dl_{N^2}}(x)dl_{N^2}(x) \\ &= \int_{\mathbb{R}^{N^2}} f(\Psi_H(x))\, \lim_{r\to 0} \frac{\Gamma(N^2/2+1)(\Psi_H^{-1})_*\nu_{r_0}(b_{\mathbb{R}^{N^2}}(x,r))}{\pi^{N^2/2}r^{N^2}}dl_{N^2}(x) \\ &= \lim_{r\to 0}\int_{\mathbb{R}^{N^2}} f(\Psi_H(x))\, \frac{\Gamma(N^2/2+1)(\nu_{r_0}(b_{\mathrm{H}(N)}(\Psi_H(x),r))}{\pi^{N^2/2}r^{N^2}}dl_{N^2}(x) \\ &= \int_{\mathrm{H}(N)} f(H)\, \lim_{r\to 0} \frac{\Gamma(N^2/2+1)(\nu_{r_0}(b_{\mathrm{H}(N)}(H,r))}{\pi^{N^2/2}r^{N^2}}d\lambda(H)\end{aligned}$$

by applying the change of variables formula, dominated convergence theorem, and so forth. The result of the lemma follows, since f was arbitrary. □

Next we find r_0 for the assumptions defined by equations 27 and 28, namely the lower and upper bounds on the Radon Nikodym derivative. Notice that the upper bound on r is superfluous, because $2 - e^{6r}$ gets smaller as r gets larger.

Lemma 5.11. *Suppose* $0 < r_0 < \frac{\log(2)}{6}$, *and* $0 \le r \le r_0$. *With* $\rho = \frac{d\nu_{r_0}}{d\lambda}$, *we have*

$$(2 - e^{6r})^{N^2} C_N \le \rho(H)$$

for λ*-almost all* $H \in b_{\mathrm{H}(N)}(0,r)$.

Proof. Fix r_0, r such that $0 \le r \le r_0 < \frac{\log(2)}{6}$. Let N_* be the λ-null set outside of which 33 holds. Let $H \in b_{\mathrm{H}(N)}(0,r) \setminus N_*$. Then

$$\rho(H) = \lim_{a\to 0} \frac{\Gamma(N^2/2+1)\nu_{r_0}(b_{\mathrm{H}(N)}(H,a))}{\pi^{N^2/2}a^{N^2}}.$$

By definition of ν_{r_0} and adding assumptions on a, we have

$$\begin{aligned}\nu_{r_0}(b_{\mathrm{H}(N)}(H,a)) &= \sigma(\psi(b_{\mathrm{H}(N)}(H,a) \cap b_{\mathrm{H}(N)}(0,3r_0))) \\ &= \sigma(\psi(b_{\mathrm{H}(N)}(H,a))) && \text{if } a \le 2r \\ &\ge \sigma(b_{\mathrm{U}(N)}(e^{iH}, (2-e^{6r})\cdot a)) && \text{if } a \le r \text{ and apply Lemma 5.8}\end{aligned}$$

Hence,

$$\begin{aligned}\rho(H) &= \lim_{a\to 0} \frac{\Gamma(N^2/2+1)\nu_{r_0}(b_{\mathrm{H}(N)}(H,a))}{\pi^{N^2/2}a^{N^2}} \\ &= \lim_{\substack{a\to 0 \\ a\le r}} \frac{\Gamma(N^2/2+1)\nu_{r_0}(b_{\mathrm{H}(N)}(H,a))}{\pi^{N^2/2}a^{N^2}} \\ &\ge \lim_{a\to 0} \frac{\Gamma(N^2/2+1)\sigma(b_{\mathrm{U}(N)}(e^{iH}, (2-e^{6r})\cdot a))}{\pi^{N^2/2}a^{N^2}}.\end{aligned}$$

By translation invariance of Haar measure, we have

$$\sigma(b_{\mathrm{U}(N)}(e^{\mathrm{i}H}, (2-e^{6r})\cdot a)) = \sigma(b_{\mathrm{U}(N)}(I, (2-e^{6r})\cdot a)).$$

Furthermore, noting that r is fixed and $r < \log(2)/6$, we can multiply by $1 = \frac{(2-e^{6r})^{N^2}}{(2-e^{6r})^{N^2}}$ inside the limit, and then pull out constants to obtain

$$\rho(H) \geq \frac{\Gamma(N^2/2+1)(2-e^{6r})^{N^2}}{\pi^{N^2/2}} \lim_{a\to 0} \frac{\sigma(b_{\mathrm{U}(N)}(I, (2-e^{6r})\cdot a))}{(2-e^{6r})^{N^2} a^{N^2}}.$$

Since $(2-e^{6r}) > 0$ is fixed, then $\delta = (2-e^{6r})\cdot a \to 0$ as $a \to 0$. Hence,

$$\begin{aligned}
&\frac{\Gamma(N^2/2+1)(2-e^{6r})^{N^2}}{\pi^{N^2/2}} \lim_{a\to 0} \frac{\sigma(b_{\mathrm{U}(N)}(I, (2-e^{6r})\cdot a))}{(2-e^{6r})^{N^2} a^{N^2}} \\
&= \frac{\Gamma(N^2/2+1)(2-e^{6r})^{N^2}}{\pi^{N^2/2}} \lim_{\delta\to 0} \frac{\sigma(b_{\mathrm{U}(N)}(I, \delta))}{\delta^{N^2}} \\
&= \frac{\Gamma(N^2/2+1)(2-e^{6r})^{N^2}}{\pi^{N^2/2}} \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2} 2^{N^2/2} \Gamma(N^2/2+1)} \\
&= (2-e^{6r})^{N^2} C_N
\end{aligned}$$

where the second equality is the formula from Theorem 3.1, and the third equality is the definition of C_N from Theorem 5.2. Therefore, we have shown that

$$(2-e^{6r})^{N^2} C_N \leq \rho(H)$$

for λ-almost all $H \in b_{\mathrm{H}(N)}(0, r)$. □

Lemma 5.12. *Take any $r > 0$. Suppose that $H \in b_{\mathrm{H}(N)}(0, 3r)$. Suppose also that $\delta \leq r$. Then*

$$\psi(b_{\mathrm{H}(N)}(H, \delta)) \subset b_{\mathrm{U}(N)}(e^{\mathrm{i}H}, e^{4r}\delta).$$

Proof. Let $\psi(G) \in \psi(b_{\mathrm{H}(N)}(H, \delta))$, for some $G \in b_{\mathrm{H}(N)}(H, \delta)$. Then $G \in b_{\mathrm{H}(N)}(0, 3r+\delta)$ because G is δ-close to H, and H is $3r$-close to 0. Therefore, by Lemma 5.3 and the fact that $H, G \in b_{\mathrm{H}(N)}(0, 3r+\delta) \subset b_{\mathrm{H}(N)}(0, 4r)$, we have

$$\begin{aligned}
\|\psi(H) - \psi(G)\| &\leq e^{4r}\|H - G\| \\
&\leq e^{4r}\cdot\delta.
\end{aligned}$$

Hence $\psi(G)$ is $e^{4r}\cdot\delta$ close to $\psi(H) = e^{\mathrm{i}H}$, which means that $\psi(G) \in b_{\mathrm{U}(N)}(e^{\mathrm{i}H}, e^{4r}\cdot\delta)$, as desired. □

Lemma 5.13. *Suppose $0 < r_0 < \frac{\log(2)}{6}$, and $0 \leq r \leq r_0$. With $\rho = \frac{d\nu_{r_0}}{d\lambda}$, we have*

$$\rho(H) \leq e^{4rN^2} C_N$$

for λ-almost all $H \in b_{\mathrm{H}(N)}(0, 3r)$.

Proof. Fix r_0, r such that $0 \leq r \leq r_0 < \frac{\log(2)}{6}$. Let N_* be the λ-null set outside of which 33 holds. Let $H \in b_{\mathrm{H}(N)}(0, 3r) \setminus N_*$. Then

$$\rho(H) = \lim_{a\to 0} \frac{\Gamma(N^2/2+1)\nu_{r_0}(b_{\mathrm{H}(N)}(H, a))}{\pi^{N^2/2} a^{N^2}}.$$

Assume $r > 0$, as the case $r = 0$ is trivial. Let $\theta = 3r - \|H\| > 0$ (as $b_{\mathrm{H}(N)}(0, 3r)$ does not include the boundary). Then if $a \leq \theta$, we have

$$\begin{aligned}
\nu_{r_0}(b_{\mathrm{H}(N)}(H, a)) &= \sigma(\psi(b_{\mathrm{H}(N)}(H, a) \cap b_{\mathrm{H}(N)}(0, 3r_0))) \\
&= \sigma(\psi(b_{\mathrm{H}(N)}(H, a))).
\end{aligned}$$

Hence, similar to Lemma 5.11, we have

$$\begin{aligned}
\rho(H) &= \lim_{\substack{a\to 0\\ a\le\theta}} \frac{\Gamma(N^2/2+1)\nu_{r_0}(b_{\mathrm{H}(N)}(H,a))}{\pi^{N^2/2}a^{N^2}} \\
&= \lim_{a\to 0} \frac{\Gamma(N^2/2+1)\sigma(\psi(b_{\mathrm{H}(N)}(H,a)))}{\pi^{N^2/2}a^{N^2}} \\
&\le \lim_{a\to 0} \frac{\Gamma(N^2/2+1)\sigma(b_{\mathrm{U}(N)}(e^{\mathrm{i}H},e^{4r}\cdot a))}{\pi^{N^2/2}a^{N^2}} && a\le r \text{ and Lemma 5.12} \\
&= \frac{\Gamma(N^2/2+1)e^{4rN^2}}{\pi^{N^2/2}} \lim_{\delta\to 0} \frac{\sigma(b_{\mathrm{U}(N)}(I,\delta))}{\delta^{N^2}} \\
&= \frac{\Gamma(N^2/2+1)e^{4rN^2}}{\pi^{N^2/2}} \frac{\prod_{j=1}^{N-1} j!}{(2\pi)^{N/2}2^{N^2/2}\Gamma(N^2/2+1)} && \text{Theorem 3.1} \\
&= e^{4rN^2}C_N,
\end{aligned}$$

as desired. □

Lemma 5.14. *Let* $r_0 \le \max\{c\in\mathbb{R} : 2c \le (2-e^{6\cdot 3c})\cdot 3c\} = \frac{\log(4/3)}{18}$. *Suppose* $a\in[0,r_0]$. *Then* $b_{\mathrm{U}(N)}(I,2a)\subset\psi(b_{\mathrm{H}(N)}(0,3a))$.

Proof. Suppose that $U\in b_{\mathrm{U}(N)}(I,2a)$. Then apply Lemma 5.8, setting $r=3a$, $\delta=3a$, and taking $H=0\in b_{\mathrm{H}(N)}(0,r)$. We obtain

$$b_{\mathrm{U}(N)}(I,(2-e^{6\cdot 3a})\cdot 3a)\subset\psi(b_{\mathrm{H}(N)}(0,3a)).$$

Since $0<a\le R$, some manipulations show that $2a\le(2-e^{6\cdot 3a})\cdot 3a$. Therefore,

$$b_{\mathrm{U}(N)}(I,2a)\subset b_{\mathrm{U}(N)}(I,(2-e^{6\cdot 3a})\cdot 3a)\subset\psi(b_{\mathrm{H}(N)}(0,3a)),$$

as desired. □

Lemma 5.15. *Take* $r_0\le\frac{\log(2)}{4}$. *Then for any* $0\le a\le r_0$, *we have*

$$\psi(b_{\mathrm{H}(N)}(0,a))\subset b_{\mathrm{U}(N)}(I,2a).$$

Proof. We apply Lemma 5.12. Let $r=\log(2)/4$ and $\delta=a$ and $H=0$. Then we see that the hypothesis of Lemma 5.12 is satisfied. In particular, we have $\delta=a\le r_0\le\log(2)/4=r$, which verifies that one condition that was at risk of being broken. The conclusion is then

$$\psi(b_{\mathrm{H}(N)}(0,a))=\psi(b_{\mathrm{H}(N)}(0,\delta))\subset b_{\mathrm{U}(N)}(e^{\mathrm{i}0},e^{4\cdot(\log(2)/4)}\cdot\delta)=b_{\mathrm{U}(N)}(I,2a),$$

as desired. □

6 Lattice Maxwell Theory

6.1 A Preparation on Gaussian Measures

We begin this section by preparing information about Gaussian measures on $\mathbb{R}^X$, where X is a finite set, say, of cardinality $k \in \{1, 2, 3, \cdots\}$.

We recall that a Gaussian measure on $\mathbb{R}^X$ is a probability measure $\mu : \mathcal{B}(\mathbb{R}^X) \to [0, 1]$ with probability density function

$$d\mu(x) = \frac{1}{(2\pi)^{k/2}(\det \Sigma)^{1/2}} \exp\left(-\frac{1}{2}\langle x - \mu, \Sigma^{-1}(x - \mu)\rangle_{\mathbb{R}^X}\right) dl_k(x)$$

for some given positive definite symmetric matrix $\Sigma \in \mathbb{R}^{X\times X}$,[2] called the covariance matrix, and a vector $\mu \in \mathbb{R}^X$, called the mean vector. Here, positive definite means $\langle x, \Sigma x\rangle_{\mathbb{R}^X} > 0$ for all $x \in \mathbb{R}^X$ satisfying $x \neq 0$. Notice that a Gaussian measure on $\mathbb{R}^X$ is uniquely recoverable from the data (Σ, μ).

Note that if Σ fails to be positive definite (but is still symmetric), then one of its eigenvalues λ is less than or equal to 0. If an eigenvalue of Σ is equal to 0, then Σ is not invertible, yielding that the Gaussian density function above is not well-defined. If an eigenvalue of Σ is negative, then diagonalizing Σ and studying an iterated integral shows that the integral

$$\int_{\mathbb{R}^X} \exp\left(-\frac{1}{2}\langle x - \mu, \Sigma^{-1}(x - \mu)\rangle_{\mathbb{R}^X}\right) dl_k(x) = \infty.$$

Therefore, symmetric matrices which fail to be positive definite do not define Gaussian measures in the sense above. There is a more general definition of Gaussian measures for positive semi-definite matrices, but such a notion is not needed for this thesis.

It turns out that in lattice Maxwell theory, it is easier to begin with 2nd degree polynomials than with the data of a covariance matrix and a mean vector.

Lemma 6.1. *Suppose $P(x) = \langle x, Ax\rangle_{\mathbb{R}^X} + \langle v, x\rangle_{\mathbb{R}^X} + c$ is a 2nd degree polynomial in the commuting variables $(x_e)_{e\in X} \in \mathbb{R}^X$, where $A \in \mathbb{R}^{X\times X}$ is a symmetric positive definite matrix, $v \in \mathbb{R}^X$ is a given vector, and $c \in \mathbb{R}$. Then the probability density function*

$$d\mu(x) = \frac{1}{Z} \exp\left(-\frac{1}{2}P(x)\right) dl_k(x)$$

where

$$Z = \int_{\mathbb{R}^d} \exp\left(-\frac{1}{2}P(x)\right) dl_k(x)$$

defines a Gaussian measure on $\mathbb{R}^X$ with covariance

$$\Sigma = A^{-1}$$

and mean

$$\mu = -\frac{1}{2}A^{-1}v.$$

Proof. We will rewrite $d\mu(x)$ in the standard format with the mean μ and covariance Σ, which suffices for the lemma, as density functions determine measures.[3] By pulling out the constant $e^{-\frac{1}{2}c}$ from the normalization constant Z and the weight $\exp\left(-\frac{1}{2}P(x)\right)$, we obtain

$$d\mu(x) = \frac{1}{\int_{\mathbb{R}^X} dl_k(y)\, \exp\left(-\frac{1}{2}\langle y, Ay\rangle_{\mathbb{R}^X} + -\frac{1}{2}\langle v, y\rangle_{\mathbb{R}^X}\right)} \exp\left(-\frac{1}{2}\langle x, Ax\rangle_{\mathbb{R}^X} + -\frac{1}{2}\langle v, x\rangle_{\mathbb{R}^X}\right) dl_k(x)$$

[2] We remark that, even though the row and column order of Σ is not specified, as X is an unordered set, the determinant does not depend on this ordering, so $\det \Sigma$ is well-defined. Furthermore, matrix multiplication for unordered matrices is given by

$$(\Sigma\Sigma')_{ij} = \sum_{l\in X} \Sigma_{il}\Sigma'_{lj},$$

as in the usual case.

[3] Alternatively, we could have computed the characteristic function of μ.

As a next step, we compute the integral

$$\int_{\mathbb{R}^X} dl_k(y) \exp\left(-\frac{1}{2}\langle y, Ay\rangle_{\mathbb{R}^X} + -\frac{1}{2}\langle v, y\rangle_{\mathbb{R}^X}\right).$$

We have the eigendecomposition $A = U^T DU$, where $U \in \mathrm{O}(k)$ has rows consisting of eigenvectors of A, corresponding to eigenvalues in the diagonal matrix D. With a change of variable $y = U^T z$, we obtain $\langle y, Ay\rangle_{\mathbb{R}^X} = \langle Uy, DUy\rangle_{\mathbb{R}^X} = \langle z, Dz\rangle_{\mathbb{R}^X}$ and $\langle v, y\rangle_{\mathbb{R}^X} = \langle Uv, z\rangle_{\mathbb{R}^X}$. Thus, we have the equivalent iterated integral

$$\prod_{j=1}^{k} \int_{\mathbb{R}} e^{-\frac{1}{2}D_{jj}z_j^2 - \frac{1}{2}(Uv)_j z_j} dz_k.$$

Completing the square and some further details show that the previous iterated integral evaluates to

$$e^{\sum_{j=1}^{k} \frac{(Uv)_j^2}{8D_{jj}}} \times (2\pi)^{k/2} \times \frac{1}{\prod_{j=1}^{k}\sqrt{D_{jj}}}.$$

Observe that $\frac{1}{\prod_{j=1}^{k}\sqrt{D_{jj}}} = \det\left(A^{-1}\right)^{1/2}$, so we have

$$d\mu(x) = \frac{1}{(2\pi)^{k/2}\det(A^{-1})^{1/2}} \exp\left(-\frac{1}{2}\langle x, Ax\rangle_{\mathbb{R}^X} + -\frac{1}{2}\langle v, x\rangle_{\mathbb{R}^X} - \sum_{j=1}^{k}\frac{(Uv)_j^2}{8D_{jj}}\right) dl_k(x).$$

At this juncture, we define $\mu := -\frac{1}{2}A^{-1}v$ and $\Sigma := A^{-1}$, and we note that Σ is positive definite because A is. Then we have

$$\begin{aligned}
-\frac{1}{2}\langle \mu, \Sigma^{-1}\mu\rangle_{\mathbb{R}^X} &= -\frac{1}{2}\langle -\frac{1}{2}A^{-1}v, A(-\frac{1}{2}A^{-1}v)\rangle_{\mathbb{R}^X} \\
&= -\frac{1}{2}\langle -\frac{1}{2}A^{-1}v, -\frac{1}{2}v\rangle_{\mathbb{R}^X} \\
&= -\frac{1}{8}\langle A^{-1}v, v\rangle_{\mathbb{R}^X} \\
&= -\frac{1}{8}\langle U^T D^{-1}Uv, v\rangle_{\mathbb{R}^X} \\
&= -\sum_{j=1}^{k}\frac{(Uv)_j^2}{8D_{jj}}.
\end{aligned}$$

Then by substitution,

$$-\frac{1}{2}\langle x, Ax\rangle_{\mathbb{R}^X} + -\frac{1}{2}\langle v, x\rangle_{\mathbb{R}^X} - \sum_{j=1}^{k}\frac{(Uv)_j^2}{8D_{jj}} = -\frac{1}{2}\langle x, \Sigma^{-1}x\rangle_{\mathbb{R}^X} - \frac{1}{2}\langle 2\Sigma^{-1}\mu, x\rangle_{\mathbb{R}^X} - \frac{1}{2}\langle \mu, \Sigma^{-1}\mu\rangle_{\mathbb{R}^X}.$$

The expression to the right of the equal sign in the previous display is equal to $-\frac{1}{2}\langle x-\mu, \Sigma^{-1}(x-\mu)\rangle_{\mathbb{R}^X}$, and with this we see that $d\mu(x)$ can be written in the standard format with covariance Σ and mean μ. □

Let $P(x) = \langle x, Ax\rangle_{\mathbb{R}^X} + \langle v, x\rangle_{\mathbb{R}^X} + c$ be a second degree polynomial as in Lemma 6.1. Note that any second degree polynomial in the commuting variables $(x_e)_{e\in X}$ can be written in this format, if we remove the positive definite condition on A. The term $\langle x, Ax\rangle_{\mathbb{R}^X}$ is called the quadratic part of P whereas the term $\langle v, x\rangle_{\mathbb{R}^X}$ is called the linear part of P. Let us record some further terminology. A quadratic map is a function $\Psi : \mathbb{R}^X \to \mathbb{R}$ such that there is a symmetric $\mathbb{R}$-bilinear form $g : \mathbb{R}^X \times \mathbb{R}^X \to \mathbb{R}$ and an $\mathbb{R}$-linear map $f : \mathbb{R}^X \to \mathbb{R}$ such that $\Psi(x) = g(x, x) + f(x)$ for all $x \in \mathbb{R}^X$. When $f = 0$, we say that Ψ is a quadratic form. In particular, the quadratic part of P is a quadratic form, and the sum of its quadratic and linear parts is a quadratic map. With regards to the standard basis of $\mathbb{R}^X$, there is to each symmetric bilinear form a symmetric matrix and vice versa:

$$\{\text{symmetric bilinear forms on } \mathbb{R}^X\} \ni g(\,\cdot\,,\,\cdot\,) \leftrightarrow [g(e_i, e_j)]_{ij} = G \in \{A \in \mathbb{R}^{X\times X} : A^T = A\},$$

where e_i denotes the standard basis vector with the entry in position i equal to 1 and zeros at the other entries. In this case, we have that $g(x,y) = \langle x, Gy\rangle_{\mathbb{R}^X}$ for all $x, y \in \mathbb{R}^X$. In other words, if we have a quadratic form $\Psi(\,\cdot\,) = g(\,\cdot\,,\,\cdot\,)$ on $\mathbb{R}^X$, then we can assume that there is a symmetric matric G such that $\Psi(x) = \langle x, Gx\rangle_{\mathbb{R}^X}$ for all $x \in \mathbb{R}^X$, where G can be computed from the symmetric bilinear form g. Moreoever, quadratic forms always correspond to the quadratic part of some 2nd degree polynomial. Last, we say a quadratic form Ψ is positive definite if $\Psi(x) = g(x,x) > 0$ for all $x \in \mathbb{R}^X$ with $x \neq 0$. This holds if and only if the associated matrix is positive definite. The terminology in this paragraph is based on [11, Chapter XIII and Chapter XV].

6.2 Lattice Maxwell Theory and Boundary Conditions

Recall that if $(x,j,k) = p \in \mathcal{P}(\Lambda_n)$, we define

$$\begin{aligned} p_1 &= (x, x+e_j) \\ p_2 &= (x+e_j, x+e_j+e_k) \\ p_3 &= (x+e_k, x+e_j+e_k) \\ p_4 &= (x, x+e_k). \end{aligned}$$

We have

$$M_n^{(\mathbb{R})} : \mathbb{R}^{E_n} \to \mathbb{R}$$

given by

$$M_n^{(\mathbb{R})}(t) = \sum_{p \in \mathcal{P}(\Lambda_n)} (t_p)^2$$

for any $t \in \mathbb{R}^{E_n}$, where

$$t_p = t_{p_1} + t_{p_2} - t_{p_3} - t_{p_4}. \tag{34}$$

It is straightforward to verify that $M_n^{(\mathbb{R})}(\cdot)$ is a quadratic form on $\mathbb{R}^{E_n}$ with associated symmetric bilinear form $M_n^{(\mathbb{R})}(t,s) = \sum_{p \in \mathcal{P}(\Lambda_n)} t_p s_p$ for all $s, t \in \mathbb{R}^{E_n}$. A simple counterexample shows that $M_n^{(\mathbb{R})}$ is not positive definite. We take $d = 2$, $n = 2$ and $t \in \mathbb{R}^{E_2}$ satisfying $t = 1$, that is, t is equal to 1 on each edge. There is only one plaquette p in this case. We have $t_{p_1} + t_{p_2} - t_{p_3} - t_{p_4} = 1 + 1 - 1 - 1 = 0$, and hence $M_2^{(\mathbb{R})}(t) = 0$, but $t \neq 0$.

In lattice Maxwell theory, it is helpful to play with different boundary conditions. For example, this is one way to obtain a positive definite quadratic form. Another reason to do this is to be able to control more than just the axial edges. For instance, in the final proof, we will split up a large box into translates of a small box, and then control the edges on the faces of the small boxes. This amounts to dealing with a particular boundary condition. Therefore, we spend the next few paragraphs setting up notation to deal with various boundary conditions. Also, motivated by Lemma 6.1, we will disect various forms into their quadratic and linear parts.

Let E be an edge set satisfying $E_n^0 \subset E \subset E_n$, where the subset relations allow set equality. Given $\theta \in \mathbb{R}^E$ and $t \in \mathbb{R}^{E_n \setminus E}$, we define the extended configuration $t^\theta \in \mathbb{R}^{E_n}$ by

$$t_e^\theta = \begin{cases} t_e & \text{if } e \in E_n \setminus E \\ \theta_e & \text{if } e \in E \end{cases}.$$

Based on the sign convention in 34, we define two functions. First,

$$\phi_{\text{sign}} : [4] \times [4] \to \{-1, 1\}$$

given by

$$\phi_{\text{sign}}(i,j) = \begin{cases} 1 & \text{if } \{i,j\} \in \{\{1,2\},\{3,4\},\{1\},\{2\},\{3\},\{4\}\} \\ -1 & \text{else.} \end{cases}$$

This definition allows us to write

$$\begin{aligned}(t_p^\theta)^2 &= \sum_{i\in[4]}\sum_{j\in[4]} \phi_{\text{sign}}(i,j) t_{p_i}^\theta t_{p_j}^\theta \\ &= \sum_{\substack{i\in[4]:\\ p_i\in E}}\sum_{\substack{j\in[4]:\\ p_j\in E}} \phi_{\text{sign}}(i,j)\theta_{p_i}\theta_{p_j} + \sum_{\substack{i\in[4]:\\ p_i\notin E}}\sum_{\substack{j\in[4]:\\ p_j\in E}} \phi_{\text{sign}}(i,j) t_{p_i}\theta_{p_j} + \\ &\sum_{\substack{i\in[4]:\\ p_i\in E}}\sum_{\substack{j\in[4]:\\ p_j\notin E}} \phi_{\text{sign}}(i,j)\theta_{p_i} t_{p_j} + \sum_{\substack{i\in[4]:\\ p_i\notin E}}\sum_{\substack{j\in[4]:\\ p_j\notin E}} \phi_{\text{sign}}(i,j) t_{p_i} t_{p_j}\end{aligned}$$

for a given plaquette $p \in \mathcal{P}(\Lambda_n)$, a given $t \in \mathbb{R}^{E_n\setminus E}$, and a given boundary condition $\theta \in \mathbb{R}^E$. This is useful because it allows us to distinguish quadratic parts and linear parts. For example, the fourth term is quadratic in t, but the second and third term on the right hand side are linear in t, with coefficients determined by θ. Next, we define

$$\Phi_{\text{sign}} : [4] \to \{-1, 1\}$$

given by

$$\Phi_{\text{sign}}(i) = \begin{cases} 1 & \text{if } i = 1, 2 \\ -1 & \text{if } i = 3, 4. \end{cases}$$

This definition will come in handy for writing quadratic parts in a useful way.

Given a boundary condition $\theta \in \mathbb{R}^E$, we define a sort of adjusted lattice Maxwell action, or lattice Maxwell action with boundary condition θ, via

$$M_{E,\theta,n} : \mathbb{R}^{E_n\setminus E} \to \mathbb{R}$$

given by

$$M_{E,\theta,n}((t_e)_{e\in E_n\setminus E}) = M_n^{(\mathbb{R})}(t^\theta) = \sum_{p\in\mathcal{P}(\Lambda_n)} (t_p^\theta)^2$$

for any $(t_e)_{e\in E_n\setminus E} \in \mathbb{R}^{E_n\setminus E}$. Note that $M_{E,\theta,n}$ is a quadratic map (in the strict sense given after Lemma 6.1) with quadratic component

$$\begin{aligned}M_{E,0,n}(t) &= \sum_{p\in\mathcal{P}(\Lambda_n)} \sum_{\substack{i\in[4]:\\ p_i\notin E}}\sum_{\substack{j\in[4]:\\ p_j\notin E}} \phi_{\text{sign}}(i,j) t_{p_i} t_{p_j} \\ &= \sum_{p\in\mathcal{P}(\Lambda_n)} \left(\sum_{\substack{i\in[4]:\\ p_i\in E^c}} \Phi_{\text{sign}}(i) t_{p_i} \right)^2 \end{aligned} \tag{35}$$

and linear component

$$\sum_{p\in\mathcal{P}(\Lambda_n)} \left(2 \sum_{\substack{i\in[4]:\\ p_i\notin E}}\sum_{\substack{j\in[4]:\\ p_j\in E}} \phi_{\text{sign}}(i,j) t_{p_i}\theta_{p_j} \right) = \langle v_{E,\theta,n}, t\rangle_{\mathbb{R}^{E_n\setminus E}}, \tag{36}$$

where $v_{E,\theta,n} \in \mathbb{R}^{E_n\setminus E}$ is a particular vector. It is worth meditating on 35. It says that, regardless of the boundary data, the quadratic component will always be the same as it is for the case of vanishing boundary data.

While it is not necessarily easy to read off the entries of the vector $v_{E,\theta,n}$ from the linear part 36, once we do a change of variable, it becomes apparent. We proceed as follows. Given an edge $e \in E_n$, let

$$\mathcal{P}_e := \mathcal{P}_e(\Lambda_n) := \{p \in \mathcal{P}(\Lambda_n) : e \text{ is an edge of } p\}.$$

Note that

$$\begin{aligned}\sum_{p\in\mathcal{P}(\Lambda_n)}\left(2\sum_{\substack{i\in[4]:\\ p_i\notin E}}\sum_{\substack{j\in[4]:\\ p_j\in E}}\phi_{\text{sign}}(i,j)t_{p_i}\theta_{p_j}\right) &= \sum_{e\in E_n\setminus E}\left(\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}}\pm 2\theta_{p_j}t_e\right)\\ &= \sum_{e\in E_n\setminus E}t_e\left(\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}}\pm 2\theta_{p_j}\right),\end{aligned} \tag{37}$$

where for the purpose of this thesis, we do not need to know whether there is a plus or a minus in front of the 2, but the upcoming footnote says how to write this rigorously. This allows us to read off the entries of the vector $v_{E,\theta,n}\in\mathbb{R}^{E_n\setminus E}$, which will be useful for the proof of Theorem 6.6 below. We include as a footnote a verification of the the first equality in 37, namely how the change of summation indices works.[4]

Lemma 6.2 below shows that $M_{E,0,n}$ is positive definite. We quote some other facts to help for interpreting the same lemma as bounds on the eigenvalues of the matrix associated to $M_{E,0,n}$. It is well known that for a symmetric matrix $A\in\mathbb{R}^{k\times k}$ with eigenvalues $\lambda_{\min}(A)\le\cdots\le\lambda_{\max}(A)$, the minimal eigenvalue satisfies

$$\lambda_{\min}(A)=\min_{\|x\|_{\mathbb{R}^k}=1}\langle x,Ax\rangle_{\mathbb{R}^k} \tag{38}$$

and the maximal eigenvalue satisfies

$$\lambda_{\max}(A)=\max_{\|x\|_{\mathbb{R}^k}=1}\langle x,Ax\rangle_{\mathbb{R}^k}. \tag{39}$$

The term that is minimized/maximized is related to the Raleigh quotient.

Let $\Sigma_{E,n}\in\mathbb{R}^{(E_n\setminus E)\times(E_n\setminus E)}$ be the symmetric matrix associated to the quadratic form $M_{E,0,n}$, that is, $M_{E,0,n}(t)=\langle t,\Sigma_{E,n}t\rangle_{\mathbb{R}^{E_n\setminus E}}$ for all $t\in\mathbb{R}^{E_n\setminus E}$. When $E=E_n^0$, we also use the notation $\Sigma_{E_n^0,n}=\Sigma_n^0$ to denote the matrix of the form $M_{E_n^0,0,n}$.

For the following lemma and elsewhere in this thesis, we will use a bound on the number of plaquettes which contain a given edge. Fix an edge $e\in E_n$. We have

$$|\mathcal{P}_e|\le(d-1)\cdot 2=:C \tag{40}$$

[4]The index set on the left of the equality is given by

$$A=\{(p,i,j)\in\mathcal{P}(\Lambda_n)\times[4]\times[4]:p_i\in E,p_j\notin E\}.$$

The index set on the right of the equality is given by

$$B=\{(e,p,k)\in(E_n\setminus E)\times\mathcal{P}(\Lambda_n)\times[4]:p_k\in E,p\in\mathcal{P}_e\}.$$

Recall that if X is a finite set, $f:X\to\mathbb{R}$ is a function, and $g:Y\to X$ is a bijection, then $\sum_{x\in X}f(x)=\sum_{y\in Y}f(g(y))$. We first establish a bijection $\phi=(\phi_1,\phi_2,\phi_3):B\to A$. Given $(e,p,k)\in B$, let $\phi(e,p,k)=(p,k,j)\in A$ where j is the unique element in $[4]$ such that $p_j=e$. It is easy to see that ϕ is well-defined because $p_k\in E$ and $p_j\notin E$. *Injectivity:* Suppose $\phi(e,p,k)=\phi(e^*,p^*,k^*)$. We have by definition of ϕ that $\phi(e,p,k)=(p,k,j)$ where $j\in[4]$ is uniquely determined by the condition $p_j=e$. And $\phi(e^*,p^*,k^*)=(p^*,k^*,j^*)$ where $j^*\in[4]$ is uniquely determined by the condition $p^*_{j^*}=e^*$. Hence, $p=p^*$, $k=k^*$, and $j=j^*$. Therefore, $e=p_j=p^*_{j^*}=e^*$. Therefore, $(e,p,k)=(e^*,p^*,k^*)$, which proves injectivity. *Surjectivity:* Suppose $(p,i,j)\in A$. The candidate in B is (p_j,p,i). One checks that this is truly in B and maps to (p,i,j) under ϕ. This concludes surjectivity.

To complete the proof, we have

$$\begin{aligned}\sum_{p\in\mathcal{P}(\Lambda_n)}\left(\sum_{\substack{i\in[4]:\\ p_i\notin E}}\sum_{\substack{j\in[4]:\\ p_j\in E}}2\phi_{\text{sign}}(i,j)t_{p_i}\theta_{p_j}\right) &= \sum_{(p,i,j)\in A}2\phi_{\text{sign}}(i,j)t_{p_i}\theta_{p_j}\\ &= \sum_{y=(e,p,k)\in B}2\phi_{\text{sign}}(\phi_2(y),\phi_3(y))t_{\phi_1(y)_{\phi_2(y)}}\theta_{\phi_1(y)_{\phi_3(y)}}\\ &= \sum_{e\in E_n\setminus E}\left(\sum_{p\in\mathcal{P}_e}\sum_{\substack{k\in[4]:\\ p_k\in E}}2\phi_{\text{sign}}(k,(\text{index of } e \text{ in } p))\theta_{p_k}t_e\right),\end{aligned}$$

where the index of e in p refers to the index $i\in[4]$ such that $e=p_i$. This concludes the verification of the desired equality.

where C depends only on d. Indeed, if $e = (x, y) = (x, x+e_j)$ for some $j \in [d]$ is an interior edge, meaning that at least one of x or y has all entries strictly between 0 and $n-1$ (i.e. does not appear on a face of the box Λ_n), then the plaquettes containing e are in bijection with the set $\{(x, x \pm e_k) : k \in [d], k \neq j\}$, which has cardinality $(d-1) \cdot 2$. If e is not an interior edge, then there are less plaquettes containing e. For an exact analysis of this particular question of how many plaquettes contain a particular edge, see [1, Page 8].

Lemma 6.2 (Lemma 13.1 [4]). *For each $t \in \mathbb{R}^{E_n \setminus E}$,*

$$\frac{C_1}{n^{d+2}} \|t\|^2_{\mathbb{R}^{E_n \setminus E}} \leq M_{E,0,n}(t) \leq C_2 \|t\|^2_{\mathbb{R}^{E_n \setminus E}},$$

where C_1 and C_2 are positive constants that depend only on d. In particular, $\lambda_{\min}(\Sigma_{E,n}) \geq \frac{C_1}{n^{d+2}}$ and $\lambda_{\max}(\Sigma_{E,n}) \leq C_2$.

Proof. First we prove the upper bound. We have

$$\begin{aligned}
M_{E,0,n}(t) &= \sum_{p \in \mathcal{P}(\Lambda_n)} \left(\sum_{\substack{i \in [4]: \\ p_i \in E^c}} \Phi_{\text{sign}}(i) t_{p_i} \right)^2 \\
&\leq \sum_{p \in \mathcal{P}(\Lambda_n)} |\{i \in [4] : p_i \in E^c\}| \cdot \sum_{\substack{i \in [4]: \\ p_i \in E^c}} t^2_{p_i} \\
&\leq 4 \sum_{p \in \mathcal{P}(\Lambda_n)} \sum_{\substack{i \in [4]: \\ p_i \in E^c}} t^2_{p_i} \\
&= 4 \cdot \sum_{e \in E_n \setminus E} \sum_{p \in \mathcal{P}_e} t^2_e \\
&= 4 \cdot \sum_{e \in E_n \setminus E} |\mathcal{P}_e| t^2_e \\
&\leq C \cdot \sum_{e \in E_n \setminus E} t^2_e = C \|t\|^2_{\mathbb{R}^{E_n \setminus E}},
\end{aligned}$$

as desired.

Now we give a proof for the lower bound. First we prove a claim, then we discuss the consequence.

Claim: For each non-negative integer k and for any $x \in \Lambda_n$ with $|x|_1 \leq k$, there exists a subset $B(x) \subset \mathcal{P}(\Lambda_n)$ such that $|B(x)| \leq |x|_1$ and

$$|t_e| \leq \sum_{p' \in B(x)} |t_{p'}|$$

for any $e = (x, y) \in E_n$.

Proof of claim. We proceed by induction on k. For the base case, if $|x|_1 = 0$, then $e = (x, y) \in E^0_n$ by definition of E^0_n. Since $E^0_n \subset E$, we have $e \in E$. Therefore, $t_e = 0$, so we conclude the base case.

Now we assume the induction hypothesis for some non-negative integer k. Suppose $x \in \Lambda_n$ with $|x|_1 = k+1$. Fix $e = (x, y) \in E_n$. By definition of E_n, there is some index $j \in [d]$ such that $y = x + e_j$. Let l be the largest index in $[d]$ such that $x_l \neq 0$. The trivial case is when $j \geq l$, because then $(x, y) \in E^0_n$, which implies that $t_e = 0$, because $E^0_n \subset E$. Suppose that $j < l$. We will find a plaquette p with enough E^0_n edges, which also has e as an edge, and has an edge related to the induction hypothesis. Let $z = x - e_l$, which implies that $(z, x) \in E^0_n$. Consider the plaquette $p = (z, j, l) \in \mathcal{P}(\Lambda_n)$ with edges

$$\begin{aligned}
p_1 &= (z, z + e_j) \\
p_2 &= (z + e_j, y) \\
p_3 &= (x, y) = e \\
p_4 &= (z, x).
\end{aligned}$$

Since $|z|_1 = k$, then by the induction hypothesis, there exists a subset $B(z) \subset \mathcal{P}(\Lambda_n)$ such that $|B(z)| \le k$ and

$$|t_{p_1}| = |t_{(z,z+e_j)}| \le \sum_{p' \in B(z)} |t_{p'}|.$$

Then by definition of E_n^0, we have $p_2, p_4 \in E_n^0$. Hence, $t_p = t_{p_1} - t_{p_3}$, as $E_n^0 \subset E$. Then we have

$$\begin{aligned} |t_e| &= |t_{p_3}| \\ &\le |t_{p_1} - t_{p_3}| + |t_{p_1}| \\ &= |t_p| + |t_{p_1}| \\ &\le |t_p| + \sum_{p' \in B(z)} |t_{p'}| \\ &= \sum_{p' \in B(z) \cup \{p\}} |t_{p'}|. \end{aligned}$$

Therefore, $B(x) = B(z) \cup \{p\}$ satisfies $|B(x)| \le k+1$, which closes the induction. □

Now we work out the consequence of the claim. Note that if $x \in \Lambda_n$, then $|x|_1 \le dn$. Therefore, for any $e = (x, y) \in E_n \setminus E$, we have

$$\begin{aligned} |t_e|^2 &\le \left(\sum_{p \in B(x)} |t_p| \right)^2 && \text{Claim above} \\ &\le \left(\sum_{p \in B(x)} 1^2 \right) \left(\sum_{p \in B(x)} t_p^2 \right) && \text{Cauchy Schwarz} \\ &\le |B(x)| \cdot M_{E,0,n}(t) \\ &\le |x|_1 \cdot M_{E,0,n}(t) \\ &\le Cn \cdot M_{E,0,n}(t). \end{aligned}$$

Then we have

$$\begin{aligned} \|t\|^2_{\mathbb{R}^{E_n \setminus E}} &= \sum_{e \in E_n \setminus E} t_e^2 \\ &\le |E_n \setminus E| \cdot Cn \cdot M_{E,0,n}(t) \\ &\le Cn^{d+1} M_{E,0,n}(t). \end{aligned}$$

Thus,

$$\frac{C}{n^{d+1}} \|t\|^2_{\mathbb{R}^{E_n \setminus E}} \le M_{E,0,n}(t),$$

which is stronger than the lower bound for the present lemma, and we thus conclude the proof. □

Given that $M_{E,0,n}$ is positive definite, and that $M_{E,0,n}$ is the quadratic component of $M_{E,\theta,n}$, then the probability measure $\tau_{E,\theta,n} : \mathcal{B}_{\mathbb{R}^{E_n \setminus E}} \to [0,1]$ with density

$$d\tau_{E,\theta,n}(t) = \frac{1}{Z} \exp\left(-\frac{1}{2} M_{E,\theta,n}(t) \right) \prod_{e \in E_n \setminus E} dl_1(t_e), \tag{41}$$

where Z denotes the normalization constant, is a Gaussian measure with covariance matrix $\Sigma^{-1}_{E,n}$, by Lemma 6.1. The measure $\tau_{E,\theta,n}$ is called *lattice Maxwell theory with boundary value* $\theta \in \mathbb{R}^E$.

Also, note that the normalization constant Z does not automatically have the usual format because the linear and constant part of $M_{E,\theta,n}$ can be nonzero. On the other hand, when $\theta = 0$, the mean vector is clearly $0 \in \mathbb{R}^{E_n \setminus E}$, so $\tau_{E,0,n}$ is a centered Gaussian probability measure. Also in this case, the density in 41 is already in standard form:

$$d\tau_{E,0,n}(t) = \frac{1}{(2\pi)^{|E_n \setminus E|/2} \det\left(\Sigma^{-1}_{E,n}\right)^{1/2}} \exp\left(-\frac{1}{2} \langle t, (\Sigma^{-1}_{E,n})^{-1} t \rangle_{\mathbb{R}^{E_n \setminus E}} \right) \prod_{e \in E_n \setminus E} dl(t_e).$$

In the particularly relevant case when the boundary edges are the axial edges, that is, $E = E_n^0$, and we have boundary value 0, then we use the shortened notation τ_n for the Gaussian measure $\tau_{E_n^0,0,n}$ whose covariance matrix is $(\Sigma_n^0)^{-1}$.

6.3 Three Key Bounds

First we prove a bound for Lattice Maxwell Theory with boundary value $0 \in \mathbb{R}^{E_n^0}$ (i.e. in the axial gauge).

We recall the following projection property of Gaussian measures. Suppose X is a finite set. Suppose $t = (t_j)_{j \in X} \in \mathbb{R}^X$ is a Gaussian distributed random vector with mean $(\mu_j)_{j \in X}$ and covariance $(\Sigma_{i,j})_{i,j \in X}$. Consider a subset $A \subset X$. Then $t_A = (t_j)_{j \in A} \in \mathbb{R}^A$ is also Gaussian distributed with mean $\mu_A = (\mu_j)_{j \in A}$ and covariance $\Sigma_A = (\Sigma_{ij})_{i,j \in A}$. Here is another way to state this. Let $P_A : \mathbb{R}^X \to \mathbb{R}^A$ be the projection $(t_j)_{j \in X} \mapsto (t_j)_{j \in A}$. Then for any Borel subset $B \in \mathcal{B}_{\mathbb{R}^A}$, we have

$$\mathbb{P}[P_A(t) \in B] = \int_B \frac{1}{(2\pi)^{|A|/2} \det(\Sigma_A)^{1/2}} e^{-\frac{1}{2}\langle t_A - \mu_A, \Sigma_A^{-1}(t_A - \mu_A)\rangle_{\mathbb{R}^{|A|}}} \prod_{j \in A} dl(t_j).$$

Notice on the other hand that

$$\mathbb{P}[P_A(t) \in B] = \mathbb{P}[t \in P_A^{-1}(B)] = \mathbb{P}[t \in B \times \mathbb{R}^{X \setminus A}].$$

Therefore,

$$\begin{gathered} \int_B \frac{1}{(2\pi)^{|A|/2} \det(\Sigma_A)^{1/2}} e^{-\frac{1}{2}\langle t_A - \mu_A, \Sigma_A^{-1}(t_A - \mu_A)\rangle_{\mathbb{R}^A}} \prod_{j \in A}(t_j) = \\ \int_{\mathbb{R}^{X \setminus A}} \int_B \frac{1}{(2\pi)^{|X|/2} \det(\Sigma)^{1/2}} e^{-\frac{1}{2}\langle x - \mu, \Sigma^{-1}(x - \mu)\rangle_{\mathbb{R}^X}} \prod_{i \in A} dl(t_i) \prod_{i \in X \setminus A} dl(t_i) \end{gathered} \tag{42}$$

because the left side is equal to $\mathbb{P}[P_A(t) \in B]$ and the right side is equal to $\mathbb{P}[t \in B \times \mathbb{R}^{X \setminus A}]$.

Recall that $E_n^1 = E_n \setminus E_n^0$.

Theorem 6.3 (Theorem 14.1 [4]). *Take any $n \geq 2$. For any nonempty $A \subseteq E_n^1$ and $\eta \in (0, 1/2]$, we have*

$$\tau_n \left(\left\{ t \in \mathbb{R}^{E_n^1} : |t(x,y)| \leq \eta \text{ for all } (x,y) \in A \right\} \right) \geq e^{-C|A|(\log(1/\eta) + \log n)},$$

where C is a positive constant that depends only on d.

Proof. We have

$$\{t \in \mathbb{R}^{E_n^1} : |t(x,y)| \leq \eta \text{ for all } (x,y) \in A\} = \mathbb{R}^{E_n^1 \setminus A} \times [-\eta, \eta]^A.$$

The probability of this set under τ_n is given by

$$\frac{1}{(2\pi)^{|E_n^1|/2} \det(\Sigma_n^0)^{-1/2}} \int_{\mathbb{R}^{E_n^1 \setminus A} \times [-\eta,\eta]^A} \exp\left(-\frac{1}{2} \langle t, \Sigma_n^0 t \rangle_{\mathbb{R}^{E_n^1}} \right) \prod_{e \in E_n^1} dl_1(t_e).$$

Recall that the covariance matrix of the measure τ_n is equal to $(\Sigma_n^0)^{-1}$. Let us denote

$$\Sigma = (\Sigma_n^0)^{-1}$$

and

$$\Sigma_A = (\Sigma_{e,e'})_{e,e' \in A}.$$

Then we have

$$\begin{gathered} \frac{1}{(2\pi)^{|E_n^1|/2} \det(\Sigma_n^0)^{-1/2}} \int_{\mathbb{R}^{E_n^1 \setminus A} \times [-\eta,\eta]^A} \exp\left(-\frac{1}{2} \langle t, \Sigma_n^0 t \rangle_{\mathbb{R}^{E_n^1}} \right) \prod_{e \in E_n^1} dl_1(t_e). = \\ \int_{[-\eta,\eta]^A} \frac{1}{(2\pi)^{|A|/2} (\det \Sigma_A)^{1/2}} \exp\left(-\frac{1}{2} \langle t_A, \Sigma_A^{-1} t_A \rangle_{\mathbb{R}^A} \right) \prod_{e \in A} dl_1(t_e) \end{gathered}$$

by equation 42. The proof is completed by using bounds on the top and bottom eigenvalue of Σ_A.

In this direction, note that

$$\lambda_{\min}(\Sigma) \leq \lambda_{\min}(\Sigma_A) \leq \lambda_{\max}(\Sigma_A) \leq \lambda_{\max}(\Sigma).$$

Furthermore,

$$\lambda_{\min}(\Sigma) = \frac{1}{\lambda_{\max}(\Sigma_n^0)} \geq \frac{1}{C_2}$$

and

$$\lambda_{\max}(\Sigma) = \frac{1}{\lambda_{\min}(\Sigma_n^0)} \leq \frac{n^{d+2}}{C_1}$$

by Lemma 6.2.

Therefore, we have

$$\det \Sigma_A \leq \lambda_{\max}(\Sigma_A)^{|A|} \leq \left(\frac{n^{d+2}}{C_1}\right)^{|A|}.$$

For the argument inside the exponential, we have

$$\begin{aligned}
\langle t_A, \Sigma_A^{-1} t_A \rangle_{\mathbb{R}^A} &\leq \lambda_{\max}(\Sigma_A^{-1}) \|t_A\|_{\mathbb{R}^A}^2 \\
&= \frac{1}{\lambda_{\min}(\Sigma_A)} \|t_A\|_{\mathbb{R}^A}^2 \\
&\leq \frac{1}{\lambda_{\min}(\Sigma)} \|t_A\|_{\mathbb{R}^A}^2 \\
&\leq C_2 \|t_A\|_{\mathbb{R}^A}^2.
\end{aligned}$$

Based on these two bounds, we have

$$\begin{aligned}
&\int_{[-\eta,\eta]^A} \frac{1}{(2\pi)^{|A|/2}(\det \Sigma_A)^{1/2}} \exp\left(-\frac{1}{2}\langle t, \Sigma_A^{-1} t\rangle_{\mathbb{R}^A}\right) \prod_{e\in A} dl_1(t_e) \geq \\
&\quad \frac{1}{(2\pi)^{|A|/2}\left(\frac{n^{d+2}}{C_1}\right)^{|A|/2}} \int_{[-\eta,\eta]^A} e^{-\frac{1}{2}C_2\|t\|_{\mathbb{R}^A}^2} \prod_{e\in A} dl_1(t_e) = \\
&\quad \frac{1}{(2\pi)^{|A|/2}\left(\frac{n^{d+2}}{C_1}\right)^{|A|/2}} \prod_{e\in A} \int_{[-\eta,\eta]} e^{-\frac{1}{2}C_2 t_e^2} dl_1(t_e) \geq \\
&\quad \frac{1}{(2\pi)^{|A|/2}\left(\frac{n^{d+2}}{C_1}\right)^{|A|/2}} \prod_{e\in A} e^{-\frac{1}{2}C_2\eta^2} \cdot 2\eta = \\
&\quad \frac{(2\eta)^{|A|}(e^{-\frac{1}{2}C_2\eta^2})^{|A|}}{(2\pi)^{|A|/2}\left(\frac{n^{d+2}}{C_1}\right)^{|A|/2}}.
\end{aligned}$$

The last step is to achieve the lower bound in the statement of the theorem. We rewrite

$$\frac{(2\eta)^{|A|}(e^{-\frac{1}{2}C_2\eta^2})^{|A|}}{(2\pi)^{|A|/2}\left(\frac{n^{d+2}}{C_1}\right)^{|A|/2}} = \frac{\sqrt{2C_1}^{|A|}}{\sqrt{\pi}^{|A|}(\sqrt{e^{C_2}})^{\eta^2|A|}} \times \frac{\eta^{|A|}}{n^{(d/2+1)|A|}}.$$

Since $\eta < 1$, we have $(\sqrt{e^{C_2}})^{\eta^2|A|} < (\sqrt{e^{C_2}})^{1\cdot|A|}$. Therefore we have

$$\frac{(2\eta)^{|A|}(e^{-\frac{1}{2}C_2\eta^2})^{|A|}}{(2\pi)^{|A|/2}\left(\frac{n^{d+2}}{C_1}\right)^{|A|/2}} \geq (C_3)^{|A|} \times \frac{\eta^{|A|}}{n^{(d/2+1)|A|}},$$

where $\frac{\sqrt{2C_1}}{\sqrt{\pi}\sqrt{e^{C_2}}} = C_3 > 0$ depends only on d and not on η nor $|A|$. Even though C_3 is possibly very small, we can find a constant C depending only on C_3 (and therefore only on d) such that $(1/2)^C \leq C_3$, whence $\eta^C \leq C_3$ because $\eta \leq 1/2$. Therefore $\eta^{C+1} \leq C_3\eta$. Thus,

$$(C_3)^{|A|} \times \frac{\eta^{|A|}}{n^{(d/2+1)|A|}} = \frac{(C_3\eta)^{|A|}}{n^{(d/2+1)|A|}} \geq \frac{\eta^{(C+1)|A|}}{n^{(d/2+1)|A|}},$$

and the right hand side is greater than or equal to $\frac{\eta^{C|A|}}{n^{C|A|}} = e^{-C|A|(\log(1/\eta)+\log n)}$ for some other constant C depending only on d. This concludes the proof. □

Next we prepare for another bound with a lemma, which states equation (12.8) [4].

Lemma 6.4. *Let X be a finite set with cardinality k and suppose that $(t_e)_{e\in X} \in \mathbb{R}^X$ is Gaussian distributed with mean $(\mu_e)_{e\in X} \in \mathbb{R}^X$ and covariance $(\Sigma_{ij})_{i,j\in X} \in \mathbb{R}^{X\times X}$. Then if $k \geq 2$, we have*

$$\mathbb{P}\left(\max_{e\in X}|t_e| \geq \max_{e\in X}|\mu_e| + \sqrt{6\lambda_{\max}(\Sigma)\log k}\right) \leq \frac{2}{k^2} \leq \frac{1}{2}.$$

Proof. For the first portion of the proof, we consider two parameters $x \geq 0$ and $\theta \geq 0$, which we will later choose in a particular way to finish the proof.

We have

$$\mathbb{P}(\max_{e\in X}|t_e - \mu_e| \geq x) \leq \mathbb{P}\left(\cup_{e\in X}\{|t_e - \mu_e| \geq x\}\right) \leq \sum_{e\in X}\mathbb{P}\{|t_e - \mu_e| \geq x\}.$$

by set inclusion and properties of $\mathbb{P}$. For each $e \in X$, by the projection property of Gaussian distributions, we know that t_e is Gaussian with mean μ_e and variance Σ_{ee}. Therefore $Y_e := t_e - \mu_e$ is Gaussian with mean 0 and variance Σ_{ee}. By symmetry of the Gaussian density function, we get $\mathbb{P}(|Y_e| \geq x) = 2\mathbb{P}(Y_e \geq x)$ for each $e \in X$. Hence,

$$\sum_{e\in X}\mathbb{P}\{|t_e - \mu_e| \geq x\} = 2\sum_{e\in X}\mathbb{P}\{t_e - \mu_e \geq x\}.$$

Next, for each $e \in X$, we have

$$\mathbb{P}(X_e - \mu_e \geq x) \leq e^{-\theta x}\mathbb{E}(e^{\theta(X_e-\mu_e)}).$$

Indeed,

$$\begin{aligned} e^{-\theta x}\mathbb{E}(e^{\theta(X_e-\mu_e)}) &= \mathbb{E}(e^{\theta(Y_e-x)}) \\ &= \mathbb{E}(1_{\{Y_e\geq x\}}e^{\theta(Y_e-x)} + 1_{\{Y_e<x\}}e^{\theta(Y_e-x)}) \\ &\geq \mathbb{E}(1_{\{Y_e\geq x\}}e^{\theta(Y_e-x)}) \\ &\geq \mathbb{E}(1_{\{Y_e\geq x\}}\cdot 1) = \mathbb{P}(X_e - \mu_e \geq x), \end{aligned}$$

where the third inequality uses that $e^{\theta(Y_e-x)}$ is bounded below by 0 on $\{Y_e < x\}$. Using that Y_e is Gaussian with mean 0 and variance Σ_{ee}, we have

$$\mathbb{E}(e^{\theta(Y_e)}) = \int_{\mathbb{R}} dy \frac{1}{(2\pi)^{1/2}(\Sigma_{ee})^{1/2}} e^{-\frac{y^2}{2\Sigma_{ee}}+\theta y} = \exp\left(\frac{\theta^2\Sigma_{ee}}{2}\right),$$

where the second equality follows from completing the square and doing the usual Gaussian integral. Note that for any $j \in X$, we have $\Sigma_{jj} = \langle e_j, \Sigma e_j\rangle_{\mathbb{R}^X}$, where $e_j \in \mathbb{R}^X$ denotes the vector with 1 in the jth entry and 0's elsewhere, so that by 39 we have $\Sigma_{jj} \leq \lambda_{\max}(\Sigma)$. In other words, the variance of t_j is bounded above by $\lambda_{\max}(\Sigma)$.

Hence,

$$2\sum_{e\in X}\mathbb{P}\{t_e - \mu_e \geq x\} \leq 2e^{-\theta x}\sum_{e\in X}\mathbb{E}(e^{\theta(X_e-\mu_e)}) = 2e^{-\theta x}\sum_{e\in X}e^{\theta^2\Sigma_{ee}/2} \leq 2ke^{-\theta x+\theta^2\lambda_{\max}(\Sigma)/2}$$

Hence,

$$\mathbb{P}(\max_{e\in X}|t_e - \mu_e| \geq x) \leq 2ke^{-\theta x+\theta^2\lambda_{\max}(\Sigma)/2}.$$

The choice $\theta = x/\lambda_{\max}(\Sigma)$ shows that

$$\mathbb{P}(\max_{e\in X}|t_e - \mu_e| \geq x) \leq 2ke^{-\frac{x^2}{2\lambda_{\max}(\Sigma)}}$$

Note also that

$$\mathbb{P}(\max_{e\in X}|t_e| \geq \max_{e\in X}|\mu_e| + x) \leq \mathbb{P}(\max_{e\in X}|t_e - \mu_e| \geq x)$$

because if $\max_{e\in X}|t_e| \geq \max_{e\in X}|\mu_e| + x$, then

$$\max_{e\in X}|t_e - \mu_e| \geq \max_{e\in X}|t_e| - \max_{e\in X}|\mu_e| \geq x,$$

which implies the associated set inclusion. When $x = \sqrt{6\lambda_{\max}(\Sigma)\log k}$, we get

$$\begin{aligned} 2ke^{-\frac{x^2}{2\lambda_{\max}(\Sigma)}} &= 2ke^{-\frac{\sqrt{6\lambda_{\max}(\Sigma)\log k}^2}{2\lambda_{\max}(\Sigma)}} \\ &= 2k\exp(-3\log k) \\ &= \frac{2}{k^2}, \end{aligned}$$

and hence

$$\mathbb{P}\left(\max_{e\in X}|t_e| \geq \max_{e\in X}|\mu_e| + \sqrt{6\lambda_{\max}(\Sigma)\log k}\right) \leq \frac{2}{k^2},$$

as desired. □

Lemma 6.5. *Let E satisfy $E_n^0 \subseteq E \subseteq E_n$, let $\theta \in \mathbb{R}^E$, and let $n \geq 1$ be given. Recall the vector $v := v_{E,\theta,n} \in \mathbb{R}^{E_n\setminus E}$ from the linear component of $M_{E,\theta,n}(\,\cdot\,)$. There exists a constant $C > 0$ depending only on d such that*

$$\|v\|^2_{\mathbb{R}^{E_n\setminus E}} \leq C\|\theta\|^2_{\mathbb{R}^E}.$$

Proof. By 37 we have

$$\begin{aligned} \|v\|^2_{\mathbb{R}^{E_n\setminus E}} &= \sum_{e\in E_n\setminus E} v_e^2 \\ &= \sum_{e\in E_n\setminus E}\left(\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}} \pm 2\theta_{p_j}\right)^2 \\ &\leq \sum_{e\in E_n\setminus E}\left(\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}} |2\theta_{p_j}|\right)^2. \end{aligned}$$

Recall that for $x = (x_1, \cdots, x_k) \in \mathbb{R}^k$ and $\|x\|_1 = \sum_{i\in[k]}|x_i|$ and $\|x\|_2 = (\sum_{i\in[k]}|x_i|^2)^{1/2}$, that $\|x\|_1 \leq \sqrt{k}\|x\|_2$, and therefore

$$(\sum_{i\in[k]}|x_i|)^2 = \|x\|_1^2 \leq k\|x\|_2^2 = k\sum_{i\in[k]} x_i^2.$$

Using this fact, we have

$$\sum_{e\in E_n\setminus E}\left(\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}} |2\theta_{p_j}|\right)^2 \leq \sum_{e\in E_n\setminus E}|\{(p,j)\in\mathcal{P}_e\times[4] : p_j\in E\}| \sum_{\substack{(p,j)\in\{(p,j)\in\\ \mathcal{P}_e\times[4]:p_j\in E\}}} |2\theta_{p_j}|^2.$$

We have for each $e \in E_n \setminus E$ by 40 that

$$|\{(p,j)\in\mathcal{P}_e\times[4] : p_j\in E\}| \leq C_1$$

for some constant C_1 depending only on d. Collecting what we have so far, this gives

$$\|v\|^2_{\mathbb{R}^{E_n\setminus E}} \leq C_1 \sum_{e\in E_n\setminus E}\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}} |2\theta_{p_j}|^2 = C_2 \sum_{e\in E_n\setminus E}\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}} |\theta_{p_j}|^2.$$

We have

$$C_2 \sum_{e\in E_n\setminus E}\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E}} |\theta_{p_j}|^2 = C_2 \sum_{e\in E}\sum_{p\in\mathcal{P}_e}\sum_{\substack{j\in[4]:\\ p_j\in E_n\setminus E}} |\theta_e|^2,$$

where the change of summation can be justified rigorously as we did in a footote for 37, but this time using the bijection ϕ between the sets

$$A = \{(e, p, j) \in (E_n \setminus E) \times \mathcal{P}(\Lambda_n) \times [4] : p \in \mathcal{P}_e, p_j \in E\},$$

and

$$B = \{(e, p, i) \in E \times \mathcal{P}(\Lambda_n) \times [4] : p \in \mathcal{P}_e, p_i \in E_n \setminus E\},$$

which sends $(e, p, i) \in B$ to $(p_i, p, j) \in A$ where j is the unique element of $[4]$ such that $p_j = e$.

Then we have

$$C_2 \sum_{e \in E} \sum_{p \in \mathcal{P}_e} \sum_{\substack{j \in [4]: \\ p_j \in E_n \setminus E}} |\theta_e|^2 = C_2 \sum_{e \in E} |\theta_e|^2 \cdot |\{(p, j) \in \mathcal{P}_e \times [4] : p_j \in E_n \setminus E\}| \le C_3 \|\theta\|_{\mathbb{R}^E}^2,$$

where the inequality follows by 40. Collecting what we've shown, we notice that the desired bound is proven. □

Next we prove another bound.

Theorem 6.6 (Theorem 14.2 [4]). *There exists a constant C depending only on d such that for any E satsifying $E_n^0 \subseteq E \subseteq E_n$, $\theta \in \mathbb{R}^E$ and $n \ge 1$,*

$$\tau_{E,\theta,n}\left(\{t \in \mathbb{R}^{E_n \setminus E} : |t_e| \le C n^{d+2}(1 + \|\theta\|_{\mathbb{R}^E}) \text{ for all } e \in E_n \setminus E\}\right) \ge \frac{1}{2}.$$

Proof. Fix E, θ, and n. Recall that the covariance matrix of $\tau_{E,\theta,n}$ is $\Sigma := \Sigma_{E,n}^{-1}$ by Lemma 6.1. Let μ denote the mean vector of $\tau_{E,\theta,n}$. We have

$$\mu = -\frac{1}{2}\Sigma v,$$

by Lemma 6.1, where $v := v_{E,\theta,n}$ is the vector in $\mathbb{R}^{E_n \setminus E}$ defining the linear component of $M_{E,\theta,n}$. Then

$$\begin{aligned}
\|\mu\|_{\mathbb{R}^{E_n \setminus E}}^2 &= \|-\frac{1}{2}\Sigma v\|_{\mathbb{R}^{E_n \setminus E}}^2 \\
&= \frac{1}{4}\|v\|_{\mathbb{R}^{E_n \setminus E}}^2 \left\langle \Sigma \frac{v}{\|v\|_{\mathbb{R}^{E_n \setminus E}}}, \Sigma \frac{v}{\|v\|_{\mathbb{R}^{E_n \setminus E}}} \right\rangle_{\mathbb{R}^{E_n \setminus E}} \\
&= \frac{1}{4}\|v\|_{\mathbb{R}^{E_n \setminus E}}^2 \left\langle \frac{v}{\|v\|_{\mathbb{R}^{E_n \setminus E}}}, \Sigma^2 \frac{v}{\|v\|_{\mathbb{R}^{E_n \setminus E}}} \right\rangle_{\mathbb{R}^{E_n \setminus E}} && (\Sigma^T = \Sigma) \\
&\le \frac{1}{4}\|v\|_{\mathbb{R}^{E_n \setminus E}}^2 \lambda_{\max}(\Sigma)^2 && (\text{see } 39) \\
&= \frac{1}{4}\|v\|_{\mathbb{R}^{E_n \setminus E}}^2 \left(\frac{1}{\lambda_{\min}(\Sigma_{E,n})}\right)^2 \\
&\le C\|\theta\|_{\mathbb{R}^E}^2 (n^{d+2})^2. && (\text{Lemma } 6.5 \text{ and Lemma } 6.2)
\end{aligned}$$

Therefore $\|\mu\|_{\mathbb{R}^{E_n \setminus E}} \le C n^{d+2} \|\theta\|_{\mathbb{R}^E}$. In particular,

$$\max_{e \in E_n \setminus E} |\mu_e| \le C n^{d+2} \|\theta\|_{\mathbb{R}^E} \tag{43}$$

because $|\mu_e| = \sqrt{\mu_e^2} \le \|\mu\|_{\mathbb{R}^{E_n \setminus E}}$ for each e. Finally, we have

$$\begin{aligned}
\frac{1}{2} &\le \tau_{E,\theta,n}\left(\left\{t \in \mathbb{R}^{E_n \setminus E} : \max_{e \in E_n \setminus E} |t_e| \le \max_{e \in E_n \setminus E} |\mu_e| + \sqrt{6\lambda_{\max}(\Sigma)\log(|E_n \setminus E|)}\right\}\right) \\
&\le \tau_{E,\theta,n}\left(\left\{t \in \mathbb{R}^{E_n \setminus E} : \max_{e \in E_n \setminus E} |t_e| \le C n^{d+2}\|\theta\|_{\mathbb{R}^E} + \sqrt{6\lambda_{\max}(\Sigma)\log(|E_n \setminus E|)}\right\}\right) \\
&\le \tau_{E,\theta,n}\left(\left\{t \in \mathbb{R}^{E_n \setminus E} : \max_{e \in E_n \setminus E} |t_e| \le C n^{d+2}\|\theta\|_{\mathbb{R}^E} + \sqrt{C_1 n^{d+2}\log(|E_n|)}\right\}\right) \\
&\le \tau_{E,\theta,n}\left(\left\{t \in \mathbb{R}^{E_n \setminus E} : \max_{e \in E_n \setminus E} |t_e| \le C n^{d+2}\|\theta\|_{\mathbb{R}^E} + C n^{d+2}\right\}\right) \\
&= \tau_{E,\theta,n}\left(\{t \in \mathbb{R}^{E_n \setminus E} : |t_e| \le C n^{d+2}(1 + \|\theta\|_{\mathbb{R}^E}) \text{ for all } e \in E_n \setminus E\}\right).
\end{aligned}$$

In the first line, we used Lemma 6.4. In the second line, we made a set larger by slackening the bound requirement on $\max_{e\in E_n\setminus E}|t_e|$, using 43. In the third line, we again made the upper bound less restrictive, because $\log(|E_n|)\geq\log(|E_n\setminus E|)$ and $\lambda_{max}(\Sigma)\leq Cn^{d+2}$, by definition of Σ and by Lemma 6.2. In the fourth line, we slackened the upper bound again, this time using a constant C depending only on d such that

$$\sqrt{C_1 n^{d+2}\log(|E_n|)}\leq Cn^{d+2}. \tag{44}$$

This holds because we have the crude upper bound $|E_n|\leq dn^d$, which follows because each vertex $x\in\Lambda_n$ can be a part of at most d edges (x,y), where $x\prec y$, and there are n^d vertices in Λ_n. Thus, if we were to map $E_n\to\Lambda_n$ by taking a positively oriented edge (x,y) to x, then this map would be at worst d to 1, implying $|E_n|\leq dn^d$. Then we have, crudely,

$$C_1 n^{d+2}\log\left(dn^d\right)\leq Cn^{(d+2)}\cdot n\leq Cn^{(d+2)\cdot 2},$$

and after taking square roots we get back 44. The fifth line is clear, and thus we conclude the proof. □

Next we prove another bound. Recall that $E_n^1=E_n\setminus E_n^0$.

Theorem 6.7 (Theorem 14.3 [4])**.** *There exist positive constants C_1 and C_2, depending only on d, such that the following is true. Take any $n\geq m\geq 2$ such that $m\leq\sqrt{n}$. Then*

$$\tau_n\left(\left\{t\in\mathbb{R}^{E_n^1}:|t_e|\leq C_1 m^{d+2}\text{ for all }e\in E_n^1\right\}\right)\geq\exp\left(-\frac{C_2 n^d\log n}{m}\right).$$

Remark 6.8. In the following proof, it is necessary to break up lattice Maxwell theory in a big box Λ_n into a product of lattice maxwell theories defined in subboxes $B_a\sim\Lambda_m$. By conditioning on the edges intersecting multiple boxes, one can actually do this, otherwise it doesn't work. I thought I would also be able to use Fubuni's theorem, where the order of integration plays the role of conditioning, but I got stuck because I couldn't break up the normalization constant. This issue does not arise when using conditioning, as the A edge variables are integrated out.

Remark 6.9. The following proof fills in the details of the proof given by Chatterjee in [4].

Proof. Let r be the largest integer such that $r(m-1)\leq n-1$. For any $a=(a_1,\cdots,a_d)\in\{0,1,\cdots,r-1\}^d$, let

$$B_a=\{(x_1,\cdots,x_d)\in\Lambda_n:a_j(m-1)\leq x_j\leq(a_j+1)(m-1),j\in[d]\}. \tag{45}$$

Let

$$\mathcal{B}=\{B_a:a\in\{0,1,\cdots,r-1\}^d\}. \tag{46}$$

In other words, each element of $\mathcal{B}$ is a translate of the box Λ_m, and two elements can intersect only at a common boundary.

Claim: There are $O(n^d m^{-1})$ edges in E_n that belong to the boundaries of elements of $\mathcal{B}$.[5]

Proof of Claim. The idea of this proof is to show that there are $O(m^{d-1})$ edges on the bottom faces of a given translate, and then to use that there are $O(n^d/m^d)$ total translates. The reason we count bottom faces is because of double counting. Now we begin the rigorous proof.

Recall 1, namely, E_Λ, the set of positively oriented edges of the subset $\Lambda\subset\mathbb{Z}^d$. Each subbox $B_a\in\mathcal{B}$ is a subset of $\mathbb{Z}^d$. We have the equivalence, in this case,

$$E_{B_a}=\{(x,y)\in E_n\mid x,y\in B_a\}.$$

With this in mind, we define for each $a\in\{0,\cdots,r-1\}^d$

$$\begin{aligned}\partial E_{B_a}=\{(x,y)\in E_{B_a}:x_j=y_j=a_j\cdot(m-1)\text{ or}\\ x_j=y_j=(a_j+1)\cdot(m-1)\text{ for some }j\in[d]\}.\end{aligned} \tag{47}$$

Hence, ∂E_{B_a} are the edges in E_n which belong to the boundary of the translate B_a. We therefore need to upper bound the cardinality of the union

$$\bigcup_{a\in\{0,\cdots,r-1\}^d}\partial E_{B_a}.$$

[5]Warning: usually boundary edges include the E_n^0 edges, but here we mean the geometric boundary of the elements of $\mathcal{B}$. This estimate is false if E_n^0 edges are included, because $|E_n^0|=n^d-1$.

Now we describe a decomposition for each ∂E_{B_a}. Let

$$\begin{aligned} A_a =&\{(x,y) \in E_n : x, y \in B_a \text{ and } y_j = x_j = a_j(m-1) \text{ for some } j \in [d] \text{ and} \\ & x_i \neq (a_i+1)(m+1) \text{ for all } i \in [d]\}. \end{aligned}$$

and

$$\begin{aligned} C_a =&\{(x,y) \in E_n : x, y \in B_a \text{ and } x_i = y_i = (a_i+1)(m-1) \\ & \text{for some } i \in [d]\}. \end{aligned}$$

Loosely speaking, A_a describes the edges appearing on the "bottom" faces of the hypercube B_a, minus the edges which intersect with "top" faces, whereas C_a describes the edges appearing on the "top" faces of B_a. It is clear that

$$\partial E_{B_a} = A_a \cup C_a$$

where the union is a disjoint union. Note moreover that for each $a = (a_1, \cdots, a_d) \in \{0, \cdots, r-1\}^d$,

$$C_a \subset A_{(a_1+1, a_2, \cdots, a_d)} \cup A_{(a_1, a_2+1, \cdots, a_d)} \cup \cdots \cup A_{(a_1, \cdots, a_{d-1}, a_d+1)}.$$

In other words, the top face of B_a is covered by the bottom faces of boxes which are one step away. This is slightly problematic if, say, $a_i + 1 = r$ for some $i \in [d]$, but since the above definitions of $B_a, \partial E_{B_a}, A_a$, and C_a are easily extended to $a \in \{0, \cdots, r\}^d$, then we need not worry. In fact, this extension is useful because we then have

$$\bigcup_{a \in \{0, \cdots, r-1\}^d} \partial E_{B_a} \subset \bigcup_{a \in \{0, \cdots, r\}^d} A_a.$$

In other words, we just need to upper bound the number of edges appearing on the bottom faces of $(r+1)^d$ translates of Λ_m.

It suffices to count the set A_0, because clearly $|A_a| = |A_0|$, where $0 = (0, \cdots, 0)$, for all $a \in \{0, \cdots, r\}^d$.

For each $0 \leq l \leq d$, let Λ_m^l be the set of all $x \in \Lambda_m$ such that l coordinates of x are equal to 0, and the remaining $d - l$ of the coordinates of x take values in the set $\{1, \cdots, m-2\}$.

It is apparent that

$$|\Lambda_m^l| = \binom{d}{l} (m-2)^{d-l}.$$

Take $x \in \Lambda_m^l$. If $l \geq 2$, then there are d possibilities for y such that $(x, y) \in A_0$, namely $y = x + e_i \in A_0$ for all $i \in [d]$. If $l = 1$, then there are $d - 1$ possibilities for y such that $(x, y) \in A_0$, namely $y = x + e_i$ for all $i \in [d] \setminus \{j\}$ where j is the coodinate satisfying $x_j = 0$. If $l = 0$, then there are no possibilities for y for which $(x, y) \in A_0$.

Hence,

$$\begin{aligned} |A_0| &= (d-1) \cdot |\Lambda_m^1| + \sum_{l=2}^{d} d \cdot |\Lambda_m^l| \\ &= (d-1) \cdot d \cdot (m-2)^{d-1} + O(m^{d-1}) \\ &= O(m^{d-1}). \end{aligned}$$

We have that the cardinality of $\bigcup_{a \in \{0, \cdots, r-1\}^d} \partial E_{B_a}$ is bounded by $(r+1)^d \cdot |A_0|$. Also, $r = O(\frac{n}{m})$, so therefore

$$(r+1)^d \cdot |A_0| = O\left(\frac{n^d}{m^d}\right) \cdot O(m^{d-1}) = O(n^d \cdot m^{-1}),$$

as desired. □

Next, observe that there are $O(n^{d-1}m)$ edges in E_n that are not contained in any element of $\mathcal{B}$. Indeed, if (x, y) is such an edge, then there must be some $j \in [d]$ so that $x_j \geq r(m-1)$. There are $O(n^{d-1} \cdot m)$ ways for this to happen because $d-1$ coordinates can be any value in $[n]$, and the remaining coordinate must fall in the range $r(m-1), r(m-1)+1, \cdots, n$, which has at most m elements. And y has $O(1)$ options, given x, which finishes the argument.

Let A be the collection of all elements of $E_n^1 = E_n \setminus E_n^0$ that belong to the boundaries of these boxes, plus all edges in E_n^1 that are not contained in any of the boxes. Then we have

$$|A| \leq \frac{Cn^d}{m} + Cn^{d-1}m \leq \frac{Cn^d}{m}, \tag{48}$$

where the first inequality follows from the above claim (which also counted E_n^0 edges contained on the geometric boundary of translates) and the fact that there are $O(n^{d-1}m)$ edges in E_n that are not contained in any element of $\mathcal{B}$. The second inequality in 48 follows from the assumption that $m^2 \leq n$. Also, we have

$$|\mathcal{B}| \leq r^d = O\left(\frac{n^d}{m^d}\right) = \frac{Cn^d}{m^d}.$$

To finish the proof, we use the language of random variables and conditional distributions. We take the canonical underlying probability space $(\Omega, \mathcal{F}, \mathbb{P}) = (\mathbb{R}^{E_n^1}, \mathcal{B}(\mathbb{R}^{E_n^1}), \tau_n)$. Let $T = (T_e)_{e \in E_n^1} : \Omega \to \mathbb{R}^{E_n^1}$ be τ_n distributed. Consider also the random vectors $T_A = (T_e)_{e \in A} : \Omega \to \mathbb{R}^{E_n^1}$ and $T_{E_n^1 \setminus A} = (T_e)_{e \in E_n^1 \setminus A} : \Omega \to \mathbb{R}^{E_n^1}$. Note that the joint density of T_A and $T_{E_n^1 \setminus A}$ is just the density of T, which is the density of the measure τ_n with respect to Lebesgue measure on $\mathbb{R}^{E_n^1}$. Let

$$\kappa(\,\cdot\,,\,\cdot\,) : \Omega \times \mathcal{B}(\mathbb{R}^{E_n^1 \setminus A}) \to [0,1]$$

be the regular conditional distribution of $T_{E_n^1 \setminus A}$ given T_A which means by definition that

$$\kappa(t, S) = \mathbb{P}[\{T_{E_n^1 \setminus A} \in S\} \mid T_A](t)$$

for all $t \in \Omega$ and all $S \in \mathcal{B}(\mathbb{R}^{E_n^1 \setminus A})$. See, for instance, [9, Definition 8.28].

The conditional distribution κ has the conditional density function $f_{T_{E_n^1 \setminus A} | T_A}(\,\cdot\,,\,\cdot\,) : \mathbb{R}^A \times \mathbb{R}^{E_n^1 \setminus A} \to \mathbb{R}$ given by[6]

$$f_{T_{E_n^1 \setminus A} | T_A}(t_A, t_{E_n^1 \setminus A}) = \frac{\frac{1}{Z_{M,\Lambda_n}} e^{-(1/2) M_{E_n^0, 0, n}(t)}}{\int_{\mathbb{R}^{E_n^1 \setminus A}} \frac{1}{Z_{M,\Lambda_n}} e^{-(1/2) M_{E_n^0, 0, n}(t)} \prod_{e \in E_n^1 \setminus A} dl(t_e)}, \tag{49}$$

for $t_A = (t_e)_{e \in A} \in \mathbb{R}^A$ and $t_{E_n^1 \setminus A} = (t_e)_{e \in E_n^1 \setminus A} \in \mathbb{R}^{E_n^1 \setminus A}$, and where the symbol t in the argument of $M_{E_n^0, 0, n}(\,\cdot\,)$ denotes the concatenation of t_A and $t_{E_n^1 \setminus A}$, namely

$$t_e = \begin{cases} (t_A)_e & \text{if } e \in A \\ (t_{E_n^1 \setminus A})_e & \text{if } e \in E_n^1 \setminus A \end{cases}.$$

The normalization constants, appearing in both numerator and denominator in 49, cancel out, and we get

$$f_{T_{E_n^1 \setminus A} | T_A}(t_A, t_{E_n^1 \setminus A}) = \frac{e^{-(1/2) \sum_{p \in \mathcal{P}(\Lambda_n)} (t_p^0)^2}}{\int_{\mathbb{R}^{E_n^1 \setminus A}} e^{-(1/2) \sum_{p \in \mathcal{P}(\Lambda_n)} (t_p^0)^2} \prod_{e \in E_n^1 \setminus A} dl(t_e)},$$

where we recall that $t_p^0 = t_{p_1}^0 + t_{p_2}^0 - t_{p_3}^0 - t_{p_4}^0$, where t^0 denotes the extension of $t \in \mathbb{R}^{E_n^1}$ to E_n by $0 \in \mathbb{R}^{E_n^0}$. To make further simplifications, we introduce (and recall) further distinguishing notation. Recall that

$$E_{B_a} = \{(x, y) \in E_n \mid x, y \in B_a\}.$$

and that ∂E_{B_a}, defined in equation 47, is the set of boundary edges of the translate B_a. Next, let

$$E_{B_a}^{\circ} = E_{B_a} \setminus \partial E_{B_a}$$

be the edges which do not lie on the boundary of the box B_a, though one vertex is allowed to lie on the boundary. Furthermore, let

$$\mathcal{P}_\partial = \{p \in \mathcal{P}(\Lambda_n) : p_i \in A \text{ or } p_i \in \partial E_{B_a} \text{ for some } B_a \in \mathcal{B}, \text{ for all } i \in [4]\} \tag{50}$$

[6] This is just the formula $f_{Y|X}(x, y) = \frac{f(x,y)}{\int f(x,y)dy}$ where f would denote the joint density of T_A and $T_{E_n^1 \setminus A}$. See [9, Example 8.31]

be the plaquettes which do not consist of any interior edges. Note that it is possible for $p_i \in \partial E_{B_a}$ but $p_i \notin A$ because A is a subset of E_n^1, and there could be axial edges on the boundary of a translate. Let

$$\mathcal{P}(\mathring{B}_a) = \{p \in \mathcal{P}(\Lambda_n) : p_j \in \mathring{E}_{B_a} \text{ for some } j \in [4]\} \tag{51}$$

be the plaquettes which contain at least one edge on the interior of one of the translates. Observe that

$$(\mathcal{P}_\partial)^c = \bigcup_{a \in \{0,\cdots,r-1\}^d} \mathcal{P}(\mathring{B}_a)$$

where the union on the right hand side is a disjoint union. We have

$$\begin{aligned} M_{E_n^0,0,n}(t) &= \sum_{p \in \mathcal{P}(\Lambda_n)} (t_p^0)^2 \\ &= \sum_{p \in \mathcal{P}_\partial} (t_p^0)^2 + \sum_{B_a \in \mathcal{B}} \sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2 \end{aligned}$$

for any $t \in \mathbb{R}^{E_n^1}$. Then we see that the conditional density satisfies

$$f_{T_{E_n^1 \setminus A}|T_A}(t_A, t_{E_n^1 \setminus A}) = \frac{e^{-(1/2)\sum_{B_a \in \mathcal{B}} \sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2}}{\int_{\mathbb{R}^{E_n^1 \setminus A}} e^{-(1/2)\sum_{B_a \in \mathcal{B}} \sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2} \prod_{e \in E_n^1 \setminus A} dl(t_e).} \tag{52}$$

because the term $e^{-(1/2)\sum_{p \in \mathcal{P}_\partial} (t_p^0)^2}$ could be treated as a constant with respect to the integral in the denominator, and thus cancelled with the same factor in the numerator. We see that the integral in the denominator of 52 factors into the product

$$\prod_{B_a \in \mathcal{B}} \int_{\mathbb{R}^{\mathring{E}_{B_a}}} e^{-(1/2)\sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2} \prod_{e \in \mathring{E}_{B_a}} dl(t_e).$$

Therefore

$$f_{T_{E_n^1 \setminus A}|T_A}(t_A, t_{E_n^1 \setminus A}) = \prod_{B_a \in \mathcal{B}} \frac{e^{-(1/2)\sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2}}{\int_{\mathbb{R}^{\mathring{E}_{B_a}}} e^{-(1/2)\sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2} \prod_{e \in \mathring{E}_{B_a}} dl(t_e)}. \tag{53}$$

Fix $t_A \in \mathbb{R}^A$ and otherwise suppose $t_{E_n^1/A} \in \mathbb{R}^{E_n^1 \setminus A}$ is allowed to vary, and use that $t \in \mathbb{R}^{E_n^1}$ is the concatenation of t_A and $t_{E_n^1 \setminus A}$. Then recall the definition of the measure $\tau_{E,\theta,m}$ on $\mathbb{R}^{E_m \setminus E}$ and the 2nd degree polynomial $M_{E,\theta,m}$. Adjusting the definition of $\tau_{E,\theta,m}$ and $M_{E,\theta,m}$ to allow for translated boxes, we see that each factor in the product 53 defines the density function of some $\tau_{E_a,\theta_a,m}$ measure, where $E_a = (E_n^0 \cap E_{B_a}) \cup (E_{B_a} \cap A)$ and $\theta_a \in \mathbb{R}^{E_a}$ satisfies

$$(\theta_a)_e = \begin{cases} t_e & \text{if } e \in E_{B_a} \cap A \\ 0 & \text{if } e \in E_n^0 \cap E_{B_a}. \end{cases} \tag{54}$$

In fact, for a given subbox B_a,

$$\sum_{p \in \mathcal{P}(\mathring{B}_a)} (t_p^0)^2 = \sum_{p \in \mathcal{P}(\mathring{B}_a)} \left(\left(t|_{\mathring{E}_{B_a} \cap E_n^1} \right)_p^{\theta_a} \right)^2$$

differs from $M_{E_a,\theta_a,m}$ only by a constant component, since the plaquettes $p \in \mathcal{P}_\partial$ have been removed. But this difference is immaterial because adding the missing plaquettes to both the numerator and denominator of each factor in the product 53 is equivalent to multiplying by one and because Gaussian measures are determined by their covariance and mean, and the constant component has no affect on either of these. The main observation, though, is that θ_a depends on t_A, and that, loosely speaking, t_A (and the axial gauge condition on E_n^0) determines the boundary values of each subbox.

Next we have

$$\begin{aligned} &\tau_n(\{t \in \mathbb{R}^{E_n^1} : |t_e| \le Cm^{d+2} \text{ for all } e \in E_n^1\}) \ge \\ &\mathbb{P}\left[|T_e| \le n^{-d/2} \, \forall e \in A \text{ and } |T_e| \le Cm^{d+2}(1 + \|T_A\|_{\mathbb{R}^A}) \, \forall e \in E_n^1 \setminus A \right] \end{aligned} \tag{55}$$

because if $|T_e| \le n^{-d/2}$ for each $e \in A$, then

$$\begin{aligned}\|T_A\|_{\mathbb{R}^A}^2 &= \sum_{e\in A} T_e^2 \le \sum_{e\in A}\left(\frac{1}{n^{d/2}}\right)^2 \\ &= \sum_{e\in A}\frac{1}{n^d} = |A|\frac{1}{n^d} \\ &\le \frac{Cn^d}{mn^d} = \frac{C}{m} \le C,\end{aligned}$$

by virtue of $m \ge 2$ and 48.

We denote the product intervals

$$S_1 = [-n^{-d/2}, n^{-d/2}]^A \subset \mathbb{R}^A$$

and

$$S_2(T_A) = [-Cm^{d+2}(1+\|T_A\|_{\mathbb{R}^A}), Cm^{d+2}(1+\|T_A\|_{\mathbb{R}^A})]^{E_n^1\setminus A} \subset \mathbb{R}^{E_n^1\setminus A},$$

noting that S_2 depends on T_A, and can thus be viewed as a random subset.

Let

$$S_3 = \{s \in \mathbb{R}^{E_n^1} : |s_e| \le Cm^{d+2}(1+\|s_A\|_{\mathbb{R}^A})\,\forall e \in E_n^1 \setminus A\}.$$

Observe that

$$1_{S_3}(T) = 1_{S_2(T_A)}(T_{E_n^1\setminus A})$$

on $\Omega = \mathbb{R}^{E_n^1}$.

Then we have

$$\begin{aligned}&\mathbb{P}\left[|T_e| \le n^{-d/2}\,\forall e \in A \text{ and } |T_e| \le Cm^{d+2}(1+\|T_A\|_{\mathbb{R}^A})\,\forall e \in E_n^1\setminus A\right] \\ &\qquad = \mathbb{E}\left[1_{S_1}(T_A)1_{S_2(T_A)}(T_{E_n^1\setminus A})\right] \\ &\qquad = \mathbb{E}\left[1_{S_1}(T_A)1_{S_3}(T)\right].\end{aligned} \tag{56}$$

Since $\{T_A \in S_1\} \in \sigma(T_A)$, then by definition of conditional expectation, we have

$$\mathbb{E}\left[1_{S_1}(T_A)1_{S_3}(T)\right] = \mathbb{E}\left[1_{S_1}(T_A)\mathbb{E}\left[1_{S_3}(T) \mid T_A\right]\right].$$

Note that for any product measurable $g : \Omega \times \mathbb{R}^{E_n^1\setminus A} \to \mathbb{R}$, we have that

$$\begin{aligned}\Omega \ni t \mapsto &\int_{\mathbb{R}^{E_n^1\setminus A}} g(T_A(t), s)\kappa(t, ds) \\ &= \int_{\mathbb{R}^{E_n^1\setminus A}} g(T_A(t), s)\, f_{T_{E_n^1\setminus A}|T_A}(T_A(t), s)\, ds \in \mathbb{R}\end{aligned}$$

is a version of the conditional expectation of $g(T)$ given T_A. [7] Therefore, with $g = 1_{S_3}$, we have

$$\mathbb{E}\left[1_{S_1}(T_A)\mathbb{E}\left[1_{S_3}(T) \mid T_A\right]\right] = \mathbb{E}\left[1_{S_1}(T_A)\int_{\mathbb{R}^{E_n^1\setminus A}} 1_{S_3}(T_A(\cdot), s)\kappa(\cdot, ds)\right].$$

[7] I previously though it was sufficient to write

$$\begin{aligned}\mathbb{E}[1_{S_1}(T_A)1_{S_2(T_A)}(T_{E_n^1\setminus A})] &= \mathbb{E}[1_{S_1}(T_A)\mathbb{E}[1_{S_2(T_A)}(T_{E_n^1\setminus A}) \mid T_A]] \\ &= \mathbb{E}[1_{S_1}(T_A)\int_{\mathbb{R}^{E_n^1\setminus A}} 1_{S_2(T_A)}(s)\kappa(\cdot, ds)]\end{aligned}$$

by using in the second equality [9, Theorem 8.38] which says $\mathbb{E}[f(X) \mid \mathcal{F}](\omega) = \int f(x)\kappa(\omega, dx)$ for almost all ω, where X takes values in a Borel space E and f is a map on E.

But then I realized that $1_{S_2(T_A)}(\cdot)$ is not a map from $\mathbb{R}^{E_n^1\setminus A}$, but rather a product map from $\mathbb{R}^{E_n^1} \times \mathbb{R}^{E_n^1\setminus A}$, because $S_2(T_A)$ is a random set. Therefore, I couldn't let $f = 1_{S_2(T_A)}(\cdot)$ and apply [9, Theorem 8.38]. This is when I asked Gemini if [9, Theorem 8.38] holds for more general f. It confirmed that it does hold and gave an argument and pointed me to [9, Section 14]. The theorem which corresponds to the result for general f is Fubini's theorem for transition kernels, [9, Theorem 14.29].

The definition of conditional expectation dictates that the footnoted fact means that

$$\mathbb{E}[1_B(\cdot)\int_{\mathbb{R}^{E_n^1\setminus A}} g(T_A(\cdot), s)\kappa(\cdot, ds)] = \mathbb{E}[1_B(\cdot)g(T(\cdot))]$$

for all Borel sets $B \in \sigma(T_A)$. Fubini's theorem for transition kernels yields this.

Furthermore, for any $t \in \Omega$, we have

$$\int_{\mathbb{R}^{E_n^1 \setminus A}} 1_{S_3}(T_A(t), s)\kappa(t, ds) = \int_{S_2(T_A(t))} \kappa(t, ds),$$

because $1_{S_3}(T_A(t), s) = 1_{S_2(T_A(t))}(s)$ for all $s \in \mathbb{R}^{E_n^1 \setminus A}$. Therefore,

$$\mathbb{E}\left[1_{S_1}(T_A) \int_{\mathbb{R}^{E_n^1 \setminus A}} 1_{S_3}(T_A(\cdot), s)\kappa(\cdot, ds)\right] = \mathbb{E}\left[1_{S_1}(T_A) \int_{S_2(T_A(\cdot))} f_{T_{E_n^1 \setminus A}|T_A}(T_A(\cdot), s) \prod_{e \in \mathbb{E}_n^1 \setminus A} dl(s_e)\right].$$

Recalling that $\mathbb{P} = \tau_n$, we have

$$\begin{aligned}
&\mathbb{E}\left[1_{S_1}(T_A) \int_{S_2(T_A(\cdot))} f_{T_{E_n^1 \setminus A}|T_A}(T_A(\cdot), s) \prod_{e \in \mathbb{E}_n^1 \setminus A} dl(s_e)\right] \\
&= \int_{\mathbb{R}^{E_n^1}} d\tau_n(t)\, 1_{S_1}(T_A(t)) \int_{S_2(T_A(t))} \prod_{e \in E_n^1 \setminus A} dl(s_e)\, f_{T_{E_n^1 \setminus A}|T_A}(T_A(t), s).
\end{aligned}$$

Note that for any $t \in \Omega$,

$$S_2(T_A(t)) = \prod_{B_a \in \mathcal{B}} \{s \in \mathbb{R}^{\mathring{E}_{B_a}} : |s_e| \le Cm^{d+2}(1 + \|T_A(t)\|_{\mathbb{R}^A})\, \forall e \in \mathring{E}_{B_a}\},$$

and we will denote $S_{B_a}(T_A(t)) = \{s \in \mathbb{R}^{\mathring{E}_{B_a}} : |s_e| \le Cm^{d+2}(1 + \|T_A(t)\|_{\mathbb{R}^A})\, \forall e \in \mathring{E}_{B_a}\}$. Recalling 53, we have for each $t \in \Omega$,

$$\begin{aligned}
&\int_{S_2(T_A(t))} \prod_{e \in E_n^1 \setminus A} dl(s_e)\, f_{T_{E_n^1 \setminus A}|T_A}(T_A(t), s) \\
&= \prod_{B_a \in \mathcal{B}} \int_{S_{B_a}(T_A(t))} \prod_{e \in \mathring{E}_{B_a}} dl(s_e) \frac{e^{-(1/2)M_{E_a,\theta_a,m}(s)}}{\int_{\mathbb{R}^{\mathring{E}_{B_a}}} \prod_{e \in \mathring{E}_{B_a}} dl(y_e) M_{E_a,\theta_a,m}(y)} \\
&= \prod_{B_a \in \mathcal{B}} \tau_{E_a,\theta_a,m}\left(\{s \in \mathbb{R}^{E_{B_a} \setminus E_a} : |s_e| \le Cm^{d+2}(1 + \|T_A(t)\|_{\mathbb{R}^A})\, \forall e \in E_{B_a} \setminus E_a\}\right) \\
&\ge (1/2)^{|\mathcal{B}|},
\end{aligned}$$

where the last inequality is an application of Theorem 6.6.

Recalling the steps, e.g. 55 and 56, we see that

$$\tau_n(\{t \in \mathbb{R}^{E_n^1} : |t_e| \le Cm^{d+2} \text{ for all } e \in E_n^1\}) \ge (1/2)^{|\mathcal{B}|} \int_{\mathbb{R}^{E_n^1}} d\tau_n(t)\, 1_{S_1}(T_A(t)).$$

Note that

$$\int_{\mathbb{R}^{E_n^1}} d\tau_n(t)\, 1_{S_1}(T_A(t)) = \tau_n(t \in \mathbb{R}^{E_n^1} : |t_e| \le n^{-d/2} \text{ for all } e \in A).$$

By Theorem 6.3, we have

$$\begin{aligned}
\tau_n(t \in \mathbb{R}^{E_n^1} : |t_e| \le n^{-d/2} \text{ for all } e \in A) &\ge e^{-C|A|(\log\left(1/(1/\sqrt{n}^d)\right) + \log n)} \\
&= e^{-C|A|(\log\left(\sqrt{n}^d\right) + \log n)} \\
&= e^{-C|A|((d/2)\log n + \log n)} \\
&= e^{-C|A|\log n}.
\end{aligned}$$

Note that

$$\begin{aligned}
(1/2)^{|\mathcal{B}|} e^{-C|A|\log n} &= e^{-O(n^d/m^d)} \times e^{-O(n^d m^{-1} \log n)} \\
&= e^{-O(n^d m^{-1} \log n + n^d/m^d)} \\
&= e^{-O(n^d m^{-1} \log n)} \\
&= \exp\left(-\frac{Cn^d \log n}{m}\right).
\end{aligned}$$

Recalling the steps, we have shown that

$$\tau_n(\{t \in \mathbb{R}^{E_n^1} : |t_e| \leq Cm^{d+2} \text{ for all } e \in E_n^1\}) \geq \exp\left(-\frac{Cn^d \log n}{m}\right),$$

where the constants C are not meant to be the same on each side of the relation. This concludes the proof. □

6.4 Infinite Volume Limit of Lattice Maxwell Theory

The partition function of lattice Maxwell theory in the box Λ_n (with boundary value $0 \in \mathbb{R}^{E_n^0}$) is

$$\begin{aligned} Z_{M,\Lambda_n} &= \int_{\mathbb{R}^{E_n^1}} e^{-(1/2)M_{E_n^0,0,n}(t)} \prod_{e\in E_n^1} dl(t_e) \\ &= \int_{\mathbb{R}^{E_n}} e^{-(1/2)M_n^{(\mathbb{R})}(t)} \prod_{e\in E_n^1} dl(t_e) \prod_{e\in E_n^0} d\delta_0(t_e), \end{aligned} \tag{57}$$

where $\delta_0 : \mathcal{B}(\mathbb{R}) \to [0,1]$ denotes the Dirac delta measure centered at $0 \in \mathbb{R}$. Furthermore, the free energy per site of lattice Maxwell theory in the box Λ_n is

$$F_{M,\Lambda_n} = \frac{\log Z_{M,\Lambda_n}}{n^d}.$$

In this section, we discuss the limit of F_{M,Λ_n} as $n \to \infty$, repeating some of the discussion from subsection 1.2.

In addition to working directly with an integral as in 57, we also have a formula using a related covariance matrix. Note that Z_{M,Λ_n} is the normalization constant of the Gaussian measure τ_n. Since the linear and constant components of $M_{E_n^0,0,n}$ vanish, the proof of Lemma 6.1 shows that the normalization constant of τ_n is already in the standard format. Therefore

$$Z_{M,\Lambda_n} = (2\pi)^{|E_n^1|/2} (\det \Sigma_n^0)^{-1/2}, \tag{58}$$

which uses that the covariance matrix of τ_n is $(\Sigma_n^0)^{-1}$. It is interesting that Z_{M,Λ_n} therefore depends only on the eigenvalues of the matrix Σ_n^0, and on the degrees of freedom, $|E_n^1|$.

Recall the definition

$$K_{n,d} = -\frac{\log\det\left(\Sigma_n^0\right)}{2n^d},$$

which, for each $n \geq 1$, is a finite real number because Σ_n^0 is a positive definite matrix, by Lemma 6.2. Taking the logarithm of 58 and dividing by n^d, we have

$$F_{M,\Lambda_n} = K_{n,d} + \frac{|E_n^1|}{2n^d} \log(2\pi).$$

By Lemma 8.2, we have

$$\lim_{n\to\infty} \frac{|E_n^1|}{2n^d} \log(2\pi) = \frac{d-1}{2} \log(2\pi).$$

Therefore, an explicit $n \to \infty$ limit of F_{M,Λ_n} entails finding an explicit formula for $K_{n,d}$ as $n \to \infty$. Instead of finding an explicit candidate for either $K_{n,d}$ or F_{M,Λ_n} as $n \to \infty$, the approach taken in [4] is to show that $(F_{M,\Lambda_n})_{n=1}^{\infty}$ is a Cauchy sequence. We unfortunately skip a proof this rather intricate result. The proof uses a lot of the estimates from the previous section, and plays around with controlling different subsets of edges. But let us simply quote the result:

Theorem 6.10 (Theorem 15.2 [4]). *As $n \to \infty$, F_{M,Λ_n} converges to a finite limit.*

On the other hand, an explicit limit for $K_{n,d}$ is obtained in [1], which yields the following[8]:

$$\begin{aligned} \lim_{n\to\infty} F_{M,\Lambda_n} =& \frac{d-1}{2} \log(2\pi) - \frac{d-1}{2} \log(2) \\ &- \frac{1}{2} \int_0^1 dx \log(1-\cos(2\pi x)) \\ &- \frac{d-2}{2} \int_{[0,1]^d} dx_1 \cdots dx_d \log \sum_{k=1}^{d} (1-\cos(2\pi x_k)) \\ =& \frac{d-1}{2} \log(\pi) + \frac{\log(2)}{2} \\ &- \frac{d-2}{2} \int_{[0,1]^d} dx_1 \cdots dx_d \log \sum_{k=1}^{d} (1-\cos(2\pi x_k)). \end{aligned}$$

[8] We used the integral-calculator.com to find the integral $\int_0^1 dx \log(1-\cos(2\pi x))$.

7 Wilson Action to Maxwell Action

7.1 Linear Algebraic Setting

Lemmas 7.2 and 7.3 below require some care with matrix spaces, the field over which a vector space is taken, associated inner products, and so forth. Therefore we remind ourselves of these notions. This section can be skipped and referred back to while reading the aforementioned lemmas.

We will consider the matrix space $\mathbb{C}^{N\times N}$ as both a vector space over $\mathbb{R}$ and a vector space over $\mathbb{C}$. Therefore, we denote $\mathbb{C}^{N\times N}_{\mathbb{R}}$ for the case when it is a real vector space, and $\mathbb{C}^{N\times N}_{\mathbb{C}}$ when it is a complex vector space.

We use the physicist's convention that a complex inner product is anti-linear in the first slot and linear in the second slot. Note that complex inner products are not taken for real vector spaces, even if the underlying elements of the real vector space are of a complex nature, e.g. $\mathbb{C}^{N\times N}_{\mathbb{R}}$.

Note also that a norm taken over a complex vector space differs from a norm taken over a real vector space only because, for homogeneity, the complex modulus is taken instead of the absolute value. Therefore, the Hilbert Schmidt norm taken over $\mathbb{C}^{N\times N}_{\mathbb{R}}$ is slightly distinct from the Hilbert Schmidt norm taken over $\mathbb{C}^{N\times N}_{\mathbb{C}}$.

Notation	Vector Space	Inner product	Induced Norm
$\mathbb{C}^{N\times N}_{\mathbb{R}}$	$(\mathbb{C}^{N\times N},\mathbb{R})$	$(A,B)\mapsto \operatorname{Re}\operatorname{Tr}(A^*B)$	$\|\cdot\|_{HS}$
$\mathbb{C}^{N\times N}_{\mathbb{C}}$	$(\mathbb{C}^{N\times N},\mathbb{C})$	$(A,B)\mapsto \operatorname{Tr}(A^*B)$	$\|\cdot\|_{HS}$
$\mathrm{H}(N)$	$(\mathrm{H}(N),\mathbb{R})$	$(A,B)\mapsto \operatorname{Re}\operatorname{Tr}(A^*B)$	$\|\cdot\|_{HS}$

Table 1: Recall that the Hilbert Schmidt norm $\|\cdot\|_{HS}$ is given by $A\mapsto\sqrt{\operatorname{Tr}(A^*A)}$. Whereas $\operatorname{Tr}(A^*B)$ might be complex, $\operatorname{Tr}(A^*A)$ is always a real number, which helps to explain why taking a real part for the inner product does not corrupt the induced norm. Furthermore, $(A,B)\mapsto\operatorname{Tr}(A^*B)$ fails to define a real inner product because $\operatorname{Tr}(A^*B)=\operatorname{Tr}(B^*A)$ is not an identity, whereas the conjugate symmetric version is.

7.2 Quadratic Approximation

We quote the following quadratic approximation result for the exponential map.

Lemma 7.1 (Lemma 16.1 [4])**.** *If H is a Hermitian matrix, then*

$$\left\| e^{\mathrm{i}H} - I - \mathrm{i}H + \frac{H^2}{2} \right\| \leq \frac{\|H\|^3}{6}.$$

The following lemma seems to have been used implicitly in [4].

Lemma 7.2. *Let $X_1,\cdots,X_k$ be $k\geq 2$ matrices in $\mathbb{C}^{N\times N}$. Then*

$$|\operatorname{Re}\operatorname{Tr} X_1\cdots X_k| \leq \|X_1\|\cdots\|X_k\|,$$

where $\|\cdot\|$ denotes the Hilbert Schmidt norm.

Note that this result does not hold when $k=1$. For a counterexample, consider the matrix $X_1=\begin{bmatrix}0.5 & 0\\ 0 & 0.5\end{bmatrix}\in\mathbb{C}^{2\times 2}$. Then $\operatorname{Re}\operatorname{Tr} X_1 = 1$. On the other hand, $\|X_1\|=\sqrt{0.5}$.

Proof. Fix $k\geq 2$. Let $X_1,\cdots,X_k$ be given. Let $X=X_1\cdots X_{k-1}$. By the Cauchy Schwarz inequality for the real Hilbert space $\mathbb{C}^{N\times N}_{\mathbb{R}}$, we have

$$|\operatorname{Re}\operatorname{Tr} XX_k| \leq \|X\|\|X_k\|.$$

Therefore

$$\begin{aligned}|\operatorname{Re}(\operatorname{Tr}\prod_{j=1}^{k} X_j)| &= |\operatorname{Re}(\operatorname{Tr} X X_k)| \\ &\le \|X\|\|X_k\| \\ &= \|\prod_{j=1}^{k-1} X_j\|\|X_k\| \\ &\le \left(\prod_{j=1}^{k-1}\|X_j\|\right)\|X_k\|,\end{aligned}$$

where the last step follows by an induction argument applied to the inequality $\|XY\| \le \|X\|\|Y\|$, which we proved in Lemma 9.4. □

Lemma 7.3 (Lemma 16.2 [4]). *Let $H \in \mathrm{H}(N)^{E_n}$. Suppose $U = e^{\mathrm{i}H}$, that is, $U_e = e^{\mathrm{i}H_e}$ for all $e \in E_n$. Furthermore, suppose that $\|H_e\| \le \eta \le 1$ for all $e \in E_n$. Then there exists a constant C depending only on N such that for any plaquette $p \in \mathcal{P}(\Lambda_n)$, we have*

$$|\operatorname{Re}(\operatorname{Tr}(I - U_p)) - \frac{1}{2}\|H_p\|^2| \le C\eta^3.$$

Proof. Let $\eta \in (0, 1]$ be given, and let $H \in b_{\mathrm{H}(N)^{E_n}}(0, \eta)$, and let $U = e^{\mathrm{i}H}$. The set up is as follows. For each $e \in E_n$, define R_e as the quadractic truncation of the exponential expansion $e^{\mathrm{i}H_e} = U_e$, namely

$$R_e = I + \mathrm{i}H_e - H_e^2/2.$$

Furthermore, define the plaquette variable $R_p = R_{p_1}R_{p_2}R_{p_3}^*R_{p_4}^*$, for any $p \in \mathcal{P}(\Lambda_n)$.[9] We have two claims. First,

$$|\operatorname{Re}(\operatorname{Tr}(I - U_p)) - \operatorname{Re}(\operatorname{Tr}(I - R_p))| \le C\eta^3.$$

Second,

$$|\operatorname{Re}(\operatorname{Tr}(I - R_p)) - \frac{1}{2}\|H_p\|^2| \le C\eta^3.$$

Once we establish these two claims, the conclusion of the lemma is easily obtained.

Proof of the first claim. [10] By additivity and homogeneity $\operatorname{Re}(\cdot)$ and $\operatorname{Tr}(\cdot)$, we have

$$\begin{aligned}\operatorname{Re}(\operatorname{Tr}(I - U_p)) - \operatorname{Re}(\operatorname{Tr}(I - R_p)) &= \operatorname{Re}(\operatorname{Tr}(R_p)) - \operatorname{Re}(\operatorname{Tr}(U_p)) \\ &= \operatorname{Re}(\operatorname{Tr}(R_p - U_p)).\end{aligned}$$

By telescoping, we have

$$\begin{aligned}R_p - U_p = &(R_{p_1} - U_{p_1})R_{p_2}R_{p_3}^*R_{p_4}^* \\ &+ U_{p_1}(R_{p_2} - U_{p_2})R_{p_3}^*R_{p_4}^* \\ &+ U_{p_1}U_{p_2}(R_{p_3}^* - U_{p_3}^*)R_{p_4}^* \\ &+ U_{p_1}U_{p_2}U_{p_3}^*(R_{p_4}^* - U_{p_4}^*).\end{aligned}$$

By additivity of $\operatorname{Re}(\cdot)$, we have

$$\begin{aligned}\operatorname{Re}(\operatorname{Tr}(R_p - U_p)) = &\operatorname{Re}\operatorname{Tr}\big((R_{p_1} - U_{p_1})R_{p_2}R_{p_3}^*R_{p_4}^*\big) \\ &+ \operatorname{Re}\operatorname{Tr}\big(U_{p_1}(R_{p_2} - U_{p_2})R_{p_3}^*R_{p_4}^*\big) \\ &+ \operatorname{Re}\operatorname{Tr}\big(U_{p_1}U_{p_2}(R_{p_3}^* - U_{p_3}^*)R_{p_4}^*\big) \\ &+ \operatorname{Re}\operatorname{Tr}\big(U_{p_1}U_{p_2}U_{p_3}^*(R_{p_4} - U_{p_4}^*)\big).\end{aligned}$$

[9] Since R_{p_3} and R_{p_4} are no longer unitary, we do not have $R_p = R_{p_1}R_{p_2}R_{p_3}^{-1}R_{p_4}^{-1}$, like we have for U_p.

[10] The first claim in the following proof differs from the proof in [4] because I think the identity

$$|\operatorname{Re}\operatorname{Tr}(I - U(x,j,k)) - \operatorname{Re}\operatorname{Tr}(I - R(x,j,k))| = \frac{1}{2}|\|U(x,j,k)\|^2 - \|I - R(x,j,k)\|^2|$$

appearing on page 39, line -10, in [4] is not true since $R(x,j,k)$ is not necessarily unitary. Therefore, we present a workaround, which is contained in the first claim.

By Lemma 7.2, the triangle inequality on $\mathbb{R}$, and what we have obtained so far, we have

$$\begin{aligned} |\operatorname{Re}\operatorname{Tr}(I-U_p)-\operatorname{Re}\operatorname{Tr}(I-R_p)| &= |\operatorname{Re}(\operatorname{Tr}(R_p-U_p))| \\ &\le \|R_{p_1}-U_{p_1}\|\|R_{p_2}\|\|R^*_{p_3}\|\|R^*_{p_4}\| \\ &+ \|U_{p_1}\|\|R_{p_2}-U_{p_2}\|\|R^*_{p_3}\|\|R^*_{p_4}\| \\ &+ \|U_{p_1}\|\|U_{p_2}\|\|R^*_{p_3}-U^*_{p_3}\|\|R^*_{p_4}\| \\ &+ \|U_{p_1}\|\|U_{p_2}\|\|U^*_{p_3}\|\|R^*_{p_4}-U^*_{p_4}\|. \end{aligned} \tag{59}$$

By Lemma 7.1 applied to the Hermitian matrix H_{p_1}, we obtain

$$\|R_{p_1}-U_{p_1}\| = \|I+\mathrm{i}H_{p_1}-H^2_{p_1}/2-e^{\mathrm{i}H_{p_1}}\| \le \frac{\|H_{p_1}\|^3}{6} \le C\eta^3.$$

By the same logic, we also have

$$\|R_{p_2}-U_{p_2}\| \le C\eta^3.$$

The argument for p_3 and p_4 is slightly different. We have

$$R^*_{p_3} = I-\mathrm{i}H_{p_3}-H^2_{p_3}/2 = I+\mathrm{i}(-H_{p_3})-(-H_{p_3})^2/2,$$

which uses that $H_{p_3}=H^*_{p_3}$. We also have

$$U^*_{p_3} = (e^{\mathrm{i}H_{p_3}})^* = e^{\mathrm{i}(-H_{p_3})}.$$

Therefore, we apply Lemma 7.1 to the Hermitian matrix $-H_{p_3}$ to obtain

$$\|R^*_{p_3}-U^*_{p_3}\| = \|I+\mathrm{i}(-H_{p_3})-(-H_{p_3})^2/2-e^{\mathrm{i}(-H_{p_3})}\| \le \frac{\|-H_{p_3}\|^3}{6} \le C\eta^3.$$

Similarly,

$$\|R^*_{p_4}-U^*_{p_4}\| \le C\eta^3.$$

The other terms in 59 are dealt with in the following way. For any unitary matrix $V\in \mathrm{U}(N)$, we have $\|V\|=\sqrt{N}\le C$.[11] Also, $\|V^*\|=\|V\|$ for unitary matrices V. Hence $\|U_{p_1}\|\le C$, $\|U_{p_2}\|\le C$, and $\|U^*_{p_3}\|\le C$. Furthermore,

$$\begin{aligned} \|R_{p_2}\| &= \|I+\mathrm{i}H_{p_2}-H^2_{p_2}/2\| \\ &\le \|I\|+\|H_{p_2}\|+\|H_{p_2}\|\|H_{p_2}\|/2 \qquad\qquad \Delta\text{ ineq} + \|XY\|\le\|X\|\|Y\| \\ &\le \sqrt{N}+\eta+\eta^2/2 \\ &\le \sqrt{N}+1+1/2 \le C. \end{aligned}$$

Similarly, $\|R^*_{p_3}\|\le C$ and $\|R^*_{p_4}\|\le C$.

It follows by 59 and the aforementioned estimates that

$$\begin{aligned} |\operatorname{Re}\operatorname{Tr}(I-U_p)-\operatorname{Re}\operatorname{Tr}(I-R_p)| &\le \sum_{j=1}^{4}(C\eta^3)\cdot C\cdot C\cdot C \\ &\le C\eta^3, \end{aligned}$$

as desired. Thus we have proved the first claim. □

The second claim says, for any plaquette p,

$$|\operatorname{Re}(\operatorname{Tr}(I-R_p))-\frac{1}{2}\|H_p\|^2| \le C\eta^3.$$

Proof of second claim. We expand:

$$\begin{aligned} \operatorname{Re}(\operatorname{Tr}(I-R_p)) = \operatorname{Re}\operatorname{Tr}\Big(&I-(I+iH_{p_1}-H^2_{p_1}/2))\times(I+iH_{p_2}-H^2_{p_2}/2)\times \\ &(I-iH_{p_3}-H^2_{p_3}/2)\times(I-iH_{p_4}-H^2_{p_4}/2)\Big). \end{aligned}$$

[11] Because $\sqrt{N}=\|I\|=\|VV^*\|=\|V\|$.

The product $R_p = R_{p_1} R_{p_2} R_{p_3}^* R_{p_4}^*$ has $3 \times 3 \times 3 \times 3 = 81$ terms, which we classify based on how many matrices H_{p_j} appear in each one.

First we begin with terms without any matrices H_{p_j}. There is one term that is equal to I.

Then we move to linear terms with one matrix H_{p_j}. There are four terms of the form $\mathrm{i}H_{p_j}$ up to a sign. The contribution is $X_{\mathrm{lin}} := \mathrm{i}(H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4})$.

Then we move to the quadratic terms with two matrices H_{p_j}. There are $\binom{4}{2} = 6$ mixed terms of the form $zH_{p_k}H_{p_j}$ where $k < j$ and $z \in \{-1, 1\}$, and then 4 homogeneous terms of the form $-\frac{1}{2}H_{p_j}^2$. The contribution is

$$X_{\mathrm{quad}} := -\frac{1}{2}(H_{p_1}^2 + H_{p_2}^2 + H_{p_3}^2 + H_{p_4}^2) - H_{p_1}H_{p_2} + H_{p_1}H_{p_3} + H_{p_1}H_{p_4} + H_{p_2}H_{p_3} + H_{p_2}H_{p_4} - H_{p_3}H_{p_4}.$$

There are also cubic, quartic, quintic, sextic, septic, and octic terms, but their contribution will be absorbed into the upper bound $C\eta^3$ after an application of the previous lemma. But let us denote by $X_{\mathrm{cubic}+} \in \mathbb{C}^{N\times N}$ the sum of these 66 higher order terms.

We have

$$R_p = I + X_{\mathrm{lin}} + X_{\mathrm{quad}} + X_{\mathrm{cubic}+}$$

and therefore

$$\operatorname{Re}\operatorname{Tr}(I - R_p) = -\operatorname{Re}\operatorname{Tr}(X_{\mathrm{lin}}) - \operatorname{Re}\operatorname{Tr}(X_{\mathrm{quad}}) - \operatorname{Re}\operatorname{Tr}(X_{\mathrm{cubic}+}).$$

First we show that $\operatorname{Re}\operatorname{Tr}(X_{\mathrm{lin}}) = 0$. Recall that the trace of a matrix is equal to the sum of its eigenvalues. Since the eigenvalues of $H_p = H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4}$ are real as H_p is a Hermitian matrix, it follows that the eigenvalues of $\mathrm{i}(H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4})$ are all imaginary. Taking the real part of the trace, we get zero, as desired.

Next, we show that

$$-\operatorname{Re}\operatorname{Tr}(X_{\mathrm{quad}}) = \frac{1}{2}\|H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4}\|^2.$$

Since H_p is Hermitian, we get that $H_p = H_p^*$. Therefore $\|H_p\|^2$ is equal to $\operatorname{Tr}\big(H_p^* H_p\big) = \operatorname{Tr} H_p^2$. Note that H_p^2 in expanded form has 16 terms. Note that 12 of these terms are mixed and of the form $zH_{p_i}H_{p_j}$ for $z \in \{-1, 1\}$, and 4 are homogeneous of the form $H_{p_i}^2$. In particular, we have

$$\begin{aligned} H_p^2 = (&H_{p_1}H_{p_2} - H_{p_1}H_{p_3} - H_{p_1}H_{p_4} - H_{p_2}H_{p_3} - H_{p_2}H_{p_4} + H_{p_3}H_{p_4}) \\ &+(H_{p_2}H_{p_1} - H_{p_3}H_{p_1} - H_{p_4}H_{p_1} - H_{p_3}H_{p_2} - H_{p_4}H_{p_2} + H_{p_3}H_{p_4})+ \\ &+(H_{p_1}^2 + H_{p_2}^2 + H_{p_3}^2 + H_{p_4}^2), \end{aligned}$$

where in the second line we've written the oppositely ordered terms compared to the first line, and the third line contains the homogeneous terms. We obtain the same value by taking the trace of the first and second lines above on the right hand side because of the identity $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$. Therefore,

$$\begin{aligned} \frac{1}{2}\operatorname{Tr}\big(H_p^2\big) = \frac{1}{2}\cdot 2\operatorname{Tr}(&H_{p_1}H_{p_2} - H_{p_1}H_{p_3} - H_{p_1}H_{p_4} - H_{p_2}H_{p_3} - H_{p_2}H_{p_4} + H_{p_3}H_{p_4}) \\ &+\frac{1}{2}\operatorname{Tr}\big(H_{p_1}^2 + H_{p_2}^2 + H_{p_3}^2 + H_{p_4}^2\big). \end{aligned}$$

The expression on the left hand side is equal to $\frac{1}{2}\|H_p\|^2$, which is a real number. Therefore, applying Re to the right hand side of the equation does nothing. That is,

$$\begin{aligned} \frac{1}{2}\|H_p\|^2 = \operatorname{Re}\operatorname{Tr}(&H_{p_1}H_{p_2} - H_{p_1}H_{p_3} - H_{p_1}H_{p_4} - H_{p_2}H_{p_3} - H_{p_2}H_{p_4} + H_{p_3}H_{p_4}) \\ &+\frac{1}{2}\operatorname{Re}\operatorname{Tr}\big(H_{p_1}^2 + H_{p_2}^2 + H_{p_3}^2 + H_{p_4}^2\big). \end{aligned}$$

Then we recognize that the right hand side is equal to $-\operatorname{Re}\operatorname{Tr} X_{\mathrm{quad}}$, which is what we wanted to show.

Then to conclude the lemma, we write

$$\begin{aligned} \left|\operatorname{Re}\operatorname{Tr}(I - R_p) - \frac{1}{2}\|H_p\|^2\right| &= |-\operatorname{Re}\operatorname{Tr} X_{\mathrm{cubic}+}| \\ &\le \sum |\operatorname{Re}\operatorname{Tr}\ zH^{k\ge 3}|. \end{aligned}$$

where in the last line, the sum is over the 66 higher order terms, and $H^{k\ge 3}$ stands for a higher order term of the form $H_{p_{i_1}}H_{p_{i_2}}\cdots H_{p_{i_k}}$ for $i_1, \cdots, i_k \in \{1, 2, 3, 4\}$ and z stands for a coefficient which satisfies

$|z| \leq 1$. Then by an application of Lemma 7.2 and the fact that $\|H_{p_j}\| \leq \eta \leq 1$ for all $j = 1, 2, 3, 4$ and that $\eta^k \leq \eta^3$ whenever $k \geq 3$, we have

$$\sum |\operatorname{Re}\operatorname{Tr} zH^{k\geq 3}| \leq 66 \times \eta^3 \leq C\eta^3.$$

Note that the constant C does not really depend on N, but it is easier to say that it depends only on N rather than to say what it does not depend on. This concludes the proof. □

Since we haven proven both claims, the proof is concluded by applying the triangle inequality. □

7.3 Closeness of Wilson action and lattice Hermitian Maxwell action

The quadratic approximation, i.e. third order error bound, developed in Lemma 7.1 and Lemma 7.3 explains the term r^3 in the following Theorem 7.4. We saw that things got messy due to the extra quadratic term, and it is natural to wonder whether the formula for the leading order term of the Yang Mills free energy (Theorem 1.2) can be obtained instead using a linear approximation, i.e. $R_{p_j} = I + \mathrm{i}H_{p_j}$.

We explain why we cannot take power lower than 3, at least using the technique developed in [4]. In other words, we explain why a simpler linear approximation, e.g. second order error bound, would not suffice for the method in [4].

Inspecting the proof of the lower bound of the main theorem, Theorem 1.2 and particularly Lemma 8.6, we collect the parameters r and β. We know what β is: it is the inverse of the square of the gauge coupling constant, and will be sent to ∞. Meanwhile, r is meant to be a small parameter which quantifies proximity to the 0 Hermitian configuration, where the machinary of exchanging integration over Lie group to integration over Lie algebra works. In particular, the integral over Lie algebra becomes a better approximator to the integral over Lie group as r gets smaller and smaller: see the coefficient of the lower bound of Theorem 5.2. This motivates why, in the proof of 8.6, r is equal to a negative power of β. Therefore, assume in the following discussion that $r = \beta^l$ for some $l < 0$.

It is easy to see that the proof of 8.6 does not work if any of the following conditions is broken:

1. $\sqrt{\beta} \cdot r$ is equal to a positive power of β.[12]
2. $\beta \cdot r^k$ is a negative power of β, where $k \in \mathbb{R}$.[13] Here, we imagine the proof of 8.6 with k replacing 3.

Condition 1 says that $\frac{1}{2} + l > 0$ and condition 2 says that $1 + lk < 0$. Together with $l < 0$, these inequalities show that $k > \frac{-1}{l} > 2$. In other words, if we take a second order error bound r^2, i.e. $k = 2$, in the following theorem, then the proof of Lemma 8.6 would break.

On a different note, we recall that

$$M_n^{\mathrm{H}}(H) = \sum_{p \in \mathcal{P}(\Lambda_n)} \|H_p\|^2,$$

as was defined in the introduction, is the lattice Maxwell action for Hermitian configurations $H \in \mathrm{H}(N)^{E_n}$.

Theorem 7.4 (Theorem 16.3 [4]). *Take any $H \in \mathrm{H}(N)^{E_n}$. Suppose that $0 < r \leq 1$ is a number such that $\|H_e\| \leq r$ for every $e \in E_n$. Then*

$$\left| S_{\Lambda_n}(e^{\mathrm{i}H}) - \frac{1}{2} M_n^{\mathrm{H}}(H) \right| \leq Cr^3 n^d$$

where C depends only on d and N.

[12] Indeed, the term $-\frac{\log \beta}{(\sqrt{\beta} r)^{1/(d+2)}}$ must go to 0 rather than $-\infty$.

[13] Indeed, the term βr^3 in the proof must go to zero, which we now imagine as the term βr^k.

Proof. We have

$$\begin{aligned}
\left|S_{\Lambda_n}(e^{\mathrm{i}H}) - \frac{1}{2}M_n^{\mathrm{H}}(H)\right| &= \left|\sum_{p\in\mathcal{P}(\Lambda_n)} \operatorname{Re}\operatorname{Tr}\bigl(I - e_p^{\mathrm{i}H}\bigr) - \frac{1}{2}\|H_p\|^2\right| \\
&\leq \sum_{p\in\mathcal{P}(\Lambda_n)} \left|\operatorname{Re}\operatorname{Tr}\bigl(I - e_p^{\mathrm{i}H}\bigr) - \frac{1}{2}\|H_p\|^2\right| \\
&\leq \sum_{p\in\mathcal{P}(\Lambda_n)} Cr^3 \qquad \text{(Lemma 7.3)} \\
&= |\mathcal{P}(\Lambda_n)|Cr^3 \\
&\leq Cn^d r^3,
\end{aligned}$$

where C depends on N and d in the last line because we bound the plaquettes per edge using d, and we had dependence on N in Lemma 7.3.[14] □

[14] Though I think there was actually no dependence on N in that lemma.

8 Proof of Main Theorem

In this section, the strings are pulled together. We begin with a small lemma that the partition function of Hermitian lattice Maxwell theory breaks up into an N^2 fold product of scalar lattice Maxwell theory partition functions. Afterward, we will move to the upper bound of Theorem 1.2, and after that, the lower bound.

Lemma 8.1. *Let n be a positive integer. We have*

$$Z_{H,\Lambda_n} = Z_{M,\Lambda_n}^{N^2}.$$

Proof. We have

$$Z_{H,\Lambda_n} = \int_{\mathrm{H}(N)^{E_n}} e^{-\frac{1}{2}M_n^{\mathrm{H}}(H)} \prod_{e\in E_n^1} d\lambda(H_e) \prod_{e\in E_n^0} d\delta_0(H_e).$$

Recall the linear isometry 13, namely $\Psi_H : (\mathbb{R}^{N^2}, \|\cdot\|_{\mathbb{R}^{N^2}}) \to (\mathrm{H}(N), \|\cdot\|_{HS})$, and the product of the isometry, namely

$$(\mathbb{R}^{N^2})^{E_n} \ni (x_e)_{e\in E_n} \mapsto \Psi_H^{E_n}((x_e)_{e\in E_n}) = (\Psi_H(x_e))_{e\in E_n} \in \mathrm{H}(N)^{E_n}. \tag{60}$$

Recall that $\lambda = (\Psi_H)_* l_{N^2}$. Hence,

$$Z_{H,\Lambda_n} = \int_{(\mathbb{R}^{N^2})^{E_n}} e^{-\frac{1}{2}M_n^{\mathrm{H}}(\Psi_H^{E_n}((x_e)_{e\in E_n}))} \prod_{e\in E_n^1} dl_{N^2}(x_e) \prod_{e\in E_n^0} d\delta_0(x_e).$$

Fix $(x_e)_{e\in E_n} \in (\mathbb{R}^{N^2})^{E_n}$ and let $\mathrm{H}(N)^{E_n} \ni H = \Psi_H^{E_n}((x_e)_{e\in E_n})$. We have

$$\begin{aligned}
M_n^{\mathrm{H}}(\Psi_H^{E_n}((x_e)_{e\in E_n})) &= \sum_{p\in\mathcal{P}(\Lambda_n)} \|H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4}\|_{HS}^2 \\
&= \sum_{p\in\mathcal{P}(\Lambda_n)} \|\Psi_H(x_{p_1}) + \Psi_H(x_{p_2}) - \Psi_H(x_{p_3}) - \Psi_H(x_{p_4})\|_{HS}^2 \\
&= \sum_{p\in\mathcal{P}(\Lambda_n)} \|\Psi_H(x_{p_1} + x_{p_2} - x_{p_3} - x_{p_4})\|_{HS}^2 \\
&= \sum_{p\in\mathcal{P}(\Lambda_n)} \|x_{p_1} + x_{p_2} - x_{p_3} - x_{p_4}\|_{\mathbb{R}^{N^2}}^2 \\
&= \sum_{p\in\mathcal{P}(\Lambda_n)} \sum_{j=1}^{N^2} (x_{p_1}^{(j)} + x_{p_2}^{(j)} - x_{p_3}^{(j)} - x_{p_4}^{(j)})^2 \\
&= \sum_{j=1}^{N^2} \sum_{p\in\mathcal{P}(\Lambda_n)} (x_{p_1}^{(j)} + x_{p_2}^{(j)} - x_{p_3}^{(j)} - x_{p_4}^{(j)})^2 \\
&= \sum_{j=1}^{N^2} M_n^{(\mathbb{R})}(x^{(j)})
\end{aligned} \tag{61}$$

We also have

$$\prod_{e\in E_n^1} dl_{N^2}(x_e) \prod_{e\in E_n^0} d\delta_0(x_e) = \prod_{j\in[N^2]} \left(\prod_{e\in E_n^1} dl(x_e^{(j)}) \prod_{e\in E_n^0} d\delta_0(x_e^{(j)}) \right) \tag{62}$$

and

$$(\mathbb{R}^{N^2})^{E_n} = \prod_{j\in[N^2]} \mathbb{R}^{E_n}.$$

By Fubini's theorem,

$$\begin{aligned}
&\int_{(\mathbb{R}^{N^2})^{E_n}} e^{-\frac{1}{2}M_n^{\mathrm{H}}(\Psi_H^{E_n}((x_e)_{e\in E_n}))} \prod_{e\in E_n^1} dl_{N^2}(x_e) \prod_{e\in E_n^0} d\delta_0(x_e) \\
&= \prod_{j\in[N^2]} \int_{\mathbb{R}^{E_n}} e^{-\frac{1}{2}\sum_{p\in\mathcal{P}(\Lambda_n)}(x_{p_1}^{(j)}+x_{p_2}^{(j)}-x_{p_3}^{(j)}-x_{p_4}^{(j)})^2} \prod_{e\in E_n^1} dl(x_e^{(j)}) \prod_{e\in E_n^0} d\delta_0(x_e^{(j)}) \\
&= \prod_{j\in[N^2]} Z_{M,\Lambda_n} = Z_{M,\Lambda_n}^{N^2},
\end{aligned}$$

as desired. □

Lemma 8.2 (Lemma 17.1 [4]). *For each $n \geq 2$,*

$$|E_n^1| = (d-1)n^d - dn^{d-1} + 1.$$

8.1 Upper Bound

Recall that C_N is the constant from Theorem 5.2 and that F_{M,Λ_n} is the free energy per site of (scalar) lattice Maxwell theory with the axial boundary condition ($0 \in \mathbb{R}^{E_n^0}$) in the box Λ_n.

Lemma 8.3 (Lemma 17.2 [4]). *Define*

$$\delta_{n,\beta} := n^{(d+1)/2}\left(\frac{\log\beta}{\beta}\right)^{1/2}.$$

Then for any n and g,

$$F_{\Lambda_n,g} \leq \frac{|E_n^1|}{n^d}\log C_N - \frac{|E_n^1|}{2n^d}N^2\log\beta + N^2F_{M,\Lambda_n} + \frac{C}{n^d} + C\delta_{n,\beta} + C\beta\delta_{n,\beta}^3,$$

where C depends only on N and d.

Proof. For each n, g, we denote

$$U(C_0\delta_{n,\beta}) = \prod_{e\in E_n} b_{\mathrm{U}(N)}(I, C_0\delta_{n,\beta}), \tag{63}$$

where C_0 is the constant from Theorem 4.11, which depends only on N and d.

By Theorem 4.11,

$$F_{\Lambda_n,g} \leq \frac{\log(2)}{n^d} + \frac{1}{n^d}\log\left(\int_{U(C_0\delta_{n,\beta})} e^{-\beta S_{\Lambda_n}(U)} \prod_{e\in E_n^1} d\sigma(U_e) \prod_{e\in E_n^0} d\delta_I(U_e)\right). \tag{64}$$

Let $r_0 = \log(4/3)/18$ be the constant from Theorem 5.2.

Fix $n \geq 2$, $g > 0$ and suppose that $\frac{1}{2}C_0\delta_{n,\beta} \leq r_0$ as case 1. Let $r = \frac{1}{2}C_0\delta_{n,\beta}$. Then

$$b_{\mathrm{U}(N)}(I, C_0\delta_{n,\beta}) = b_{\mathrm{U}(N)}(I, 2r),$$

and

$$U(C_0\delta_{n,\beta}) = b_{\mathrm{U}(N)}(I, 2r)^{E_n}.$$

By Theorem 5.2 applied to all of the E_n^1 edges, we have

$$\begin{aligned}
&\int_{U(C_0\delta_{n,\beta})} e^{-\beta S_{\Lambda_n}(U)} \prod_{e\in E_n^1} d\sigma(U_e) \prod_{e\in E_n^0} d\delta_I(U_e) \\
&\leq (e^{4r})^{N^2|E_n^1|} C_N^{|E_n^1|} \int_{b_{\mathrm{H}(N)}(0,3r)^{E_n}} e^{-\beta S_{\Lambda_n}(e^{\mathrm{i}H})} \prod_{e\in E_n^1} d\lambda(H_e) \prod_{e\in E_n^0} d\delta_0(H_e).
\end{aligned} \tag{65}$$

Note that $(e^{4r})^{N^2|E_n^1|} \leq e^{C_2 r n^d}$ for some C_2 depending only on N and d, which uses that $|E_n^1| \leq dn^d$. For any $H \in b_{\mathrm{H}(N)}(0,3r)^{E_n}$, and using $3r \leq 1$, we have

$$-\beta S_{\Lambda_n}(e^{\mathrm{i}H}) \leq -\frac{\beta}{2} M_n^{\mathrm{H}}(H) + \beta C_3 r^3 n^d \tag{66}$$

by Theorem 7.4, for a constant C_3 depending only on N and d. Then by monotonicity of the integral, we have

$$\begin{aligned} &\int_{U(C_0\delta_{n,\beta})} e^{-\beta S_{\Lambda_n}(U)} \prod_{e\in E_n^1} d\sigma(U_e) \prod_{e\in E_n^0} d\delta_I(U_e) \\ &\leq e^{C_2 r n^d + \beta C_3 r^3 n^d} C_N^{|E_n^1|} \int_{b_{\mathrm{H}(N)}(0,3r)^{E_n}} e^{-\frac{\beta}{2} M_n^{\mathrm{H}}(H)} \prod_{e\in E_n^1} d\lambda(H_e) \prod_{e\in E_n^0} d\delta_0(H_e). \end{aligned} \tag{67}$$

We have

$$\begin{aligned} \beta M_n^{\mathrm{H}}(H) &= \beta \sum_{p\in\mathcal{P}(\Lambda_n)} \|H_{p_1} + H_{p_2} - H_{p_3} - H_{p_4}\|^2 \\ &= \sum_{p\in\mathcal{P}(\Lambda_n)} \|\sqrt{\beta}H_{p_1} + \sqrt{\beta}H_{p_2} - \sqrt{\beta}H_{p_3} - \sqrt{\beta}H_{p_4}\|^2. \end{aligned} \tag{68}$$

Instead of integrating over $(H_e)_{e\in E_n^1}$, we will integrate over $(G_e)_{e\in E_n^1} = (\sqrt{\beta}H_e)_{e\in E_n^1}$, recalling that the Dirac measure centered at $0 \in \mathrm{H}(N)$ forces $H_e = 0$ for all $e \in E_n^0$. For example, if we perform this change of variable one edge at a time, then the Jacobian for each edge is $\left(\frac{1}{\sqrt{\beta}}\right)^{N^2}$ and the region of integration on a fixed edge changes from $b_{\mathrm{H}(N)}(0,3r)$ to $b_{\mathrm{H}(N)}(0,3r\sqrt{\beta})$. The result is

$$\begin{aligned} &\int_{b_{\mathrm{H}(N)}(0,3r)^{E_n}} e^{-\frac{\beta}{2} M_n^{\mathrm{H}}(H)} \prod_{e\in E_n^1} d\lambda(H_e) \prod_{e\in E_n^0} d\delta_0(H_e) \\ &= \int_{b_{\mathrm{H}(N)}(0,3r\beta)^{E_n}} e^{-\frac{1}{2} M_n^{\mathrm{H}}(G)} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2|E_n^1|} \prod_{e\in E_n^1} d\lambda(G_e) \prod_{e\in E_n^0} d\delta_0(G_e). \end{aligned} \tag{69}$$

By integrating a non-negative function over a larger region, we get that the right hand side of 69 is less than or equal to

$$\left(\frac{1}{\sqrt{\beta}}\right)^{N^2|E_n^1|} \int_{\mathrm{H}(N)^{E_n}} e^{-\frac{1}{2} M_n^{\mathrm{H}}(G)} \prod_{e\in E_n^1} d\lambda(G_e) \prod_{e\in E_n^0} d\delta_0(G_e) = \left(\frac{1}{\sqrt{\beta}}\right)^{N^2|E_n^1|} Z_{H,\Lambda_n}. \tag{70}$$

By 65, 66, 67, 68, 69, and 70, we get

$$\begin{aligned} &\int_{U(C_0\delta_{n,\beta})} e^{-\beta S_{\Lambda_n}(U)} \prod_{e\in E_n^1} d\sigma(U_e) \prod_{e\in E_n^0} d\delta_I(U_e) \\ &\leq e^{C_2 r n^d + \beta C_3 r^3 n^d} C_N^{|E_n^1|} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2|E_n^1|} Z_{H,\Lambda_n} \\ &= e^{C_2 r n^d + \beta C_3 r^3 n^d} C_N^{|E_n^1|} (\beta)^{-\frac{1}{2}N^2|E_n^1|} Z_{M,\Lambda_n}^{N^2}, \end{aligned} \tag{71}$$

where the last equality is Lemma 8.1. Recalling 64, we have

$$\begin{aligned} F_{\Lambda_n,g} &\leq \frac{\log(2)}{n^d} + \frac{1}{n^d} \log\left(e^{C_2 r n^d + \beta C_3 r^3 n^d} C_N^{|E_n^1|} (\beta)^{-\frac{1}{2}N^2|E_n^1|} Z_{M,\Lambda_n}^{N^2}\right) \\ &= \frac{\log(2)}{n^d} + C_2 r + \beta C_3 r^3 + \frac{|E_n^1|}{n^d} \log(C_N) - \frac{N^2|E_n^1|}{2n^d} \log(\beta) + N^2 \frac{\log Z_{M,\Lambda_n}}{n^d}. \end{aligned} \tag{72}$$

Recalling that $r = \frac{1}{2}C_0\delta_{n,\beta}$ and $\frac{\log Z_{M,\Lambda_n}}{n^d} = F_{M,\Lambda_n}$ and choosing some constant C depending only on N and d large enough, we get the result of the present lemma, in the case where $r \leq r_0$.

For case 2, suppose $\frac{1}{2}C_0\delta_{n,\beta} > r_0$. Recall 8, which says that $Z_{\Lambda_n,g} \leq 1$. Hence

$$F_{\Lambda_n,g} \leq 0. \tag{73}$$

Case 2 then becomes trivial once we show that the upper bound of the present lemma is larger than 0, for C chosen large enough depending only on N and d.

Claim: There exists a constant $C_4 > 0$ depending only on N and d such that

$$\frac{|E_n^1|}{n^d} \log C_N + N^2 F_{M,\Lambda_n} \geq -C_4. \tag{74}$$

Proof of claim. We have that F_{M,Λ_n} converges to a finite limit by Theorem 6.10. Eventually all elements of the sequence are close to the limit, which can be lower bounded irrespective of n. The possibly finite number of sequence elements far away from the limit can also be lower bounded by a constant depending only on N and d (in particular, independent of n and β). On the other hand, the term $\frac{|E_n^1|}{n^d}$ is at most $d-1$ for $n \geq 2$, so $\frac{|E_n^1|}{n^d} \log C_N$ is lower bounded by a constant depending only on N and d. □

There are two remaining observations. First,

$$C_6 \delta_{n,\beta} - C_4 \geq 0 \tag{75}$$

if C_6 is chosen large enough depending only on N and d. This follows by the assumption that $\frac{1}{2} C_0 \delta_{n,\beta} > r_0$.

Second,

$$C_7 \beta \delta_{n,\beta}^3 - \frac{|E_n^1|}{2n^d} N^2 \log \beta \geq 0 \tag{76}$$

for C_7 depending only on N and d chosen large enough. To see this, assume for simplicity that $\log \beta > 0$. We have

$$-\frac{|E_n^1|}{2n^d} N^2 \log \beta \geq -\frac{dN^2}{2} \log \beta \geq -C_8 \log \beta.$$

Multiplying the assumption $\delta_{n,\beta} > \frac{2}{C_0} r_0$ by $\log \beta$ and using that $n^{3(d+1)/2} \geq n^{(d+1)/2}$ yields

$$\begin{aligned}
\beta \delta_{n,\beta}^3 &= n^{3(d+1)/2} \frac{(\log \beta)^{3/2}}{\sqrt{\beta}} \\
&\geq n^{(d+1)/2} \frac{(\log \beta)^{3/2}}{\sqrt{\beta}} \\
&= \delta_{n,\beta} \log \beta \\
&\geq \frac{2}{C_0} r_0 \log \beta.
\end{aligned}$$

There is some $C_7 > 0$ such that $C_7 \cdot \frac{2}{C_0} r_0 \geq C_8$, and hence that

$$C_7 \beta \delta_{n,\beta}^3 \geq C_7 \cdot \frac{2}{C_0} r_0 \log \beta \geq C_8 \log \beta.$$

Hence,

$$C_7 \beta \delta_{n,\beta}^3 - \frac{|E_n^1|}{2n^d} N^2 \log \beta \geq C_7 \beta \delta_{n,\beta}^3 - C_8 \log \beta \geq 0,$$

which is 76.

Taking C to be the maximum of C_6 and C_7, we get together with 76, 75, and 74, that

$$\frac{|E_n^1|}{n^d} \log C_N - \frac{|E_n^1|}{2n^d} N^2 \log \beta + N^2 F_{M,\Lambda_n} + \frac{C}{n^d} + C\delta_{n,\beta} + C\beta\delta_{n,\beta}^3 \geq 0 \geq F_{\Lambda_n,g}.$$

This concludes case 2. □

In lattice Maxwell theory, when given $2 \leq m \leq n$, we considered r^d translates of Λ_m which (nearly) filled the box Λ_n, and the translates B_a, see 45, were allowed to intersect at a common boundary. We denoted the collection of these translates by $\mathcal{B}$, see 46. This time, we will not allow the translates to intersect. Let us introduce the new definitions.

Let r be the unique integer such that $n = rm + k$ for $0 \leq k \leq m-1$. For any $a = (a_1, \cdots, a_d) \in \{0, \cdots, r-1\}^d$, let

$$B'_a = \{(x_1, \cdots, x_d) \in \Lambda_n : a_j \cdot m \leq x_j \leq (a_j + 1) \cdot m - 1 \, \forall j \in [d]\}. \tag{77}$$

Let

$$\mathcal{B}' = \{B'_a \subset \Lambda_n : a \in \{0, \cdots, r-1\}^d\}. \tag{78}$$

Therefore, $\mathcal{B}'$ is a collection of r^d disjoint translates of Λ_m. It is apparent that

$$\Lambda_{rm} = \bigcup_{B'_a \in \mathcal{B}'} B'_a, \tag{79}$$

where the union on the right hand side is a disjoint union.

Recall the definition of the positively oriented edge set $E_{B'_a}$, see 1, the definition of the plaquette set $\mathcal{P}(B'_a)$, see 3, the definition of the Wilson action $S_{B'_a}(\,\cdot\,) : \mathrm{U}(N)^{E_{B'_a}} \to [0, \infty)$,

$$S_{B'_a}(U) = \sum_{p \in \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p)),$$

see 4, and the definition of the partition function

$$Z_{B'_a, g} = \int_{\mathrm{U}(N)^{E_{B'_a}}} e^{-\beta S_{B'_a}(U)} \prod_{e \in E_{B'_a}} d\sigma(U_e),$$

see 8. Furthermore, it is apparent that

$$Z_{B_a, g} = Z_{\Lambda_m, g}$$

if $B_a \in \mathcal{B}'$.

Lemma 8.4 (Lemma 17.3 [4])**.** *For any $2 \le m \le n$,*

$$F_{\Lambda_n, g} \le \left(1 - \frac{Cm}{n}\right) F_{\Lambda_m, g},$$

where C depends only on d.

Proof. Assume the notation in the preface to this lemma. In case m does not divide n, we first switch to the possibly smaller box Λ_{rm}, and then we switch to the collection of translates that subdivide Λ_{rm}. We have

$$\begin{aligned}
Z_{\Lambda_n, g} &= \int_{\mathrm{U}(N)^{E_n}} e^{-\beta \sum_{p \in \mathcal{P}(\Lambda_n)} \frac{1}{2}\|I - U_p\|^2} \prod_{e \in E_n} d\sigma(U_e) \\
&\le \int_{\mathrm{U}(N)^{E_n}} e^{-\beta \sum_{p \in \mathcal{P}(\Lambda_{rm})} \frac{1}{2}\|I - U_p\|^2} \prod_{e \in E_n} d\sigma(U_e) \\
&= \int_{\mathrm{U}(N)^{E_{\Lambda_{rm}}}} e^{-\beta \sum_{p \in \mathcal{P}(\Lambda_{rm})} \frac{1}{2}\|I - U_p\|^2} \prod_{e \in E_{\Lambda_{rm}}} d\sigma(U_e) \\
&\le \int_{\mathrm{U}(N)^{E_{\Lambda_{rm}}}} \exp\left(-\beta \sum_{p \in \bigcup_{B'_a \in \mathcal{B}'} \mathcal{P}(B'_a)} \frac{1}{2}\|I - U_p\|^2\right) \prod_{e \in E_{\Lambda_{rm}}} d\sigma(U_e) \\
&= \prod_{B'_a \in \mathcal{B}'} \int_{\mathrm{U}(N)^{E_{B'_a}}} e^{-\beta \sum_{p \in \mathcal{P}(B'_a)}} \prod_{e \in E_{B'_a}} d\sigma(U_e) \\
&= \prod_{B'_a \in \mathcal{B}'} Z_{B'_a, g} \\
&= Z^{r^d}_{\Lambda_m, g}.
\end{aligned}$$

The second line and fourth line above use that dropping plaquettes increases the Gibbs weight $e^{-\beta S}$. For instance, note that

$$\bigcup_{B'_a \in \mathcal{B}'} \mathcal{P}(B'_a) \subset \mathcal{P}(\Lambda_{rm}), \tag{80}$$

where this is a strict subset relation except in possibly trivial cases. The third line and fifth line use that we can integrate out edges which are not contained in any plaquettes in of the "Hamiltonian," i.e. the argument of $e^{-\beta(\,\cdot\,)}$, which uses the fact that σ is a probability measure.

Recall that

$$F_{\Lambda_m,g} = \frac{\log Z_{\Lambda_m,g}}{m^d}.$$

Also, recall that $n = mr + k$, where $0 \le k \le m-1$. We obtain from $Z_{\Lambda_n,g} \le Z_{\Lambda_m,g}^{r^d}$ upon taking the logarithm, dividing by n^d, and multipling by $1 = \frac{m^d}{m^d}$ that

$$\begin{aligned} F_{\Lambda_n,g} &\le \frac{r^d m^d}{n^d} F_{\Lambda_m,g} \\ &= \frac{(n-k)^d}{n^d} F_{\Lambda_m,g}. \end{aligned}$$

Next, we have

$$\begin{aligned} \frac{(n-k)^d}{n^d} &\ge \frac{(n-m)^d}{n^d} && k \le m \\ &\ge 1 - \frac{Cm}{n} && \text{binomial formula.} \end{aligned}$$

Multiplying this relation by $F_{\Lambda_m,g} \le 0$ (see 73), yields

$$F_{\Lambda_n,g} \le \frac{(n-k)^d}{n^d} F_{\Lambda_m,g} \le (1 - \frac{Cm}{n}) F_{\Lambda_m,g},$$

as desired. □

Lemma 8.5 (Lemma 17.4 [4])**.**

$$\limsup_{\substack{n\to\infty \\ g\to 0}} \left(F_{\Lambda_n,g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta \right) \le (d-1)\log C_N + N^2 \lim_{n\to\infty} F_{M,\Lambda_n}.$$

Proof. Let $\mathbb{R}_+ = (0,\infty)$. By Lemma 9.12, it suffices to show that

$$\limsup_{k\to\infty} \left(F_{\Lambda_{n_k},g_k} + \frac{|E_{n_k}^1|}{2{n_k}^d} N^2 \log \beta_k \right) \le (d-1)\log C_N + N^2 \lim_{n\to\infty} F_{M,\Lambda_n}. \tag{81}$$

for any sequence $(n_k,\beta_k)_{k\in\mathbb{N}} \in (\mathbb{N}\times\mathbb{R}_+)^{\mathbb{N}}$ where $n_k \to \infty$ and $\beta_k \to \infty$ as $k\to\infty$. Note that $\beta_k = \frac{1}{g_k^2}$.

We organize the set of all pairs of sequences (n_k,β_k) who tend to infinity, which we denote by A, based on how

$$\delta_{n_k,\beta_k} := n_k^{(d+1)/2} \left(\frac{\log \beta_k}{\beta_k} \right)^{1/2}$$

and $\beta_k \delta_{n_k,\beta_k}^3$ behave, and then we work on each case separately. We have

$$\begin{aligned} A &= \{((n_j)_{j\in\mathbb{N}},(\beta_j)_{j\in\mathbb{N}}) \in \mathbb{N}^{\mathbb{N}} \times \mathbb{R}_+^{\mathbb{N}} : n_j \to \infty, \beta_j \to \infty \text{ as } j\to\infty\} && (82) \\ &= \{(n_j,\beta_j) : n_j \to \infty, \beta_j \to \infty \text{ and both } \delta_{n_k,\beta_k} \to 0, \beta_k \delta_{n_k,\beta_k}^3 \to 0 \text{ as } j\to\infty\} && (83) \\ &\bigcup_{M\in(0,\infty)} \bigcup \{(n_j,\beta_j) : n_j \to \infty, \beta_j \to \infty \text{ as } j\to\infty \text{ and } \exists \text{ subsequence } j_k \text{ such that } \delta_{n_{j_k},\beta_{n_{j_k}}} \ge M \,\forall k\} \\ & && (84) \\ &\bigcup_{M\in(0,\infty)} \bigcup \{(n_j,\beta_j) : n_j \to \infty, \beta_j \to \infty \text{ as } j\to\infty \text{ and } \exists \text{ subsequence } j_k \text{ such that } \beta_{j_k}\delta_{n_{j_k},\beta_{n_{j_k}}}^3 \ge M \,\forall k\} \\ & && (85) \\ &=: A_1 \cup A_2 \cup A_3. && (86) \end{aligned}$$

We have defined three sets after the second equality. First, A_1, the sequences which are associated with $\delta_{n,\beta}$ and $\beta\delta_{n,\beta}^3$ tending to 0. Second, A_2, the sequences for which $\delta_{n,\beta}$ is bounded away from zero along a subsequence. Third, A_3, the sequences for which $\beta\delta_{n,\beta}^3$ is bounded away from zero along a subsequence.

In the remainder of the proof, it is helpful to write β instead of β_j and n instead of n_j, but to still understand n and β sequentially. For example, we can think of $n : \mathbb{N} \to \mathbb{N}$ as $j \mapsto n_j$.

Note that A_1 is disjoint from $A_2 \cup A_3$. Furthermore, $A_2 \subsetneq A_3$. Indeed, suppose $\delta_{n,\beta} \geq M$ along a subsequence. We can assume β is large enough for $\sqrt{\log \beta} \leq \sqrt{\log \beta}^3$. Also, $n^{(d+1)/2} \leq (n^{(d+1)/2})^3$. Hence, $\beta\delta_{n,\beta}^3 \geq \delta_{n,\beta} \geq M$, showing that $\beta\delta_{n,\beta}^3$ is also bounded away from zero along the same subsequence. On the other hand, if β grows like $n^{2(d+1)}$, then we see that, ignoring the logarithmic contribution, $\beta\delta_{n,\beta}^3 \approx n^{(d+1)/2}$ is bounded away from zero, but $\delta_{n,\beta} \approx \frac{1}{n^{(d+1)/2}}$ is not.

Suppose n and β are sequences in A_1. Recall that $|E_n^1| = (d-1)n^d - dn^{d-1} + 1$ by Lemma 8.2. Recall also the upper bound on the free energy per site of $\mathrm{U}(N)$ lattice gauge theory from Lemma 8.3. And also recall that the limit of the free energy per site of lattice Maxwell theory exists, as $n \to \infty$. Then we have

$$\begin{aligned}
&\limsup_{k\to\infty} \left(F_{\Lambda_n,g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta \right) \\
&\leq \limsup_{k\to\infty} \left(\frac{|E_n^1|}{n^d} \log C_N - N^2 F_{M,\Lambda_n} + \frac{C}{n^d} + C\delta_{n,\beta} + C\beta\delta_{n,\beta}^3 \right) \\
&= (d-1)\log C_N - N^2 \lim_{n\to\infty} F_{M,\Lambda_n},
\end{aligned}$$

as desired.

The remaining case is when n and β are sequences in A_3, since $A_2 \subset A_3$. By definition, there is a subsequence j_k such that $\beta_{j_k}\delta_{n_{j_k},\beta_{j_k}}^3 \geq M$ for all k. Note that

$$\mathbb{N} \setminus \{j_k : k \in \mathbb{N}\}$$

is another possibly finite index set. When finite, this means that j_k eventually describes the entire sequence (n_j, β_j). If infinite, then the sequence $(n_j, \beta_j)_{j \in \mathbb{N}\setminus\{j_k : k\in\mathbb{N}\}}$ is simply another sequence which falls into A_1, A_2 or A_3. Therefore, it suffices to suppose that n and β are such that $\beta\delta_{n,\beta}^3 \geq M$ along the whole sequence.

Since $\beta\delta_{n,\beta}^3 \geq M$,

$$n \geq \frac{M^{\frac{2}{3(d+1)}} \beta^{\frac{1}{3(d+1)}}}{(\log \beta)^{\frac{1}{(d+1)}}} \tag{87}$$

This allows us to define another parameter m with the decay conditions that n lacks. We proceed as follows.

Let us denote $\alpha = \frac{1}{3(d+1)}$ and let us take a small parameter ϵ satisfying

$$0 < \epsilon < \alpha.$$

Define the sequence $m : \mathbb{N} \to \mathbb{N}$ by

$$m_j = \lfloor \beta_j^\epsilon \rfloor,$$

where $\lfloor x \rfloor$ denotes the largest integer less than or equal to x.

Claim: We have

$$\delta_{m_j,\beta_j} \to 0 \text{ as } j \to \infty. \tag{88}$$

Proof of claim. We have

$$\begin{aligned}
\delta_{m_j,\beta_j} &= (\lfloor \beta_j^\epsilon \rfloor)^{(d+1)/2} \frac{(\log \beta_j)^{1/2}}{\beta_j^{1/2}} \\
&\leq (\beta_j^\alpha)^{(d+1)/2} \frac{(\log \beta_j)^{1/2}}{\beta_j^{1/2}} && \epsilon < \alpha, \text{ and assume } \beta_j \geq 1 \\
&= \frac{(\log \beta_j)^{1/2}}{\beta_j^{1/2-1/6}} \to 0 \text{ as } j \to \infty,
\end{aligned}$$

as desired. □

Claim: We have

$$\beta_j \delta_{m_j,\beta_j}^3 \to 0 \text{ as } j \to \infty. \tag{89}$$

Proof of claim. Note the strict inequality $\frac{\epsilon(d+1)\cdot 3}{2} < \frac{\alpha(d+1)\cdot 3)}{2} = \frac{1}{2}$. Then observe

$$\begin{aligned}\beta_j \delta_{m_j,\beta_j}^3 &= \left(\lfloor \beta_j^\epsilon \rfloor^{(d+1)/2}\right)^3 \frac{(\log \beta_j)^{3/2}}{\beta_j^{1/2}} \\ &\leq \beta_j^{\frac{\epsilon(d+1)\cdot 3}{2}} \frac{(\log \beta_j)^{3/2}}{\beta_j^{1/2}} \\ &= \frac{(\log \beta_j)^{3/2}}{\beta_j^{1/2-\frac{\epsilon(d+1)\cdot 3}{2}}} \to 0 \text{ as } j \to \infty,\end{aligned}$$

as desired. □

Claim: We have

$$m_j \log \beta_j = o(n_j) \text{ as } j \to \infty. \tag{90}$$

Proof of claim. By 87 we have

$$\lim_{j\to\infty} \frac{m_j \log \beta_j}{n_j} \leq \lim_{j\to\infty} \frac{\beta_j^\epsilon \log \beta_j (\log \beta_j)^{\frac{1}{d+1}}}{M^{2(d+1)/3} \beta_j^{\frac{1}{3(d+1)}}}.$$

Since $\epsilon < \frac{1}{3(d+1)}$, the denominator has a positive power of β_j, and thus the quotient goes to 0 as $j \to \infty$, as desired. □

In addition, the computation as in the proof of the previous claim without the term $\log \beta_j$ shows that

$$m_j = o(n_j) \text{ as } j \to \infty. \tag{91}$$

Since m is smaller than n, Lemma 8.4 shows that

$$F_{\Lambda_n,g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta \leq \left(1 - \frac{Cm}{n}\right) F_{\Lambda_m,g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta.$$

Then using the bound for $F_{\Lambda_m,g}$ given by Lemma 8.3, we have

$$\begin{aligned}&F_{\Lambda_n,g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta \\ &\leq \left(1 - \frac{Cm}{n}\right)\left(\frac{|E_m^1|}{m^d} \log C_N - \frac{|E_m^1|}{2m^d} N^2 \log \beta + N^2 F_{M,\Lambda_m} + \frac{C}{m^d} + C\delta_{m,\beta} + C\beta\delta_{m,\beta}^3,\right) + \frac{|E_n^1|}{2n^d} N^2 \log \beta \\ &\leq \left(1 - \frac{Cm}{n}\right)\left(\frac{|E_m^1|}{m^d} \log C_N + N^2 F_{M,\Lambda_m} + \frac{C}{m^d} + C\delta_{m,\beta} + C\beta\delta_{m,\beta}^3,\right) \\ &\quad + \left(\frac{|E_n^1|}{2n^d} - \frac{|E_m^1|}{2m^d}\left(1 - \frac{Cm}{n}\right)\right) N^2 \log \beta.\end{aligned}$$

There are four lines in the display above. In the third line, by 91, $(1 - \frac{Cm_j}{n_j}) \to 1$ as $j \to \infty$, and the term bounded by parenthesis after it contains the terms $\frac{C}{m^d} \to 0$, $C\delta_{m,\beta} \to 0$, $C\beta\delta_{m,\beta}^3 \to 0$, and (by Lemma 8.2)

$$\frac{|E_m^1|}{m^d} \log C_N = (d-1)\log C_N - \frac{d \log C_N}{m} + \frac{\log C_N}{m^d} \to (d-1)\log C_N$$

as $j \to \infty$ (recalling that $m : j \mapsto m_j$ and $\beta : j \mapsto \beta_j$). Therefore, in the $j \to \infty$ limit, the third line is

$$(d-1)\log C_N + N^2 \lim_{m\to\infty} F_{M,\Lambda_m}.$$

On the other hand, expanding the fourth line yields

$$\left(\frac{1}{2n^d} - \frac{1}{2m^d} + \frac{d}{2m} - \frac{d}{2n} + C(d-1)\frac{m}{n} - \frac{Cd}{n} + \frac{C}{nm^{d-1}}\right) N^2 \log \beta. \tag{92}$$

In 92, we recall that n grows faster than a small power of β, see 90 and 87, so $\frac{N^2 \log \beta}{2n^d} \to 0$, $\frac{dN^2 \log \beta}{2m} \to 0$, $-\frac{CdN^2 \log \beta}{n} \to 0$ in the limit. Furthermore, m is the floor function of a small power of β, so in the limit behaves like a small power of β, and thus $\frac{N^2 \log \beta}{2m^d} \to 0$, $\frac{dN^2 \log \beta}{2m} \to 0$, $\frac{CN^2 \log \beta}{nm^{d-1}} \to 0$. The last term is $C(d-1)N^2 \frac{m \log \beta}{n}$, which goes to zero by 90.

Since $N^2 \lim_{m\to\infty} F_{M,\Lambda_m} = N^2 \lim_{n\to\infty} F_{M,\Lambda_n}$, we have shown that

$$\limsup_{j\to\infty} F_{\Lambda_{n_j},g_j} + \frac{|E^1_{n_j}|}{2n_j^d} N^2 \log \beta_j \leq (d-1)\log C_N + N^2 \lim_{n\to\infty} F_{M,\Lambda_n}$$

for sequences $N : j \mapsto n_j$, $\beta : j \mapsto \beta_j = \frac{1}{g_j^2}$ in the set A_3 (recall 85).

Since we proved the upper bound also for sequences in A_1, we have completed the proof of this lemma. □

8.2 Lower Bound

Lemma 8.6 (Lemma 17.5 [4]). *For any m and n,*

$$|F_{\Lambda_m,g} - F_{\Lambda_n,g}| \leq \frac{C\beta}{m} + \frac{C\beta}{n},$$

where C depends only on N and d.

Proof. The strategy is to show that the small box free energy $F_{\Lambda_n,g}$ is $\frac{C\beta}{n}$-close to the large box free energy $F_{\Lambda_{nm},g}$, then to appeal to symmetry in the m and n to also get that $F_{\Lambda_m,g}$ is $\frac{C\beta}{m}$-close to $F_{\Lambda_{nm},g}$. Applying the triangle inequality then finishes the proof.

Fix n and m arbitrary positive integers. Let $l = nm$. Note that Λ_l is a disjoint union of m^d translates of Λ_n. In the notation prior to to Lemma 8.4 given by equations 77, 78, we have

$$B'_a = \{(x_1, \cdots, x_d) \in \Lambda_l : a_j \cdot n \leq x_j \leq (a_j+1)\cdot n - 1 \, \forall j \in [d]\},$$

whenever $a = (a_1, \cdots, a_d) \in \{0, \cdots, m-1\}^d$, and $\mathcal{B}' = \{B'_a : a \in \{0, \cdots, m-1\}^d\}$ is the collection of disjoint translates, and

$$\Lambda_l = \Lambda_{mn} = \bigcup_{B'_a \in \mathcal{B}} B'_a.$$

Furthermore, $B'_{(0,\cdots,0)} = \Lambda_n$.

There is a bound, depending only on N, on the contribution of each plaquette to the Wilson action. Given a plaquette p, recall that the plaquette variable U_p is an element of $\mathrm{U}(N)$. Therefore its eigenvalues (possibly repeated) are $e^{\mathrm{i}\theta_j}$ for $j = 1, \cdots, N$, where θ_j can be taken in $[-\pi, \pi)$. Note that $-1 \leq \operatorname{Re} e^{\mathrm{i}\theta_j} \leq 1$ for each $j \in [N]$. Therefore,

$$\begin{aligned}\operatorname{Re}(\operatorname{Tr}(I - U_p)) &= \operatorname{Re}(\operatorname{Tr}(I)) - \operatorname{Re}(\operatorname{Tr}(U_p)) \\ &= N - \sum_{j=1}^{N} \operatorname{Re} e^{i\theta_j} \in [0, 2N].\end{aligned}$$

Next, note that

$$\left| \mathcal{P}(\Lambda_l) \setminus \bigcup_{B'_a \in \mathcal{B}'} \mathcal{P}(B'_a) \right| \leq C m^d n^{d-1},$$

for some C depending only on d. Indeed, each translate is a copy of Λ_n which has Cn^{d-1} boundary vertices (here we mean the geometric boundary, not the axial gauge boundary) because each face of Λ_n is a box of width n of one lower dimension, and there are C faces, depending on d. To each of these vertices, there are at most C edges. Furthermore, each plaquette which does not appear in any of the plaquette subsets $\mathcal{P}(B'_a)$ must have an edge appearing in one of the boxes B'_a, and since there are at most C plaquettes per edge, then we see that there are at most $Cm^d n^{d-1}$ plaquettes not appearing in any of the translates.

Therefore, for any $U \in \mathrm{U}(N)^{E_l}$, we have

$$\begin{aligned} 0 &\leq \sum_{p \in \mathcal{P}(\Lambda_l)} \mathrm{Re}(\mathrm{Tr}(I - U_p)) - \sum_{B'_a \in \mathcal{B}'} \sum_{p \in \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p)) \\ &= \sum_{p \in \mathcal{P}(\Lambda_l) \setminus \bigcup_{B'_a \in \mathcal{B}'} \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p)) \\ &\leq 2N \cdot C m^d n^{d-1} = C m^d n^{d-1}. \end{aligned} \tag{93}$$

The rest of the proof has computations which are similar to those in the proof of Lemma 8.4. The first line in the above display shows that

$$-\beta \sum_{B'_a \in \mathcal{B}'} \sum_{p \in \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p)) \geq -\beta \sum_{p \in \mathcal{P}(\Lambda_l)} \mathrm{Re}(\mathrm{Tr}(I - U_p)).$$

Then taking the exponential and integrating against $\prod_{e \in E_l} d\sigma(U_e)$, we get

$$\int_{\mathrm{U}(N)^{E_l}} e^{-\beta \sum_{B'_a \in \mathcal{B}'} \sum_{p \in \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p))} \prod_{e \in E_l} d\sigma(U_e) \geq \int_{\mathrm{U}(N)^{E_l}} e^{-\beta \sum_{p \in \mathcal{P}(\Lambda_l)} \mathrm{Re}(\mathrm{Tr}(I - U_p))} \prod_{e \in E_l} d\sigma(U_e),$$

and we notice that the right hand side is equal to $Z_{\Lambda_l, g}$, the partition function for the box Λ_l. By integrating out edges which don't appear in any plaquettes in elements of $\mathcal{B}'$, the left hand side breaks up into the product integral

$$\prod_{B'_a \in \mathcal{B}'} \int_{\mathrm{U}(N)^{E_{B'_a}}} e^{-\beta \sum_{p \in \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p))} \prod_{e \in E_{B'_a}} d\sigma(U_e),$$

which we recognize is equal to $Z_{\Lambda_n, g}^{m^d}$, because $Z_{B'_a, g} = Z_{\Lambda_n, g}$ for any $a \in \{0, \cdots, m-1\}^d$, and $|\mathcal{B}'| = m^d$. Therefore,

$$Z_{\Lambda_n, g}^{m^d} \geq Z_{\Lambda_l, g}.$$

The more meaningful use of the inqualities in 93 is to obtain that

$$-\beta \sum_{p \in \mathcal{P}(\Lambda_l)} \mathrm{Re}\,\mathrm{Tr}(I - U_p) \geq -C\beta m^d n^{d-1} - \beta \sum_{B'_a \in \mathcal{B}'} \sum_{p \in \mathcal{P}(B'_a)} \mathrm{Re}(\mathrm{Tr}(I - U_p)),$$

and then to take the exponential and integrate against $\prod_{e \in E_l} d\sigma(U_e)$ to obtain

$$Z_{\Lambda_l, g} \geq e^{-C\beta m^d n^{d-1}} Z_{\Lambda_n, g}^{m^d}.$$

Now we have

$$Z_{\Lambda_n, g}^{m^d} \geq Z_{\Lambda_l, g} \geq e^{-C\beta m^d n^{d-1}} Z_{\Lambda_n, g}^{m^d}.$$

Taking the logarithm is no problem by Theorem 3.3) (or even simpler, by the $2N$ bound on the plaquette variables above), and then dividing by $l^d = m^d n^d$ after that yields

$$\frac{\log Z_{\Lambda_n, g}}{n^d} \geq \frac{\log Z_{\Lambda_l, g}}{l^d} \geq -\frac{C\beta}{n} + \frac{\log Z_{\Lambda_n, g}}{n^d}.$$

This shows that

$$|F_{\Lambda_n, g} - F_{\Lambda_l, g}| \leq \frac{C\beta}{n},$$

and by the remark at the beginning of the proof, we just switch the roles of n and m to get the analagous inequality, and then apply the triangle inequality to finish the proof. □

Lemma 8.7 (Lemma 17.6 [4]). *If β is large enough (depending on N and d) and $n \leq \beta^2$, then*

$$F_{\Lambda_n, g} \geq \frac{|E_n^1|}{n^d} \log C_N - \frac{|E_n^1|}{2n^d} N^2 \log \beta + N^2 F_{M, \Lambda_n} - \frac{C \log \beta}{\beta^{1/(10d+20)}} - \frac{C \log n}{\sqrt{n}},$$

where C is a positive constant that depends only on N and d.

Proof. Here is how we will proceed with the proof. First we will introduce a parameter $r \leq r_0$, where r_0 is from Theorem 5.2. Then we will apply our machinary to the integral which defines the partition function: (1) switch to the axial gauge (Lemma 4.8), (2) switch from Lie Group to Lie Algebra (Theorem 5.2), (3) exchange Wilson action for Maxwell action (Theorem 7.4), (4) make a change of variable in $\mathrm{H}(N)^{E_n^1} \cong \mathbb{R}^{N^2|E_n^1|}$ to get β out of the Maxwell action, (5) break up the Hermitian Maxwell configuration into N^2 independent scalar Maxwell configurations, (6) apply known bounds from lattice Maxwell theory (Theorem 6.7). At step (6), an ambiguity will arise in the case that n grows too slowly. A clever trick resolves the issue, and then we can proceed to step (7) of taking the logarithm and dividing by n^d, and seeing which terms come out. Hopefully they are the same terms as in the upper bound. At the end, we will retroactively decide how r should depend on β and n, based on which terms we want to vanish.

Let $r < r_0$ be fixed. We have by Lemma 4.8 and then by non-negativity of the integrand that

$$\begin{aligned} Z_{\Lambda_n,g} &= \int_{\mathrm{U}(N)^{E_n^1} \times \{I\}^{E_n^0}} e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n^1} d\sigma(U_e) \prod_{e \in E_n^0} d\delta_I(U_e) \\ &\geq \int_{b_{\mathrm{U}(N)}(0,2r)^{E_n^1} \times \{I\}^{E_n^0}} e^{-\beta S_{\Lambda_n}(U)} \prod_{e \in E_n^1} d\sigma(U_e) \prod_{e \in E_n^0} d\delta_I(U_e). \end{aligned} \tag{94}$$

Then switching from Lie group to Lie algebra, we have by Theorem 5.2 (iterated $|E_n^1|$ times) that

$$Z_{\Lambda_n,g} \geq (2 - e^{6r})^{N^2|E_n^1|} C_N^{|E_n^1|} \int_{b_{\mathrm{H}(N)}(0,r)^{E_n^1} \times \{0\}^{E_n^0}} e^{-\beta S_{\Lambda_n}(e^{iH})} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e). \tag{95}$$

Note that there is a constant C depending only on N and d such that $2 - e^{6r} \geq e^{-Cr}$ for all $0 \leq r \leq r_0$. Also, r_0, and hence r, is far less than 1, so we can switch from Wilson action to Maxwell action using Theorem 7.4. It implies that

$$-\beta S_{\Lambda_n}(e^{iH}) \geq -\beta C r^3 n^d - \frac{1}{2} \beta M_n^{\mathrm{H}}(H)$$

for all $H \in b_{\mathrm{H}(N)}(0,r)^{E_n}$, and in particular for $H \in b_{\mathrm{H}(N)}(0,r)^{E_n^1} \times \{0\}^{E_n^0}$. Then we get by pulling the constant $e^{-\beta C r^3 n^d}$ outside of the integral that

$$Z_{\Lambda_n,g} \geq e^{-Cr|E_n^1|} C_N^{|E_n^1|} e^{-\beta C r^3 n^d} \int_{b_{\mathrm{H}(N)}(0,r)^{E_n^1} \times \{0\}^{E_n^0}} e^{-\frac{1}{2}\beta M_n^{\mathrm{H}}(H)} \prod_{e \in E_n^1} d\lambda(H_e) \prod_{e \in E_n^0} d\delta_0(H_e). \tag{96}$$

Recall from 68 that $\beta M_n^{\mathrm{H}}(H) = M_n^{\mathrm{H}}(\sqrt{\beta} H)$. Now we make the change of variable $G = \sqrt{\beta} H$, which means that on each edge $e \in E_n^1$, we incur a Jacobian of $\left(\frac{1}{\sqrt{\beta}}\right)^{N^2}$ (recalling that the Lebesgue measure on $\mathrm{H}(N)$ is equal to the pushforward of N^2 dimensional Lebesgue measure by an isometry), and the region of integration for each edge in E_n^1 changes from $b_{\mathrm{H}(N)}(0,r)$ to $b_{\mathrm{H}(N)}(0,\sqrt{\beta} r)$. On the axial edges, the change of variables can be ignored because scaling the zero matrix by $\sqrt{\beta}$ yields the zero matrix. This yields

$$\begin{aligned} Z_{\Lambda_n,g} \geq\ & e^{-Cr|E_n^1|} C_N^{|E_n^1|} e^{-\beta C r^3 n^d} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2|E_n^1|} \\ & \int_{b_{\mathrm{H}(N)}(0,\sqrt{\beta} r)^{E_n^1} \times \{0\}^{E_n^0}} e^{-\frac{1}{2} M_n^{\mathrm{H}}(G)} \prod_{e \in E_n^1} d\lambda(G_e) \prod_{e \in E_n^0} d\delta_0(G_e). \end{aligned} \tag{97}$$

Now we try to switch to N^2 independent Maxwell configurations, which is different from how Chatterjee proceeded. Let

$$b^{l_2}_{\mathbb{R}^{N^2}}(0,c) = \{x \in \mathbb{R}^{N^2} : \sqrt{x_1^2 + \cdots + x_{N^2}^2} < c\}$$

denote the Euclidean ball of radius c in $\mathbb{R}^{N^2}$. Recalling the product isometry $\Psi_H^{E_n}$, see 60, we have

$$\begin{aligned} & \int_{b_{\mathrm{H}(N)}(0,\sqrt{\beta} r)^{E_n^1} \times \{0\}^{E_n^0}} e^{-\frac{1}{2} M_n^{\mathrm{H}}(G)} \prod_{e \in E_n^1} d\lambda(G_e) \prod_{e \in E_n^0} d\delta_0(G_e) \\ &= \int_{b^{l_2}_{\mathbb{R}^{N^2}}(0,\sqrt{\beta} r)^{E_n^1} \times \{0\}^{E_n^0}} e^{-\frac{1}{2} M_n^{\mathrm{H}}(\Psi_H^{E_n}(x))} \prod_{e \in E_n^1} dl_{N^2}(x_e) \prod_{e \in E_n^0} d\delta_0(x_e), \end{aligned} \tag{98}$$

where $x = (x_e)_{e\in E_n} \in (\mathbb{R}^{N^2})^{E_n}$. Recall 61 and 62, which say that the Hermitian Maxwell action splits up into N^2 scalar Maxwell actions, and that the product measure can be rearranged appropriately. We just need to find a suitable product subset of the Euclidean ball $b^{l_2}_{\mathbb{R}^{N^2}}(0,\sqrt{\beta}r)$, on each edge. Let

$$b^{l_\infty}_{\mathbb{R}^{N^2}}(0,c) = \{x \in \mathbb{R}^{N^2} : \max_j |x_j| \le c\}.$$

Stemming from the inequality $\sqrt{N^2} l_\infty \ge l_2$, we have

$$\prod_{j\in[N^2]} b_{\mathbb{R}}\left(0, \frac{\sqrt{\beta}r}{\sqrt{N^2}}\right) = b^{l_\infty}_{\mathbb{R}^{N^2}}\left(0, \frac{\sqrt{\beta}r}{\sqrt{N^2}}\right) \subset b^{l_2}_{\mathbb{R}^{N^2}}\left(0, \sqrt{\beta}r\right),$$

where $b_{\mathbb{R}}(0,c) = \{x \in \mathbb{R} : |x| \le c\}$. This is the product set we will use to switch to N^2 scalar configurations using Fubini's theorem. We have, first integrating over a smaller set, and then by the remarks above,

$$\begin{aligned}
&\int_{b^{l_2}_{\mathbb{R}^{N^2}}(0,\sqrt{\beta}r)^{E_n^1}\times\{0\}^{E_n^0}} e^{-\frac{1}{2}M_n^{\mathrm{H}}(\Psi_H^{E_n}(x))} \prod_{e\in E_n^1} dl_{N^2}(x_e) \prod_{e\in E_n^0} d\delta_0(x_e) \\
&\ge \int_{\left(\prod_{j\in[N^2]} b_{\mathbb{R}}\left(0,\frac{\sqrt{\beta}r}{N}\right)\right)^{E_n^1}\times\{0\}^{E_n^0}} e^{-\frac{1}{2}M_n^{\mathrm{H}}(\Psi_H^{E_n}(x))} \prod_{e\in E_n^1} dl_{N^2}(x_e) \prod_{e\in E_n^0} d\delta_0(x_e) \\
&= \prod_{j\in[N^2]} \int_{b_{\mathbb{R}}\left(0,\frac{\sqrt{\beta}r}{N}\right)^{E_n^1}\times\{0\}^{E_n^0}} e^{-\frac{1}{2}M_n^{(\mathbb{R})}(x^{(j)})} \prod_{e\in E_n^1} dl_1(x_e^{(j)}) \prod_{e\in E_n^0} d\delta_0(x_e^{(j)}) \\
&= \prod_{j\in[N^2]} Z_{M,\Lambda_n} \tau_n\left(\{t \in \mathbb{R}^{E_n^1} : |t_e| \le \frac{\sqrt{\beta}r}{N} \text{ for all } e \in E_n^1\}\right),
\end{aligned} \tag{99}$$

where the last step is just recalling the Gaussian measure τ_n, whose normalization constant is Z_{M,Λ_n}. The next step is to apply Theorem 6.7, so with C_1, C_2 as in that theorem, we need to introduce an integer m satisfying $2 \le m \le \sqrt{n}$ and $C_1 m^{d+2} \le \frac{\sqrt{\beta}r}{N}$. Then we obtain

$$\begin{aligned}
&\tau_n\left(\{t \in \mathbb{R}^{E_n^1} : |t_e| \le \frac{\sqrt{\beta}r}{N} \text{ for all } e \in E_n^1\}\right) \\
&\ge \tau_n\left(\{t \in \mathbb{R}^{E_n^1} : |t_e| \le C_1 m^{d+2} \text{ for all } e \in E_n^1\}\right) \\
&\ge \exp\left(-C_2 \frac{n^d \log n}{m}\right)
\end{aligned} \tag{100}$$

We don't want m in the final result, and we can't lower bound the exponential by replacing m with its upper bound, so let m_L be a lower bound on m, in terms of n and β, to be specified later. Then we have

$$\exp\left(-C_2 \frac{n^d \log n}{m}\right) \ge \exp\left(-C_2 \frac{n^d \log n}{m_L}\right).$$

Combining 97, 99, and 100, we now have

$$Z_{\Lambda_n,g} \ge e^{-Cr|E_n^1|} C_N^{|E_n^1|} e^{-\beta C r^3 n^d} \left(\frac{1}{\sqrt{\beta}}\right)^{N^2|E_n^1|} \exp\left(-C_2 \frac{n^d \log n}{m_L}\right)^{N^2}. \tag{101}$$

Taking the logarithm and dividing by n^d yields

$$F_{\Lambda_n,g} \ge \frac{-Cr|E_n^1|}{n^d} + \frac{|E_n^1|}{n^d} \log C_N - C\beta r^3 + \frac{N^2|E_n^1|}{n^d} \log\left(\frac{1}{\sqrt{\beta}}\right) + N^2 F_{M,\Lambda_n} - \frac{C \log n}{m_L}. \tag{102}$$

Some of the terms above are already what we saw in the upper bound. So it remains to write r and the lower bound on m in terms of n and β such that the remaining terms will vanish when the limit is taken. Using Lemma 8.2, $|E_n^1| = (d-1)n^d - dn^{d-1} + 1$, the first extraneous term is $\frac{-Cr|E_n^1|}{n^d} = -Cr + \frac{Cr}{n} - \frac{Cr}{n^d}$. Therefore r must go to 0 as n and β go to infinity. The next extraneous term is $C\beta r^3$. In order for this term to vanish, r must decay like β^{-c} where $c > \frac{1}{3}$. For instance, n could grow arbitrarily slow compared to β, so we cannot set r in terms of n here. The last extraneous term is $\frac{C \log n}{m_L}$. A simple way to pick a

lower bound on m is to divide the upper bound on m by 2. But the upper bound depends on the growth rate of n, because if n grows slow enough, then $C_1 m^{d+2} \leq \frac{\sqrt{\beta} r}{N}$ might not imply that $m \leq \sqrt{n}$. On the other hand, note that $C_1 m^{d+2} \leq \min\{\frac{\sqrt{\beta} r}{N}, C_1 n^{(d+2)/2}\} =: \eta$ implies that $m \leq \sqrt{n}$ and $C_1 m^{d+2} \leq \frac{\sqrt{\beta} r}{N}$. Setting $m_L = (1/2)\left(\frac{\eta}{C_1}\right)^{1/(d+2)}$, which is one half of the upper bound on m, the aforementioned clever trick (from the first paragraph of this proof) is that

$$-\frac{C \log n}{m_L} = -\frac{C \log n}{(1/2)\left(\frac{\eta}{C_1}\right)^{1/(d+2)}} \geq -\frac{C \log n}{(1/2)\left(\frac{\frac{\sqrt{\beta} r}{N}}{C_1}\right)^{1/(d+2)}} - \frac{C \log n}{(1/2)\left(\frac{C_1 n^{(d+2)/2}}{C_1}\right)^{1/(d+2)}}, \tag{103}$$

which says that replacing η in the second term by its two possibilities produces a lower bound. Rewriting 103 to absorb the constants, we have

$$-\frac{C \log n}{m_L} \geq -\frac{C \log n}{(\sqrt{\beta} r)^{1/(d+2)}} - \frac{C \log n}{\sqrt{n}}, \tag{104}$$

which indicates that if r decays like β^{-c} for $c \geq \frac{1}{2}$, then there would be an unwanted divergence, i.e., the lower bound of this lemma would be useless. Combining this with the earlier observation about r, we see that $r = \beta^{-c}$ for some $c \in (\frac{1}{3}, \frac{1}{2})$. Chattejee takes $c = 2/5$, so we let that be the case, which implies $\sqrt{\beta} r = \beta^{1/10}$. This is a convenient choice because it means that two of the extraneous terms are dominated by the third:

$$\begin{aligned} &-\frac{C \log n}{(\sqrt{\beta} r)^{1/(d+2)}} - \frac{C r |E_n^1|}{n^d} - C\beta r^3 \\ &= -\frac{C \log n}{\beta^{1/(10d+20)}} - \frac{C |E_n^1|}{\beta^{2/5} n^d} - C\beta^{-(1/5)} \geq -\frac{C_3 \log n}{\beta^{1/(10d+20)}}, \end{aligned} \tag{105}$$

for some constant C_3 depending only on N and d. This is one place where we use that β is large enough depending only on N and d.

To be sure, n must also not grow exponentially compared to β, or else $\frac{\log n}{\beta^\epsilon}$ for small ϵ could diverge in the limit. But the next Lemma 8.8, which is an application of the previous Lemma 8.6, shows that the case $n \geq \beta^2$ is easy, so we can just as well assume that $n \leq \beta^2$, as is seen in the statment of this present lemma.

Combining displays 104, 102, and 105, we obtain the formula seen in the statement of the present lemma, as desired. □

Lemma 8.8 (Lemma 17.7 [4])**.**

$$\liminf_{\substack{n\to\infty \\ g\to 0}} \left(F_{\Lambda_n, g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta \right) \geq (d-1) \log C_N + N^2 \lim_{n\to\infty} F_{M, \Lambda_n}.$$

Proof. If $n : j \mapsto n_j$ and $\beta : j \mapsto \beta_j$ go to infinity such that $n_j \leq \beta_j^2$ for all j sufficiently large, then we are done, by the previous lemma and by recalling that the free energy per site of lattice Maxwell theory has a finite limit.

Otherwise, there is a subsequence j_k such that $n_{j_k} > \beta_{j_k}^2$ for all k. Let m_{j_k} be the integer part of β_{j_k}. We have, adding 0 and suppressing j_k from the notation,

$$\begin{aligned} &F_{\Lambda_n, g} + \frac{|E_n^1|}{2n^d} N^2 \log \beta \\ &= (F_{\Lambda_n, g} - F_{\Lambda_m, g}) + \left(F_{\Lambda_m, g} + \frac{|E_m^1|}{2m^d} N^2 \log \beta \right) + \left(\frac{|E_n^1|}{2n^d} N^2 \log \beta - \frac{|E_m^1|}{2m^d} N^2 \log \beta \right). \end{aligned}$$

Since the middle term, by the previous lemma, already satisfies the conclusion of the present lemma in the subsequential limit, then we get the desired result if a lower bound of $(F_{\Lambda_n, g} - F_{\Lambda_m, g})$ vanishes and if a lower bound of $\left(\frac{|E_n^1|}{2n^d} N^2 \log \beta - \frac{|E_m^1|}{2m^d} N^2 \log \beta \right)$ vanishes in the subsequential limit. By Lemma 8.6, we have

$$F_{\Lambda_n, g} - F_{\Lambda_m, g} \geq -\frac{C\beta}{n} - \frac{C\beta}{m}.$$

Since n grows faster than β^2, we have $-\frac{C\beta}{n} \to 0$. Since m grows like β^2, we have $-\frac{C\beta}{m} \to 0$. By Lemma 8.2, the other difference expands to

$$\frac{dN^2 \log \beta}{m} - \frac{d \log \beta}{n} - \frac{\log \beta}{m^d} + \frac{\log \beta}{n^d}.$$

These terms all vanish because both n and m are polynomial in β. □

9 Appendix

9.1 $\mathrm{U}(N)$ as topological group, and Haar Measure and Integration on $\mathrm{U}(N)$

Let N be a positive integer. Given a matrix $X \in \mathbb{C}^{N\times N}$, the conjugate transpose of X is denoted by X^*. Let $\mathrm{U}(N) = \{U \in \mathbb{C}^{N\times N} : U^*U = I\}$ be the complex unitary matrices with N^2 entries, where I denotes the $N \times N$ identity matrix. For example, $\mathrm{U}(1)$ is the unit circle in the complex plane. To define a probability measure on $\mathrm{U}(N)$, we need to speak of a sigma algebra. For that, we will equip $\mathrm{U}(N)$ with a topology induced by a metric, and then take the Borel sigma algebra. We did not say that its topology is induced by a *norm* because $\mathrm{U}(N)$ is not an $\mathbb{R}$ or $\mathbb{C}$ vector space as it is not closed under scalar multiplication nor matrix addition.

The Hilbert-Schmidt norm of X is

$$\|X\| = \mathrm{Tr}(X^*X)^{1/2} = \left(\sum_{1\le i,j\le N} |X_{ij}|^2 \right)^{1/2}, \tag{106}$$

where $|X_{ij}| = \sqrt{(\mathrm{Re}\, X_{ij})^2 + (\mathrm{Im}\, X_{ij})^2}$ denotes the complex modulus of the complex number X_{ij} appearing in row i and column j of X. Depending on context and for convenience, we will also use upper indices to denote matrix entries. On the linear algebraic side, it can be verified that $(\mathbb{C}^{N\times N}, \|\cdot\|)$ is a normed vector space over $\mathbb{R}$ or $\mathbb{C}$. Note that the Hilbert-Schmidt norm is usually not an operator norm because

$$\|I\|_{op} = \sup\{\ \|Iv\| : v \in \mathbb{C}^N, \|v\| = 1\} = 1,$$

whereas we have that the Hilbert-Schmidt norm of I is $\sqrt{N}$.

The function $d_{\mathbb{C}^{N\times N}} : (\mathbb{C}^{N\times N})^2 \to [0,\infty)$ given by $d_{\mathbb{C}^{N\times N}}(A,B) = \|A - B\|$ defines a metric on the set $\mathbb{C}^{N\times N}$. The metric topology $\tau_{\mathbb{C}^{N\times N}}$ on $\mathbb{C}^{N\times N}$ is the collection of subsets $C \subset \mathbb{C}^{N\times N}$ such that for any matrix $A \in C$, there is an $\epsilon > 0$ such that

$$\{B \in \mathbb{C}^{N\times N} : \|B - A\| < \epsilon\} \subset C.$$

Meanwhile, the function $d_{\mathrm{U}(N)} : \mathrm{U}(N)^2 \to [0,\infty)$ given by $d_{\mathrm{U}(N)}(U,V) = \|U - V\|$ is the restriction of $d_{\mathbb{C}^{N\times N}}$ to the subset $\mathrm{U}(N)$. As such, $d_{\mathrm{U}(N)}$ defines a metric on the set $\mathrm{U}(N)$. Furthermore, the metric topology on $\mathrm{U}(N)$, which we denote by $\tau_{\mathrm{U}(N)}$, is equal to the subspace topology $\{\mathrm{U}(N) \cap C : C \in \tau_{\mathbb{C}^{N\times N}}\}$.

The Borel sigma algebra on $\mathrm{U}(N)$, which we denote by $\mathcal{B}(\mathrm{U}(N))$ (or $\mathcal{B}_{\mathrm{U}(N)}$ to reduce the number of parentheses in the notation), is equal to the smallest sigma algebra containing $\tau_{\mathrm{U}(N)}$ as a subcollection:

$$\mathcal{B}_{\mathrm{U}(N)} = \bigcap_{\substack{\gamma \supset \tau_{\mathrm{U}(N)} \\ \gamma \text{ sigma algebra}}} \gamma.$$

Then $(\mathrm{U}(N), \mathcal{B}_{\mathrm{U}(N)})$ becomes a measurable space. However, for the sake of defining Haar measure on $\mathrm{U}(N)$, we need further background on $\mathrm{U}(N)$. We need to exhibit $\mathrm{U}(N)$ as a compact topological group.

Note that $\mathrm{U}(N)$ is a group when equipped with matrix multiplication. We will show that this group operation is continuous as a map $\mathrm{U}(N)^2 \to \mathrm{U}(N)$, where $\mathrm{U}(N)^2$ is equipped with the product topology $\tau^{\otimes 2}_{\mathrm{U}(N)}$. The aforementioned product topology is induced by the metric $d_{\mathrm{U}(N)^2} : \mathrm{U}(N)^2 \times \mathrm{U}(N)^2 \to [0,\infty)$ given by

$$d_{\mathrm{U}(N)^2}((U_1,U_2),(V_1,V_2)) = \max_{i=1,2}(d_{\mathrm{U}(N)}(U_i,V_i))$$

for all $(U_1,U_2),(V_1,V_2) \in \mathrm{U}(N)^2$.

The following lemma appears as Lemma 7.3 in [4] and it shows that the isometry property of $\mathrm{U}(N)$ extends from $\mathbb{C}^N$ to $\mathbb{C}^{N\times N}$.

Lemma 9.1. *For any $U \in \mathrm{U}(N)$ and $M \in \mathbb{C}^{N\times N}$, $\|UM\| = \|M\|$.*

Lemma 9.2. *Matrix multiplication is continuous on $\mathrm{U}(N)$.*

Proof. In this proof we will denote the matrix multiplication map by $m : \mathrm{U}(N)^2 \to \mathrm{U}(N)$. Fix $(U_1,U_2) \in \mathrm{U}(N)^2$. It suffices to prove that m is continuous at (U_1,U_2). Fix $\epsilon > 0$. Choose $\delta = \epsilon/2$. Fix

$(V_1, V_2) \in \mathrm{U}(N)^2$ such that $\max_{i=1,2}(d_{\mathrm{U}(N)}(U_i, V_i)) < \delta$. Then

$$\begin{aligned} d_{\mathrm{U}(N)}(m(U_1,U_2), m(V_1,V_2)) &= \|U_1U_2 - V_1V_2\| \\ &= \|U_1U_2 + (-V_1U_2 + V_1U_2) - V_1V_2\| \\ &= \|(U_1 - V_1)U_2 + V_1(U_2 - V_2)\| \\ &\leq \|(U_1 - V_1)U_2\| + \|V_1(U_2 - V_2)\| \end{aligned}$$

where the last inequality follows from the triangle inequality. Then we apply Lemma 9.1 taking into account that U_2 and V_1 are unitary matrices. This gives

$$\begin{aligned} d_{\mathrm{U}(N)}(m(U_1,U_2), m(V_1,V_2)) &\leq \|U_1 - V_1\| + \|U_2 - V_2\| \\ &\leq 2 \max_{i=1,2}(d_{\mathrm{U}(N)}(U_i, V_i)) \\ &\leq 2\delta = \epsilon. \end{aligned}$$

Thus, m is continuous at (U_1, U_2), which concludes the proof. □

We will use the following formula to prove that matrix inversion on $\mathrm{U}(N)$ is continuous:

$$\|X\|^2 = \mathrm{Tr}(X^*X),$$

where $\mathrm{Tr}\, X$ is the sum of the diagonal entries of X.

Lemma 9.3. *The map $U \mapsto U^{-1}$ on $\mathrm{U}(N)$ is continuous.*

Proof. In this proof, $f : \mathrm{U}(N) \to \mathrm{U}(N)$ will denote matrix inversion. It suffices to prove that f is continuous at $U \in \mathrm{U}(N)$. Let $\epsilon > 0$. Choose $\delta = \epsilon$. Take $V \in \mathrm{U}(N)$ such that $d_{\mathrm{U}(N)}(U, V) \leq \delta$. Before executing the final procedure, notice that by the identity $\mathrm{Tr}(XY) = \mathrm{Tr}(YX)$, we have

$$\|X^*\|^2 = \mathrm{Tr}((X^*)^*X^*) = \mathrm{Tr}(XX^*) = \mathrm{Tr}(X^*X) = \|X\|^2$$

whenever $X \in \mathbb{C}^{N\times N}$. Then

$$\begin{aligned} d_{\mathrm{U}(N)}(f(U), f(V)) &= \|U^{-1} - V^{-1}\| = \|U^* - V^*\| \\ &= \|U - V\| \\ &= d_{\mathrm{U}(N)}(U, V) = \delta = \epsilon, \end{aligned}$$

which shows that matrix inversion is continuous at $U \in \mathrm{U}(N)$. □

Lemmas 9.2 and 9.3 are the main ingredients for exhibiting $\mathrm{U}(N)$ as a topological group. Next we show that $\mathrm{U}(N)$ is compact. It suffices to show that $\mathrm{U}(N)$ is a closed and bounded subset of $\mathbb{C}^{N\times N}$. That $\mathrm{U}(N)$ is bounded follows from the computation

$$\|U\|^2 = \mathrm{Tr}(U^*U) = \mathrm{Tr}(I) = N$$

for $U \in \mathrm{U}(N)$. To show that $\mathrm{U}(N)$ is closed, we notice that it is the 0-level set of the map $M \mapsto M^*M - I$ on $\mathbb{C}^{N\times N}$. That this map is continuous is the direction of the following lemmas.

Lemma 9.4. *If $X, Y \in \mathbb{C}^{N\times N}$, then $\|XY\| \leq \|X\|\|Y\|$.*

Proof. Let $x_1, \cdots, x_N$ denote the rows of X and let $y_1, \cdots, y_N$ denote the columns of Y. For a vector $x \in \mathbb{C}^N$, we note that $\|x\|^2 = \sum_{i=1}^N |x^i|^2$ is the sum of the real and imaginary parts of the components of x. By (106), we have

$$\|X\|^2 = \|x_1\|^2 + \cdots + \|x_N\|^2, \qquad \|Y\|^2 = \|y_1\|^2 + \cdots + \|y_N\|^2.$$

On the other hand, if $a \cdot b = \sum_{i=1}^N a^i b^i$ denotes the dot product (not the complex inner product) of two vectors $a, b \in \mathbb{C}^N$, then

$$\|XY\|^2 = \sum_{1\leq i,j\leq N} |x_i \cdot y_j|^2.$$

It suffices to prove that $|x_i \cdot y_j|^2 \leq \|x_i\|^2 \|y_j\|^2$ for all $1 \leq i, j \leq N$. Fix $1 \leq i, j \leq N$. By the triangle inequality for the complex modulus, we have

$$|x_i \cdot y_j| \leq \sum_{k=1}^{N} |x_i^k||y_j^k|.$$

The Cauchy-Schwarz inequality $a \cdot b \leq \|a\|_{\mathbb{R}^N} \|b\|_{\mathbb{R}^N}$ on $\mathbb{R}^N$ applied to the vectors $a = (|x_i^k|)_{k=1}^N$ and $b = (|y_i^k|)_{k=1}^N$ yields

$$\sum_{k=1}^{N} |x_i^k||y_j^k| \leq \left(\sum_{k=1}^{N} |x_i^k|^2 \right)^{1/2} \left(\sum_{k=1}^{N} |y_i^k|^2 \right)^{1/2} = \|x_i\| \|y_j\|,$$

which after taking squares is enough to conclude the proof. □

Lemma 9.5. *The map $M \mapsto M^*M - I$ is continuous on $\mathbb{C}^{N \times N}$. In particular, $\mathrm{U}(N)$ is compact.*

Proof. In this proof we let f denote the map $M \mapsto M^*M - I$. Fix $M \in \mathbb{C}^{N \times N}$. It suffices to prove that f is continuous at M. Let $R > 0$ be an upper bound on $\|M\|$.

Let $\epsilon > 0$. Choose $\delta = \min(R, \frac{\epsilon}{3R})$. Let $X \in \mathbb{C}^{N \times N}$ such that $\|X - M\| \leq \delta$. Then

$$\begin{aligned} \|f(X) - f(M)\| &= \|X^*X - M^*M\| \\ &\leq \|(X^* - M^*)X\| + \|M^*(X - M)\| \\ &\leq \|X^* - M^*\| \|X\| + \|M^*\| \|X - M\|. \end{aligned}$$

Notice that the previous display follows similar lines as the proof of Lemma 9.2, but in the third inequality in the present proof we do not have that X and M^* are unitary, so we needed to invoke Lemma 9.4. Next, we have $\|M^*\| = \|M\| < R$ and $\|X\| \leq R + \delta \leq 2R$. Also, $\|X^* - M^*\| = \|X - M\| \leq \delta$, and then

$$\|f(X) - f(M)\| \leq \delta R + 2R\delta \leq 3R \frac{\epsilon}{3R} = \epsilon.$$

This shows that f is continuous at M.

Finally note that $\mathrm{U}(N) = \{M \in \mathbb{C}^{N \times N} : f(M) = 0\} = f^{-1}(\{0\})$. Since $\{0\} \subset \mathbb{C}^{N \times N}$ is closed and f is continuous, $\mathrm{U}(N)$ is a closed subset of $\mathbb{C}^{N \times N}$. By previous remarks, $\mathrm{U}(N)$ is a compact topological space. □

To pin down the notion of Haar probability measure on $\mathrm{U}(N)$, we follow the treatment in [8]. Before delving into the rigorous details, we pause for some commentary. Oddly enough, one defines a left Haar measure *after* defining a so-called left Haar integral. The left Haar integral encapsulates the notion of the average effect on a function of left multiplying its argument by all group elements. One then obtains a right Haar integral, which is not in general equal to the left Haar integral, even in reasonable cases. Fortunately, for compact topological groups such as $\mathrm{U}(N)$, the left and right Haar integral coincide, and the resulting Haar measure is finite, which allows us to pin down one probability Haar measure.

Furthermore, the development of Haar integral and Haar measure as expoounded in [8] is *abstract* in the sense that it does not involve computing explicit infinitesimal volume elements for the integration of functions on the given topological group. This means that we do not require any knowledge of the coordinates of $\mathrm{U}(N)$ for defining Haar integral and Haar measure on $\mathrm{U}(N)$. On the other hand, an explicit approach is possible and is taken up for the group $\mathrm{U}(N)$ on pages 194 to 198 in [13]. Later on, when computing small ball probabilities near the identity in $\mathrm{U}(N)$, we will rely on the explicit formulation.

We proceed now with the rigorous details. Let

$$C(\mathrm{U}(N), \mathbb{C}) = \{f : \mathrm{U}(N) \to \mathbb{C} : f \text{ continuous}\}$$

denote the $\mathbb{C}$-vector space of all complex-valued continuous functions on $\mathrm{U}(N)$. Since $\mathrm{U}(N)$ is compact, we have

$$C_c(\mathrm{U}(N), \mathbb{C}) = C(\mathrm{U}(N), \mathbb{C}),$$

where $f \in C_c(\mathrm{U}(N), \mathbb{C})$ means that there exists a compact subset $K \subset \mathrm{U}(N)$ such that $f(x) = 0$ for all $x \in K^c$.

Let

$$C^+(\mathrm{U}(N), \mathbb{C}) = \{f \in C(\mathrm{U}(N), \mathbb{C}) : \mathrm{Im}(f) \subset [0, \infty)\}.$$

be the set of nonnegative continuous functions on $\mathrm{U}(N)$ which does not form a vector space because closure under scalar multiplication does not hold.

For any $V \in \mathrm{U}(N)$, let $L_V : \mathrm{U}(N) \to \mathrm{U}(N)$ denote left multiplication by V, namely, $L_V(U) = VU$ for all $U \in \mathrm{U}(N)$. Continuity of the map L_V follows from the observation that $\|V(U_1 - U_2)\| = \|U_1 - U_2\|$ for $U_1, U_2 \in \mathrm{U}(N)$. In the following result, we use continuity of L_V to see that $f \circ L_V \in C(\mathrm{U}(N), \mathbb{C})$ for all $f \in C(\mathrm{U}(N), \mathbb{C})$.

Lemma 9.6. *There exists a function $I : C(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ with the following properties.*

(i) I is a complex linear functional. In other words, $I(f+g) = I(f) + I(g)$ and $I(\alpha f) = \alpha I(f)$ for all $f, g \in C(\mathrm{U}(N), \mathbb{C})$ and $\alpha \in \mathbb{C}$, and finally, I takes values in $\mathbb{C}$.

(ii) I is left invariant. This means that $I(f) = I(f \circ L_V)$ for all $V \in \mathrm{U}(N)$ and all $f \in C(\mathrm{U}(N), \mathbb{C})$.

(iii) I is non-negative. In other words, $I(f) \geq 0$ if $f \in C^+(\mathrm{U}(N), \mathbb{C})$.

(iv) $I(f) > 0$ if $f \in C^+(\mathrm{U}(N), \mathbb{C})$ and $f(U) \neq 0$ for some $U \in \mathrm{U}(N)$.

(v) I is unique up to a constant multiple. That is, if $J : C(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ satisfies properties (i), (ii), (iii), and J does not vanish identically, then there exists a constant $c > 0$ such that $J(f) = cI(f)$ for all $f \in C(\mathrm{U}(N), \mathbb{C})$.

Proof. Since $\mathrm{U}(N)$ is compact, it is also locally compact. Thus we can apply Theorem 15.5 on page 185 in [8] to obtain $I : C^+(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ with the following properties.

(i)' $I(f) \in (0, \infty)$ if $f \in C^+(\mathrm{U}(N), \mathbb{C})$ and $f(U) \neq 0$ for some $U \in \mathrm{U}(N)$.

(ii)' $I(f+g) = I(f) + I(g)$ for all $f, g \in C^+(\mathrm{U}(N), \mathbb{C})$.

(iii)' $I(\alpha f) = \alpha I(f)$ for all $\alpha \geq 0$ and $f \in C^+(\mathrm{U}(N), \mathbb{C})$.

(iv)' $I(f \circ L_V) = I(f)$ for all $f \in C^+(\mathrm{U}(N), \mathbb{C})$ and all $V \in \mathrm{U}(N)$.

(v)' If $J : C^+(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ satisfies $J(f) \geq 0$ for all $f \in C^+(\mathrm{U}(N), \mathbb{C})$ and properties (ii)', (iii)', and (iv)', and J does not vanish identically, then there exists some $c > 0$ such that $J(f) = cI(f)$ for all $f \in C^+(\mathrm{U}(N), \mathbb{C})$.

Let

$$C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C}) = \{f \in C(\mathrm{U}(N), \mathbb{C}) : \mathrm{Im}(f) \subset \mathbb{R}\},$$

which is closed under $\mathbb{R}$-scalar multiplication, and is thus an $\mathbb{R}$-vector space. We first extend I uniquely to $C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$ and verify that the various properties remain.

For $f \in C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$, let $|f| \in C^+(\mathrm{U}(N), \mathbb{C})$ be the function defined by $|f|(U) = |f(U)|$ for all $U \in \mathrm{U}(N)$. Note that

$$f = \frac{|f| + f}{2} - \frac{|f| - f}{2}, \tag{107}$$

where $\frac{|f|+f}{2}, \frac{|f|-f}{2} \in C^+(\mathrm{U}(N), \mathbb{C})$.

Let $f \in C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$. Then we let

$$I_{\mathbb{R}}(f) := I\left(\frac{|f| + f}{2}\right) - I\left(\frac{|f| - f}{2}\right).$$

Let $f, g \in C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$. Assume without loss of generality that $|f| \geq |g|$. Then the fact that $|f| + |g| = |f + g|$ follows from a case analysis on the sign of f and g. On the other hand, if $|g| \geq |f|$, then a symmetric argument holds.

Then we have

$$\begin{aligned}
I_{\mathbb{R}}(f+g) &= I\left(\frac{|f+g| + f + g}{2}\right) - I\left(\frac{|f+g| - f - g}{2}\right) \\
&= I\left(\frac{|f| + |g| + f + g}{2}\right) - I\left(\frac{|f| + |g| - f - g}{2}\right) \\
&= I\left(\frac{|f| + f}{2}\right) - I\left(\frac{|f| - f}{2}\right) + \\
&\quad I\left(\frac{|g| + g}{2}\right) - I\left(\frac{|g| - g}{2}\right) \\
&= I_{\mathbb{R}}(f) + I_{\mathbb{R}}(g),
\end{aligned}$$

where the third equality follows from property (ii)'.

We note that $I_{\mathbb{R}}(\alpha f) = \alpha I_{\mathbb{R}}(f)$ for all $f \in C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$ and $\alpha \in \mathbb{R}$ follows from considering the cases $\alpha \geq 0$ and $\alpha < 0$.

Let $V \in \mathrm{U}(N)$. Then

$$\begin{aligned} I_{\mathbb{R}}(f \circ L_V) &= I\left(\frac{|f \circ L_V| + f \circ L_V}{2}\right) - I\left(\frac{|f \circ L_V| - f \circ L_V}{2}\right) \\ &= I\left(\frac{|f| + f}{2}\right) - I\left(\frac{|f| - f}{2}\right) \\ &= I_{\mathbb{R}}(f), \end{aligned}$$

where the second equality follows from property (iv)'.

$I_{\mathbb{R}}$ is the unique $\mathbb{R}$-linear extension of I to $C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$. Indeed, if $I'_{\mathbb{R}}$ is some other linear extension, then for $f \in C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$,

$$\begin{aligned} I'_{\mathbb{R}}(f) &= I'_{\mathbb{R}}(\frac{|f| + f}{2} - \frac{|f| - f}{2}) \\ &= I'_{\mathbb{R}}\left(\frac{|f| + f}{2}\right) - I'_{\mathbb{R}}\left(\frac{|f| - f}{2}\right) \\ &= I\left(\frac{|f| + f}{2}\right) - I\left(\frac{|f| - f}{2}\right) = I_{\mathbb{R}}(f). \end{aligned}$$

Now we reformulate property (v)' to the present extension. Suppose $J' : C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ satisfies $J'(f) \geq 0$ for all $f \in C^{+}(\mathrm{U}(N), \mathbb{C})$ and that J' is $\mathbb{R}$-linear and does not vanish identically. Let J denote the restriction of J' to $C^{+}(\mathrm{U}(N), \mathbb{C})$. Then $J' = cI$, where c is given in property (v)'. Note that the unique $\mathbb{R}$-linear extension of cI to $C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$ is $cI_{\mathbb{R}}$. This implies that the unique $\mathbb{R}$-linear extension of J' to $C^{\mathbb{R}}(\mathrm{U}(N), \mathbb{C})$ is also $cI_{\mathbb{R}}$. Hence, $cI_{\mathbb{R}} = J$, as desired.

An application of Theorem B.38 on page 463 of [8] extends $I_{\mathbb{R}}$ to $C(\mathrm{U}(N), \mathbb{C})$, and similar computations show that the other properties carry through. Thus we conclude the proof. □

The functional I in Lemma 9.6 is called a left Haar integral on $\mathrm{U}(N)$. The reason for this name is as follows. Pretending that I is defined on all functions from $\mathrm{U}(N)$ to $\mathbb{C}$, we obtain a left Haar measure σ_L by defining

$$\sigma_L(B) = I(\chi_B)$$

for any Borel set $B \in \mathcal{B}_{\mathrm{U}(N)}$, where $\chi_B(U) = \begin{cases} 1 & U \in B \\ 0 & U \notin B \end{cases}$. Then for any Borel measurable function $f : \mathrm{U}(N) \to \mathbb{C}$, we have

$$I(f) = \int_{\mathrm{U}(N)} f(U) d\sigma_L(U),$$

where the right hand side is defined using the Lebesgue theory of integration.

In addition, we note that the complex linear functional I described in Lemma 9.6 is amenable to the discussion of integration on locally compact Hausdorff spaces in Section 11 of [8] because of properties (i) and (iii) in the same lemma. We rely on many results from Section 11 in the following lemma, where we have removed L from the notation σ_L in foresight of the equality between left and right Haar measure on $\mathrm{U}(N)$.

Lemma 9.7 (Left Haar Probability Measure)**.** *There exists a unique set function*

$$\sigma : \mathcal{B}_{\mathrm{U}(N)} \to [0, 1]$$

with the following properties.

(i) σ is a probability measure.

(ii) $\sigma(B) < \infty$ if B is compact in $\mathrm{U}(N)$.

(iii) $\sigma(B) > 0$ for some open set B in $\mathrm{U}(N)$.

(iv) (Left-invariance) $\sigma(UB) = \sigma(B)$ for all $B \in \mathcal{B}_{\mathrm{U}(N)}$ and $U \in \mathrm{U}(N)$.

(v) (Inner Regularity) For every open set B of $\mathrm{U}(N)$, we have

$$\sigma(B) = \sup\{\sigma(F) : F \text{ is compactly contained in } B\}.$$

(vi) (Outer Regularity) For every $B \in \mathcal{B}_{\mathrm{U}(N)}$ we have

$$\sigma(B) = \inf\{\sigma(V) : V \text{ is open and } B \subset V\}.$$

Proof. The first step is the existence of σ. We extend I from Lemma 9.6 to an appropriately large class of functions, then define σ as I applied to indicator functions, and finally check that σ has all the properties in the present lemma. Here are the details.

We begin with Definition (11.8) from [8]. A function $f : \mathrm{U}(N) \to [0, \infty]$ is said to be lower semicontinuous if the following holds for every matrix $U_0 \in \mathrm{U}(N)$. If $f(U_0) < \infty$, then for every $\epsilon > 0$, there is a neighborhood B of U_0 in $\mathrm{U}(N)$ such that $f(U) > f(U_0) - \epsilon$ for all $U \in B$. If $f(U_0) = \infty$, the for every positive number A, there is a neighborhood B of U_0 such that $f(U) > A$ for all $U \in B$. Then we define

$$\mathcal{M}^+ = \{f : \mathrm{U}(N) \to [0, \infty] : f \text{ is lower semicontinuous}\}.$$

We continue with Definition (11.11) from [8]. Let $\bar{I} : \mathcal{M}^+ \to [0, \infty]$ be given by

$$\bar{I}(f) = \sup\{I(g) : g \in C^+(\mathrm{U}(N), \mathbb{C}), g \leq f\}$$

for any $f \in \mathcal{M}^+$. That is, we approximate from below by the left Haar integral of non-negative continuous functions.

Next we write down Definition (11.15) from [8]. Let $[0, \infty]^{\mathrm{U}(N)}$ denote the set of all functions from $\mathrm{U}(N)$ to $[0, \infty]$. Then we define $\bar{\bar{I}} : [0, \infty]^{\mathrm{U}(N)} \to [0, \infty]$ by

$$\bar{\bar{I}}(f) = \inf\{\bar{I}(g) : g \in \mathcal{M}^+, g \geq f\}$$

for all $f \in [0, \infty]^{\mathrm{U}(N)}$. That is, we approximate from above by lower semicontinuous functions.

Then we proceed to define σ as a Carathéodory outer measure on the class of all subsets of $\mathrm{U}(N)$: for any subset $B \in 2^{\mathrm{U}(N)}$, we let

$$\sigma(B) = \bar{\bar{I}}(\chi_B).$$

That σ under this definition is truly a Carathéodory outer measure is the content of Theorem (11.21). In particular, $\sigma(\emptyset) = 0$.

Following Definition (11.28) and invoking Theorem (11.29), we define the sigma algebra

$$\mathcal{F}_\sigma = \{B \subset \mathrm{U}(N) : \sigma(S) \geq \sigma(S \cap B) + \sigma(S \cap B^c), \forall S \in 2^{\mathrm{U}(N)}\},$$

and we have that $\sigma : \mathcal{F}_\sigma \to [0, \infty]$ is a sigma additive set function on a sigma algebra taking the empty set to 0, which implies that σ is a measure on the space $(\mathrm{U}(N), \mathcal{F}_\sigma)$.

Theorem (11.30) says that $\mathcal{B}_{\mathrm{U}(N)} \subset \mathcal{F}_\sigma$ so that by restricting σ to $\mathcal{B}_{\mathrm{U}(N)}$, we obtain a measure on $(\mathrm{U}(N), \mathcal{B}_{\mathrm{U}(N)})$.

Next, we argue that $\sigma(\mathrm{U}(N)) < \infty$. We have $\chi_{\mathrm{U}(N)} \in C^+(\mathrm{U}(N), \mathbb{C})$. Next, we apply Lemma 9.6 part (iii) to obtain $I(\chi_{\mathrm{U}(N)}) \in (0, \infty)$. We make the following note about how this uses compactness of $\mathrm{U}(N)$. We formulated Lemma 9.6 in terms of continuous functions because they coincide with the class of compactly supported continuous functions. In other words, for a noncompact group G, we would have formulated Lemma 9.6 in terms of compactly supported continuous functions on G, and then χ_G would not be amenable to the revised Lemma 9.6.

Since $\sigma(\mathrm{U}(N)) < \infty$, let us assume without loss of generality that $\sigma(\mathrm{U}(N)) = 1$ by normalizing by $1/\sigma(\mathrm{U}(N))$. Therefore, we obtain property (i) of the present lemma.

Let K be a compact subset of $\mathrm{U}(N)$. Then we have $\sigma(K) \leq \sigma(\mathrm{U}(N)) < \infty$. Thus we obtain property (ii) of the present lemma.

Let B be a non-empty open subset of $\mathrm{U}(N)$. Let $U \in B$. There exists a continuous function $f \in C^+(\mathrm{U}(N), \mathbb{C})$ such that $f(U) = 1$ and $f \leq \chi_B$. Indeed, B^c is closed, and by the metric nature of $\tau_{\mathrm{U}(N)}$, there is is some closed ball B' such that $U \in B' \subset B$. Then by Urysohn's lemma, there is a continuous function $f : \mathrm{U}(N) \to [0, 1]$ for which $f(B') = \{1\}$ and $f(B^c) = \{0\}$. Then we see that $f \leq \chi_B$. Then by Lemma 9.6 and the definition of $\bar{\bar{I}}$, we see that

$$\sigma(B) = \bar{\bar{I}}(\chi_B) \geq I(f) > 0.$$

Here is the inequality $\bar{\bar{I}}(\chi_B) \geq I(f)$ in more detail. Let $g \in \mathcal{M}_+$ majorize χ_B. Since f minorizes χ_B, then by definition of $\bar{I}(g)$, we have $\bar{I}(g) \geq I(f)$. In particular, g being arbitary in this argument implies that $I(f)$ is a lower bound of the set $\{\bar{I}(g) : g \in \mathcal{M}^+, g \geq \chi_B\}$. Thus, $I(f) \leq \inf\{\bar{I}(g) : g \in \mathcal{M}^+, g \geq \chi_B\} = \bar{\bar{I}}(\chi_B)$. Thus we obtain a stronger version of property (iii) of the present lemma, as we have proven it for any non-empty open subset of $\mathrm{U}(N)$.

Now we come to the key property of σ, its left invariance. Fix $U \in \mathrm{U}(N)$ and let $B \in \mathcal{B}_{\mathrm{U}(N)}$. We want to show that $\sigma(UB) = \sigma(B)$, which is equivalent to proving that $\bar{\bar{I}}(\chi_{UB}) = \bar{\bar{I}}(\chi_B)$. Next, the equality $\chi_{UB} = \chi_B \circ L_U$ reduces the problem to proving that $\bar{\bar{I}}(\chi_B \circ L_U) = \bar{\bar{I}}(\chi_B)$. A short argument shows that $\mathcal{M}_+ = \{f \circ L_U : f \in \mathcal{M}_+\}$. We can use this to unravel the definition of $\bar{\bar{I}}(\chi_B \circ L_U)$ as follows:

$$\begin{aligned}
\bar{\bar{I}}(\chi_B \circ L_U) &= \inf\{\bar{I}(g) : g \in \mathcal{M}^+, g \geq \chi_B \circ L_U\} \\
&= \inf\{\bar{I}(g) : g \in \{f \circ L_U : f \in \mathcal{M}_+\}, g \geq \chi_B \circ L_U\} \\
&= \inf\{\bar{I}(f \circ L_U) : f \in \mathcal{M}_+, f \circ L_U \geq \chi_B \circ L_U\} \\
&= \inf\{\bar{I}(f \circ L_U) : f \in \mathcal{M}_+, f \geq \chi_B\},
\end{aligned}$$

where the last equality uses that $f \circ L_U \geq \chi_B \circ L_U$ if and only if $f \geq \chi_B$. By the definition of $\bar{\bar{I}}(\chi_B)$ compared to the last expression in the previous display, it suffices to prove that $\bar{I}(f \circ L_U) = \bar{I}(f)$ for all $f \in \mathcal{M}^+$. This follows from an application of Lemma 9.6 and the observation that $\{g \circ L_U : g \in C^+(\mathrm{U}(N), \mathbb{C})\} = C^+(\mathrm{U}(N), \mathbb{C})$. Hence, we obtain property (iv) of the present lemma.

Additionally, inner regularity, or property (v) is the content of Theorem (11.23). Meanwhile, outer regularity, or property (vi), is the content of Theorem (11.22). This concludes the existence step of the proof.

The second step is uniqueness. Suppose $\mu : \mathcal{B}_{\mathrm{U}(N)} \to [0, 1]$ satisfies properties (i) - (vi) of the present lemma. We define $J : C(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ as Lebesgue integration against μ:

$$J(\phi) = \int_{\mathrm{U}(N)} \phi \, d\mu$$

for all $\phi \in C(\mathrm{U}(N), \mathbb{C})$. One can check that J is a complex linear functional, is non-negative, and is strictly positive for non-zero and non-negative continuous functions. That is, J satisfies properties (i), (iii), and (iv) of Lemma 9.6. Let us study property (ii) of Lemma 9.6, which is left invariance of J.

Fix $\phi \in C(\mathrm{U}(N), \mathbb{C})$. Then $\phi = \mathrm{Re}(\phi) + i\,\mathrm{Im}(\phi)$, and both of these real and imaginary components are continuous. Note that the Lebesgue integral $J(\phi)$ is defined as $J(\mathrm{Re}(\phi)) + iJ(\mathrm{Im}(\phi))$. Furthermore, since any continuous function f taking values in $\mathbb{R}$ can be written as a difference of two non-negative continuous functions $f_+ = \max(f, 0)$ and $f_- = -\min(f, 0)$, which is the same as (107), then by definition of the Lebesgue integral of a function taking values in $\mathbb{R}$, we obtain

$$J(\phi) = J(\mathrm{Re}(\phi)_+) - J(\mathrm{Re}(\phi)_-) + iJ(\mathrm{Im}(\phi)_+) - iJ(\mathrm{Im}(\phi)_-).$$

This shows that it suffices to prove the following version of translation invariance for J: if $f \in C^+(\mathrm{U}(N), \mathbb{C})$ and $U \in \mathrm{U}(N)$, then $J(f) = J(f \circ L_U)$.

Fix $f \in C^+(\mathrm{U}(N), \mathbb{C})$ and $U \in \mathrm{U}(N)$. We have

$$\begin{aligned}
J(f) = \int_{\mathrm{U}(N)} f \, d\mu &= \sup\{\sum_{k=1}^{n} [\inf(f(V) : V \in A_k)]\mu(A_k) : \\
&\qquad \{A_1, \cdots, A_n\} \text{ is a partition of } \mathrm{U}(N) \text{ with each } A_k \in \mathcal{B}_{\mathrm{U}(N)}\}. \\
&= \sup\{\sum_{k=1}^{n} [\inf((f \circ L_U)(V) : V \in U^* A_k)]\mu(U^* A_k) : \\
&\qquad \{U^* A_1, \cdots, U^* A_n\} \text{ is a partition of } \mathrm{U}(N) \text{ with each } U^* A_k \in \mathcal{B}_{\mathrm{U}(N)}\}. \\
&= \int_{\mathrm{U}(N)} f \circ L_U \, d\mu = J(f \circ L_U),
\end{aligned}$$

where the second equality used left invariance of the measure μ.

Note (11.38) in [8] gives us the following statement. If (1) σ and μ both satisfy inner and outer regularity, (2) σ and μ are both finite on compact sets, and (3) $\int_{\mathrm{U}(N)} \phi \, d\sigma = \int_{\mathrm{U}(N)} \phi \, d\mu$ for all $\phi \in C^+(\mathrm{U}(N), \mathbb{C})$, then $\sigma = \mu$. We have already obtained assumptions (1) and (2). Let us check that assumption (3) holds. Fix $\phi \in C^+(\mathrm{U}(N), \mathbb{C})$. By (11.37), the Riesz Representation Theorem, we have

$$\int_{\mathrm{U}(N)} \phi \, d\sigma = I(\phi).$$

Next, since I and J both satisfy the conditions in Lemma 9.6, there is a positive constant $c > 0$ such that $I = cJ$. We solve for c by choosing the continuous function $1 := \chi_{\mathrm{U}(N)} \in C^+(\mathrm{U}(N), \mathbb{C})$. On the one hand $I(1) = \bar{\bar{I}}(1) = \sigma(\mathrm{U}(N)) = 1$. On the other hand, $cJ(1) = c\int_{\mathrm{U}(N)} d\mu = c\mu(\mathrm{U}(N)) = c$. Therefore, $c = 1$. Thus,

$$\int_{\mathrm{U}(N)} \phi \, d\sigma = I(\phi) = J(\phi) = \int_{\mathrm{U}(N)} \phi \, d\mu,$$

which is assumption (3). Applying the statement (11.38), we obtain $\sigma = \mu$, which concludes the proof of the present lemma. □

Now, using the Lebesgue theory of integration, we can integrate any Borel measurable function $f : \mathrm{U}(N) \to \mathbb{C}$ against left Haar measure σ. Up to this point however, the key formula

$$\int_{\mathrm{U}(N)} f(VU) \, d\sigma(U) = \int_{\mathrm{U}(N)} f(U) \, d\sigma(U) \tag{108}$$

stands for any given $V \in \mathrm{U}(N)$ only when $f \in C(\mathrm{U}(N), \mathbb{C})$. Here we used Lemma 9.6 (ii) and the Riesz representation theorem: $I(f) = \int_{\mathrm{U}(N)} f(U) \, d\sigma(U)$. But we can extend this formula by noting that for Borel measurable $f : \mathrm{U}(N) \to [0, \infty]$, we have by (11.36) in [8] the equality

$$\bar{\bar{I}}(f) = \int_{\mathrm{U}(N)} f(U) \, d\sigma(U).$$

Then arguing as we did for left invariance of σ in the proof of Lemma 9.7, we see that $\bar{\bar{I}}(f \circ L_U) = \bar{\bar{I}}(f)$ for any $U \in \mathrm{U}(N)$, so that the formula 108 holds for any Borel measurable $f : \mathrm{U(N)} \to [0, \infty]$. To obtain the formula for any Borel measurable $f : \mathrm{U(N)} \to \mathbb{C}$, we decompose f into real and imaginary parts, then into positive and negative parts, and apply formula 108 to each part separately.

Next, we discuss right invariance. For each $U \in \mathrm{U}(N)$, let $R_U : \mathrm{U}(N) \to \mathrm{U}(N)$ be the map $V \mapsto VU$. These maps are continuous for the same reason that the left multiplication maps are continuous. A right Haar integral on $\mathrm{U}(N)$ is a function $I' : C(\mathrm{U}(N), \mathbb{C}) \to \mathbb{C}$ which is a non-negative, non-zero, complex linear functional (properties (i), (iii), and (iv) of Lemma 9.6), and satisfies right invariance:

$$I'(f \circ R_U) = I'(f)$$

for all $f \in C(\mathrm{U}(N), \mathbb{C})$ and $U \in \mathrm{U}$. Uniqueness up to a multiplicative constant also holds for right Haar integral I'. We can obtain a right Haar integral I' from normalized left Haar integral I as follows. We set

$$I'(f) = I(f \circ (\cdot)^{-1}).$$

It turns out that for compact topological groups, $I(f \circ (\cdot)^{-1}) = I(f)$, so that $I' = I$. That is, left Haar integral and right Haar integral coincide, given this way of obtaining a right Haar integral. See Theorems (15.13) and (15.15) in [8]. The relevant consequences are that the left Haar probability measure σ is also right invariant:

$$\sigma(BV) = \sigma(B)$$

for all $B \in \mathcal{B}_{\mathrm{U}(N)}$ and $V \in \mathrm{U}(N)$ and that for all Borel measurable functions $f : \mathrm{U}(N) \to \mathbb{C}$, we have

$$\int_{\mathrm{U}(N)} f \, d\sigma = \int_{\mathrm{U}(N)} f \circ R_V \, d\sigma$$

for all $V \in \mathrm{U}(N)$. This concludes our discussion on Haar measure.

9.2 $\mathrm{U}(N)$ as a Smooth Manifold

A smooth map $F : M \to N$ between smooth manifolds M and N is called a smooth immersion if $\operatorname{rank} F = \dim M$ [12, Section 4]. If M and N are smooth manifolds, a smooth embedding $F : M \to N$ is a smooth immersion that is also a topological embedding, i.e. a homeomorphism onto its image in the subspace topology [12, Section 4]. If U is an open subset of $\mathbb{R}^m$ (for some non-negative integer m) and $k \in \{0, \cdots, m\} = \{0\} \cup [m]$, then a k-slice is any subset $S \subset U$ of the form

$$S = \{(x^1, \cdots, x^k, x^{k+1}, \cdots, x^m) \in U : x^{k+1} = c^{k+1}, \cdots, x^m = c^m\}$$

for some constants $c^{k+1}, \cdots, c^m$ [12, Section 5]. We will consider k-slices where some other $m-k$ element subset of the coordinates besides the last $m-k$ coordinates are fixed, and the definition for this is similar.

Next we record the analagous definition of k slices for manifolds. Let M be a smooth m-manifold, and let (U, ϕ) be a smooth chart on M. If S is a subset of U such that $\phi(S) \subset \mathbb{R}^m$ is a k-slice of $\phi(U) \subset \mathbb{R}^m$, then we say that S is a k-slice of U [12, Section 5].

Next we set down the definition of the so called local k-slice condition, which is a property that can be ascribed to a given subset of a manifold. Given a subset $S \subset M$ and a non-negative integer k, we say that S satisfies the local k-slice condition if each point of S is contained in the domain of a smooth chart (U, ϕ) for M such that $S \cap U$ is a single k-slice in U [12, Section 5].

Next we state the following theorem, which is what we will employ after demonstrating that $\mathrm{U}(N)$ as a subset of $\mathbb{C}^{N\times N}$ (which we view as a $2N^2$ dimensional real manifold) satisfies the local $k = N^2$ slice condition.

Theorem 9.8. *Let M be a smooth m-manifold. If $S \subset M$ is a subset that satisfies the local k-slice condition, then with the subspace topology, S is a topological manifold of dimension k, and it has a smooth structure making it into a k-dimensional embedded submanifold of M [12, Section 5, Theorem 5.8].*

For $X \in \mathbb{C}^{N\times N}$ and $r \geq 0$, let $b(X, r) = \{Y \in \mathbb{C}^{N\times N} : \|Y - X\| < r\}$ be the open ball of radius r centered at X in the space of $N \times N$ complex matrices.

For $X \in b(I, 1)$, let

$$\log(X) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(X-I)^n}{n}$$

where the right hand side is a limit of partial sums taken in the space $\mathbb{C}^{N\times N}$ equipped with the Hilbert Schmidt matrix norm.

Lemma 9.9. *The function $\log : b(I, 1) \to \mathbb{C}^{N\times N}$ is well-defined and smooth as a map from the $2N^2$ dimensional real manifold $b(I, 1)$ to the $2N^2$ dimensional real manifold $\mathbb{C}^{N\times N}$.*

Proof. First we prove that log is well-defined on $b(I, 1)$. Fix $X \in b(I, 1)$. Then $\|X - I\| \leq r < 1$ for some real number r. Let $S_M = \sum_{n=1}^{M} (-1)^{n+1} \frac{(X-I)^n}{n}$, which is an element of $\mathbb{C}^{N\times N}$. Then $(S_M)_{M\geq 1}$ is a Cauchy sequence in $(\mathbb{C}^{N\times N}, \|\cdot\|)$. To see this, we assume without loss of generality that $M_1 \leq M_2$ and then observe that

$$\|S_{M_1} - S_{M_2}\| \leq \sum_{j=M_1+1}^{\infty} \frac{\|X - I\|^j}{j}.$$

The term on the right hand side is bounded using

$$\sum_{j=M_1+1}^{\infty} \frac{\|X - I\|^j}{j} \leq \sum_{j=M_1+1}^{\infty} \frac{r^j}{j}.$$

The radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{x^n}{n}$ is equal to $\frac{1}{\limsup_{n\to\infty} (1/n)^{1/n}} = 1$. Since r falls within the radius of convergence of the series in the previous sentence, we see that $\sum_{n=1}^{\infty} \frac{r^n}{n} < \infty$, which implies that its tail goes to zero. That is, as $M_1 \to \infty$, we have $\sum_{j=M_1+1}^{\infty} \frac{r^j}{j} \to 0$. Thus, S_M is a Cauchy sequence. Since $\mathbb{C}^{N\times N}$ is complete and limits in this space are unique, there is a unique matrix, call it $\log(X)$, such that $\lim_{M\to\infty} S_M = \log(X)$. This proves that log is well-defined on $b(I, 1)$.

Next we study smoothness. Note that for each input X, the output $\log(X)$ has N^2 complex entries. Fix one of these entries, say entry (j, k) for some $1 \leq j, k \leq N$. We study the associated multi-variable complex function $F : b(I, 1) \subset \mathbb{C}^{N^2} \to \mathbb{C}$ given by $F(X) = (\log(X))_{jk}$ for all $X \in b(I, 1)$. The first obervation is that F is a multi-variate power series expanded at the point $I \in \mathbb{C}^{N\times N}$. Since F is defined in this way, it is automatically holomorphic as the limit of the partial sums, which converge uniformly on compact sets. We conclude that F is smooth as a function from $\mathbb{R}^{2N^2} \to \mathbb{R}^2$. (For a more detailed argument, see [10, p. 100].) Since j, k were arbitary, we see that $\log(X)$ is smooth in all of its entries, and is therefore jointly smooth. □

Let $\mathfrak{u}(N) = \{X \in \mathbb{C}^{N\times N} : X^* + X = 0\}$ be the skew-Hermitian matrices. Note that $\mathfrak{u}(N) = i \cdot \mathrm{H}(N)$.

Lemma 9.10. *The logarithm of any unitary matrix which is 1-close to the identity matrix is a skew Hermitian matrix. That is, for any $U \in \mathrm{U}(N) \cap b(I, 1)$, we have $\log(U) \in \mathfrak{u}(N)$.*

Proof. Let $U = VDV^*$ be an eigendecomposition for U, where $D = \operatorname{diag}(e^{i\theta_1}, \cdots, e^{i\theta_N})$, for $\theta_1, \cdots, \theta_N \in (-\pi, \pi)$, and $V \in \mathrm{U}(N)$.

Here is why $e^{i\theta_j} \neq -1$ for all $j = 1, \cdots, N$. First note that $(VDV^* - I) = V(DV^* - V^*) = V(D-I)V^*$ by unitarity of V. Hence, $\|I - U\| = \|I - D\|$. We have $\|I - D\|^2 = \sum_{jk} |(I-D)_{jk}|^2$. Suppose $e^{i\theta_m} = -1$ at entry mm in D. Then $|(I-D)_{mm}|^2 = |2|^2 = 4$. Then we have

$$\|I - D\|^2 = \sum_{jk} |(I-D)_{jk}|^2 \geq 4.$$

But $U \in b(I, 1)$, and we have a contradiction. This will be helpful when we consider the principal branch of the logarithm, which is defined on $\mathbb{C} \setminus (-\infty, 0]$.

For seeing how matrices V pass through the logarithm, we note that $(V(D-I)V^*)^n = V(D-I)^n V^*$, for n a positive integer. Then we have

$$\begin{aligned} \log(VDV^*) &= \lim_{N \to \infty} \sum_{n=1}^{N} (-1)^{n+1} \frac{V(D-I)^n V^*}{n} \\ &= V \log(D) V^*. \end{aligned}$$

We know that $\log(D) = \operatorname{diag}(\log(\lambda_1), \cdots, \log(\lambda_N))$, where $\lambda_j = e^{i\theta_j}$. Using the principal branch of the logarithm, and employing the fact that $\theta_j \neq \pi$ (since $U \in b(I, 1)$), we have $\log(\lambda_j) = i\theta_j$ for all $j = 1, \cdots, N$.

Note that $\log(D)^* = \operatorname{diag}(-i\theta_1, \cdots, -i\theta_n)$, which, in turn, is equal to $-\log(D)$. Therefore, we have

$$\begin{aligned} \log(U)^* &= (V \log(D) V^*)^* \\ &= V \log(D)^* V^* \\ &= -V \log(D) V^* \\ &= -\log(U). \end{aligned}$$

This shows that $\log(U)$ is skew-Hermitian whenever $U \in \mathrm{U}(N) \cap b(I, 1)$. □

Lemma 9.10 can also be seen by noting that log is the inverse of $\exp_{|\log(b(I,1))}$ and that $U = e^{iH}$ for some Hermitian matrix $H \in \mathrm{H}(N)$.

Let $\Psi : \mathbb{C}^{N \times N} \to \mathbb{R}^{2N^2}$ denote an identification between $\mathbb{C}^{N \times N}$ and $\mathbb{R}^{2N^2}$. In the case $N = 2$, for instance, we suppose that

$$\Psi\left(\begin{bmatrix} a+ib & c+id \\ e+if & g+ih \end{bmatrix}\right) = (a, b, c, d, e, f, g, h)^T.$$

$\mathrm{H}(N)$ is nearly an N^2 slice of $\mathbb{C}^{N \times N}$. An analagous situation is the question of writing $\{x = y\} \subset \mathbb{R}^2$ as a 1-slice of $\mathbb{R}^2$. To do so, we just rotate the coordinate axes by $\pi/4$ radians so that the line $x = y$ falls on the x axis. We only need to modify this idea slightly to see that $\mathrm{H}(N)$, in real coordinates, is an N^2 slice of $\mathbb{R}^{2N^2}$.

Let us see how to do this when $N = 2$. Let $\varphi : \mathbb{R}^8 \to \mathbb{R}^8$ be the linear map given by

$$\varphi\left(\begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \cos(\pi/4) & 0 & -\sin(\pi/4) & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos(\pi/4) & 0 & -\sin(\pi/4) & 0 & 0 \\ 0 & 0 & \sin(\pi/4) & 0 & \cos(\pi/4) & 0 & 0 & 0 \\ 0 & 0 & 0 & \sin(\pi/4) & 0 & \cos(\pi/4) & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \\ e \\ f \\ g \\ h \end{bmatrix}.$$

Let $H \in \mathrm{H}(N)$ be a Hermitian matrix, so that

$$H = \begin{bmatrix} a+ib & c+id \\ e+if & g+ih \end{bmatrix} = \begin{bmatrix} a & c+id \\ c-id & g \end{bmatrix}.$$

Then we have

$$\varphi(\Psi(H)) = \varphi\left(\begin{bmatrix} a \\ 0 \\ c \\ d \\ c \\ -d \\ g \\ 0 \end{bmatrix}\right) = \begin{bmatrix} a \\ 0 \\ 0 \\ \sqrt{2}d \\ \sqrt{2}c \\ 0 \\ g \\ 0 \end{bmatrix} \in \{x \in \mathbb{R}^8 : x_2, x_3, x_6, x_8 = 0\}.$$

Generalizing these ideas to any N, we obtain a linear map $\varphi : \mathbb{R}^{2N^2} \to \mathbb{R}^{2N^2}$ such that $\varphi(\Psi(\mathrm{H}(N)))$ is an N^2 slice of $\mathbb{R}^{2N^2}$.

By Lemma 9.9, the map

$$\varphi \circ \Psi \circ (-i \cdot \log) : b(I, 1) \to \mathbb{R}^{2N^2} \tag{109}$$

is smooth. Let us name this map ϕ and note that ϕ is compatible with the standard smooth structure on $\mathbb{C}^{N\times N}$, so that $(b(I, 1), \phi)$ constitutes a smooth chart for $\mathbb{C}^{N\times N}$. It follows from Lemma 9.10 and the fact that $\varphi(\Psi(\mathrm{H}(N)))$ is an N^2 slice of $\mathbb{R}^{2N^2}$ that

$$\phi(\mathrm{U}(N) \cap b(I, 1))$$

is an N^2 slice of $\phi(b(I, 1)) \subset \mathbb{R}^{2N^2}$. In other words, $\mathrm{U}(N) \cap b(I, 1)$ is an N^2 slice of $b(I, 1)$.

Lemma 9.11. *The subset $b(I, 1) \cap \mathrm{U}(N)$ is an N^2 dimensional embedded submanifold of $\mathbb{C}^{N\times N}$. Furthermore, $\mathrm{U}(N)$ is an N^2 dimensional embedded submanifold of $\mathbb{C}^{N\times N}$.*

Proof. The first claim follows from the discussion above and Theorem 9.8, because for each matrix $V \in b(I, 1)\cap\mathrm{U}(N)$, we have that $(b(I, 1)\cap\mathrm{U}(N), \phi)$, where ϕ is defined by (109), is an N^2-slice of $b(I, 1)$.

For the second claim, let $V \in \mathrm{U}(N)$ be arbitrary. By Lemma 9.2, which says that matrix multiplication is continuous, we have that $V \cdot b(I, 1) = b(V, 1)$ is open in $\mathbb{C}^{N\times N}$ and contains the matrix V.

Next we need a smooth chart $\phi_V : b(V, 1) \to \mathbb{R}^{2N^2}$ of the ambient manifold $\mathbb{C}^{N\times N}$ such that $\phi_V(b(V, 1) \cap \mathrm{U}(N))$ is an N^2 slice of $\phi_V(b(V, 1))$.

The first device is a translation of elements of $b(V, 1)$ within the domain of the log power series. Note that $X \in b(V, 1)$ implies $V^*X \in b(I, 1)$. In particular, the map $\log \circ L_{V^*} : V \cdot b(I, 1) \to \mathbb{C}^{N\times N}$, where L_{V^*} is left multiplication by V^*, is well-defined. It satisfies the formula

$$\log(L_{V^*}(X)) = \log(V^*X) = \sum_{k=1}^{\infty}(-1)^{k+1}\frac{(V^*X - I)^k}{k}$$

for $X \in b(V, 1)$. Note also that L_{V^*} is smooth with regards to the standard smooth structure on $\mathbb{C}^{N\times N}$.

Consider the map $\phi_V : b(V, 1) \to \mathbb{R}^{2N^2}$ given by

$$\phi_V = \varphi \circ \Psi \circ (-i \cdot \log) \circ L_{V^*} = \phi \circ L_{V^*},$$

where φ is the rotation linear map from earlier, Ψ is the identification of $\mathbb{C}^{N\times N}$ with $\mathbb{R}^{2N^2}$, and ϕ is defined by (109). Then $(b(V, 1), \phi_V)$ is a smooth chart in $\mathbb{C}^{N\times N}$, and $b(V, 1) \cap \mathrm{U}(N)$ is an N^2 slice of $b(V, 1)$ with regards to the chart ϕ_V.

Since V was arbitrary, we see that $\mathrm{U}(N)$ satisfies the local N^2 slice condition. By Theorem 9.8, we conclude the second claim. □

9.3 Limits involving two parameters

The goal of this subsection is to distill the meaning of the term

$$\limsup_{\substack{n\to\infty \\ \beta\to\infty}} f(n, \beta)$$

and the associated growth rates of n and β. This is intended as a direct aid to section 8. This section has been thought of independently by the author, and may or may not be "standard" material, e.g. Analysis 1. Let $\mathbb{R}_+ = (0, \infty)$. Let $f : \mathbb{N} \times \mathbb{R}_+ \to \mathbb{R}$ be a function. Let $\mathbb{N} = \{1, 2, 3, \cdots\}$. A natural definition is

$$\limsup_{\substack{n\to\infty \\ \beta\to\infty}} f(n, \beta) = \inf_{(j^*,\beta^*)\in\mathbb{N}\times\mathbb{R}_+} \sup\{f(n, \beta) \in \mathbb{R} : (n, \beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\}. \tag{110}$$

Note that with respect to the lexicographic ordering on $\mathbb{N} \times \mathbb{R}_+$, the supremum on the right hand side decreases as (j^*, β^*) increases, because the index set gets smaller. This is why the right hand side corresponds to $\beta \to \infty$ and $n \to \infty$. The definition is natural because as (j^*, β^*) increases, parameter space gets "sweeped over." For example, any growth relationship between n and β in which they both tend to infinity will be covered. It is also analagous to the definition

$$\limsup_{k\to\infty} a_k = \inf_{j\in\mathbb{N}} \sup_{k\geq n} a_k$$

for sequences with one parameter.

However, 110 is not directly used in section 8. A sequential notion takes its place.

Lemma 9.12. *Suppose $L \in [-\infty, \infty]$. Then we have*

$$\inf_{(j^*,\beta^*)\in\mathbb{N}\times\mathbb{R}_+} \sup\{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\} \leq L, \tag{111}$$

that is, $\limsup_{\substack{n\to\infty \\ \beta\to\infty}} f(n,\beta) \leq L$, if and only if, for any pair of sequences $(n_j)_{j\in\mathbb{N}} \in \mathbb{N}^{\mathbb{N}}$, $(\beta_j)_{j\in\mathbb{N}} \in \mathbb{R}_+^{\mathbb{N}}$ satisfying $n_j \to \infty$, $\beta_j \to \infty$ as $j \to \infty$, we obtain

$$\limsup_{k\to\infty} f(n_k, \beta_k) \leq L. \tag{112}$$

Proof. First we show that 111 implies 112. Let $(n_j)_{j\in\mathbb{N}} \in \mathbb{N}^{\mathbb{N}}$, $(\beta_j)_{j\in\mathbb{N}} \in \mathbb{R}_+^{\mathbb{N}}$ satisfy $n_j \to \infty$, $\beta_j \to \infty$ as $j \to \infty$. Fix $(j^*, \beta^*) \in \mathbb{N} \times \mathbb{R}_+$. Since n_j and β_j go to infinity, there is some k sufficiently large so that

$$\{f(n_j, \beta_j) : j \geq k\} \subset \{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\},$$

and hence

$$\sup_{j\geq k} f(n_j, \beta_j) \leq \sup\{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\},$$

and hence

$$\inf_{k\in\mathbb{N}} \sup_{j\geq k} f(n_j, \beta_j) \leq \sup\{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\}.$$

Since (j^*, β^*) is arbitrary, this shows that $\inf_{k\in\mathbb{N}} \sup_{j\geq k} f(n_j, \beta_j)$ is a lower bound of the set

$$\{\sup\{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\} : (j^*, \beta^*) \in \mathbb{N} \times \mathbb{R}_+\}.$$

Therefore by definition of infimum, we have

$$\inf_{k\in\mathbb{N}} \sup_{j\geq k} f(n_j, \beta_j) \leq \inf_{(j^*,\beta^*)\in\mathbb{N}\times\mathbb{R}_+} \sup\{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\} \leq L.$$

This is 112, as desired.

Now we show that 112 implies 111. We write a proof by contradiction. Suppose there exists $\epsilon > 0$ such that

$$\inf_{(j^*,\beta^*)\in\mathbb{N}\times\mathbb{R}_+} \sup\{f(n,\beta) \in \mathbb{R} : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+ : n \geq j^*, \beta \geq \beta^*\} = L + \epsilon.$$

Let us inductively construct a pair of sequences to break 112. Suppose $j = 1$. Observe that

$$\sup\{f(n,\beta) : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+, n \geq 1, \beta \geq 1\} \geq L + \epsilon.$$

Hence, there is some pair $(n_1, \beta_1) \in \mathbb{N} \times \mathbb{R}_+$ such that $f(n_1, \beta_1) \geq L + \frac{\epsilon}{2}$ and $n_1 \geq 1$, $\beta_1 \geq 1$. Now suppose that $(n_j, \beta_j) \in \mathbb{N} \times \mathbb{R}_+$ have been assigned for all $1 \leq j \leq N$ such that $n_j \geq j$, $\beta_j \geq j$, and $f(n_j, \beta_j) \geq L + \frac{\epsilon}{2}$. We have

$$\sup\{f(n,\beta) : (n,\beta) \in \mathbb{N} \times \mathbb{R}_+, n \geq N+1, \beta \geq N+1\} \geq L + \epsilon.$$

so there exists some $(n_{N+1}, \beta_{N+1}) \in \mathbb{N} \times \mathbb{R}_+$ with $n_{N+1} \geq N+1$ and $\beta_{N+1} \geq N+1$ such that $f(n_{N+1}, \beta_{N+1}) \geq L + \frac{\epsilon}{2}$. This completes the induction. Therefore $n_j \to \infty$ and $\beta_j \to \infty$, but $\limsup_{j\to\infty} f(n_j, \beta_j) \geq L + \frac{\epsilon}{2}$, a contradiction. □

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